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This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. $N$] shows where page $N$ of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM13.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proof of (2.1), page 69: the proof is added.
2. **Missing proof.** Proof of (2.2), page 69: the proof is added.
3. **Missing proof.** Proof of (2.3), page 70: the proof is added.
4. **Missing proof.** Algorithm (3.3), step (3), page 73: the check that (i)–(iii) still hold is added.
5. **Missing proof.** Algorithm (3.3), step (6), page 74: the proof that deleting an edge lowers the branch-width by at most one is added.
6. **Proof gap.** Proof of (4.3), step (1), page 77: $G_1$ may have up to $\delta + 1$ mutually parallel edges, not $\delta$, and a reason is given for "as is easily seen".
7. **Missing proof.** Claim (1) in the proof of (5.3), page 80: the check is added.

8. **Missing proof.** Section 7, page 89: the construction of a wall from a model of an elementary wall, in time $O(\varepsilon)$, is added.
9. **Missing proof.** Proof of (7.3), page 90: the proof is added.
10. **Proof gap.** Proof of (7.6), page 92: a reason is added why the walls $H_1, \dots, H_k$ exist, that is, why $H$ is high enough.
11. **Proof gap.** Claim (1) in the proof of (7.6), page 92: a bound is added on the distance between the new ends $w_j$ and $w_{j'}$.
12. **Proof gap.** Claim (1) in the proof of (7.6), page 92: the case where the center $v$ lies on $Q_j$ is added.
13. **Proof gap.** Proof of (7.6), page 92: the proof of Claim (1) is used once, from $H_0$ to $H_k$, instead of Claim (1) at each step.
14. **Missing proof.** Proof of (7.7), page 93: the algorithm is described, with its running time.
15. **Missing proof.** Proof of (8.4), page 95: the proof is added.
16. **Missing proof.** Proof of (9.2), page 99: the method is added by which the algorithm finds a graph that may replace $F$, and searches it.
17. **Proof gap.** Algorithm (10.1), page 107: the width $\omega + q$ becomes $\max(\omega, 2) + q$.
18. **Proof gap.** Algorithm (10.5), step (2), page 109: in the width, $\omega + \xi$ becomes $\max(\omega, 2) + \xi$.

# [p. 65] Graph Minors. XIII. The Disjoint Paths Problem

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We describe an algorithm, which for fixed $k \geq 0$ has running time $O(|V(G)|^3)$, to solve the following problem: given a graph $G$ and $k$ pairs of vertices of $G$, decide if there are $k$ mutually vertex-disjoint paths of $G$ joining the pairs.

## 1. Introduction

Consider the following algorithmic problem:

*Disjoint Paths (DP).* Instance: A graph $G$, and pairs $(s_1, t_1), \ldots, (s_k, t_k)$ of vertices of $G$. Question: Do there exist paths $P_1, \ldots, P_k$ of $G$, mutually vertex-disjoint, such that $P_i$ joins $s_i$ and $t_i$ $(1 \leq i \leq k)$?

This problem was shown to be NP-complete by Karp [8], if $k$ is a variable part of the input. For fixed $k$, however, the problem is more tractable. For instance, in [26, 27, 29] there are polynomial algorithms to solve DP with $k = 2$. (In contrast, the corresponding question for directed graphs $G$, where we seek directed paths $P_1, \ldots, P_k$, is NP-complete even with $k = 2$ [5].) The main result of this paper is that for any fixed $k$ there is a polynomial algorithm to solve DP. It has running time $O(v^3)$, where [p. 66] $v = v(G)$ is defined to be $|V(G)| + 1$. The algorithm is not practically feasible, since it involves the manipulation of enormous constants.

Although the algorithm is presented in this paper in full, the proof of its correctness is not, for that requires a lemma, (10.2), that is not proved in this paper. We postpone proving (10.2) to [24], because its proof is an application of the main results of this series, which will also appear in later papers. A draft version of the full proof of (10.2) has been made available to the referees, but due to its great length ($> 300$ pages) it is unreasonable to expect them to guarantee its correctness.

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As a consequence of our DP algorithm we have a polynomial algorithm for one of Garey and Johnson's [6] open problems, which they call Subgraph Homeomorphism (for Fixed Graph $H$), the following. A *subdivision* of a graph $H$ is a graph which may be obtained from a graph isomorphic to $H$ by repeatedly replacing an edge by two edges in series and a new vertex of valency 2. Let $H$ be a fixed graph. The problem is:

*$H$-Subdivision Subgraph.* Instance: A graph $G$. Question: Is there a subgraph of $G$ which is a subdivision of $H$?

Clearly one can use the algorithm for DP to obtain an algorithm for *$H$-Subdivision Subgraph* with running time $O(v^{|V(H)|+3})$.

The following problem is more general than DP, but for fixed $k$ there is an $O(v^3)$ algorithm for its solution as well:

*Disjoint Connected Subgraphs.* Instance: A graph $G$ and non-null subsets $Z_1, \ldots, Z_t \subseteq V(G)$ with $\Sigma|Z_i| \leq k$. Question: Do there exist connected subgraphs $G_1, \ldots, G$, of $G$, mutually vertex-disjoint, with $Z_i \subseteq V(G_i)$ $(1 \leq i \leq t)$?

(This does generalize DP; for given the $(s_i, t_i)$'s of DP, we set $Z_i = \{s_i, t_i\}$.)

A graph $H$ is a minor of a graph $G$ if $H$ can be obtained from a subgraph of $G$ by contracting edges. A further result of this paper is that for every fixed graph $H$ there is an $O(v^3)$ algorithm to solve the following:

*$H$-Minor Containment.* Instance: A graph $G$. Question: Is some minor of $G$ isomorphic to $H$?

The fact that there is a polynomial algorithm for this for all $H$ can be deduced from the solvability for all $H$ of *$H$-Subdivision Subgraph*; because for any $H$ there is a finite list $H_1, \ldots, H_k$ of graphs such that a general graph $G$ has a minor isomorphic to $H$ if and only if it contains a subdivision of one of $H_1, \ldots, H_k$. To make the degree of the polynomial independent of $H$ needs a different method, however.

[p. 67] As an application, let $\mathscr{F}$ be a class of graphs such that if $G \in \mathscr{F}$ and $H$ is isomorphic to a minor of $G$ then $H \in \mathscr{F}$. (Such a class is *minor-closed*.) We shall show in [23] that there is a finite list $H_1, \ldots, H_k$ of graphs, such that a general graph $G$ is in $\mathscr{F}$ if and only if it has no minor isomorphic to any of $H_1, \ldots, H_k$. We deduce that there is an $O(v^3)$ algorithm for

*$\mathscr{F}$-Membership.* Instance: A graph $G$. Question: Is $G \in \mathscr{F}$?

For we use the algorithm for *$H$-Minor Containment* to test for the presence of $H_1, \ldots, H_k$ as minors. (This proves the existence of the algorithm; however, we do not know how to *find* the algorithm, even for concrete classes of $\mathscr{F}$, because we do not know how to find $H_1, \ldots, H_k$.) We show also that if $\mathscr{F}$ does not contain every planar graph, there is an $O(v^2)$ algorithm to

determine membership.

For example,

(i) Let $\Sigma$ be a surface; there is an $O(v^3)$ algorithm to test if a graph $G$ may be drawn in $\Sigma$. This improves a result of Filotti, Miller, and Reif [4].

(ii) Let $\omega \geq 0$ be an integer; there is an $O(v^2)$ algorithm to test if a graph $G$ has tree-width $\leq \omega$ (or branch-width $\leq \omega$, similarly). (For definitions, see [15] and Section 3.) This improves a result of Arnborg, Corneil, and Proskurowski [1].

We remark that for both the above examples, the finiteness of the list $H_1, \ldots, H_k$ of excluded minors does not depend on the result of [23]; it is shown in [17, 14], respectively.

A clarification: how can the above algorithms have running time just dependent on $v$ and not on the number of edges? Certainly, for simple graphs, if the number of edges is large enough in terms of $v$ then the problems (i) and (ii) are both easily answered (the answer is "no"). But what about graphs with multiple edges? We would like to be able to construct the underlying simple graph $H$ of a graph $G$ in time which depends only on $|E(H)|$, not on $|E(G)|$. This can of course be done if our graphs are presented with an appropriate data structure, and we assume this is so. Similarly, we shall always assume that we can decide, for a graph $G$ and vertices $u, v$ of $G$, whether there are at least $k$ edges of $G$ joining $u$ and $v$, in time only dependent on $k$.

## 2. Folios

To solve the problems of the last section, we shall solve a common generalization of *Disjoint Connected Subgraphs* and *H-Minor Containment*, [p. 68] which we describe in this section. For maximum generality we work with directed graphs, although that is not necessary for the applications. Thus each edge has a head and a tail (the same, for a loop). The underlying graph of a digraph is obtained by forgetting the directions of its edges. We shall work almost completely in terms of the underlying graph, and shall often fail to distinguish it from the digraph, for notational and other convenience. In particular, a digraph is *connected* if the underlying graph is connected.

Let $G, H$ be both graphs or both digraphs. A *model* $\phi$ of $H$ in $G$ assigns to each edge $e$ of $H$ an edge $\phi(e)$ of $G$, and to each vertex $v$ of $H$ a non-null connected subgraph $\phi(v)$ of $G$, such that

(i) the graphs $\phi(v)$ ($v \in V(H)$) are mutually vertex-disjoint, the edges $\phi(e)$ ($e \in E(H)$) are all distinct, and for $v \in V(H)$ and $e \in E(H)$, $\phi(e) \notin E(\phi(v))$

(ii) for $e \in E(H)$, if $e$ has head $u$ and tail $v$ (or, in the undirected case, has ends $u, v$) then $\phi(e)$ has a head in $V(\phi(u))$ and a tail in $V(\phi(v))$ (respectively, $\phi(e)$ has one end in $V(\phi(u))$ and the other in $V(\phi(v))$).

Thus, $H$ is isomorphic to a minor of $G$ if and only if there is a model of $H$ in $G$; indeed, this could be taken as the definition of when $H$ is isomorphic to a minor of $G$.

We would like to extend the idea of a minor to "rooted" digraphs. Intuitively, a rooted digraph is a digraph $G$, of which some vertices, $v_1, \ldots, v_k$ say, are selected and called roots. If we take minors in an unrestricted sense, however, some of these roots may be deleted and some pairs of them may be identified, and this is notationally too complicated to be satisfactory. The best compromise seems to be to permit roots to become identified, but not to permit them to be deleted.

Thus, let us say that $(G, v_1, \ldots, v_k)$ is a *rooted digraph* if $G$ is a digraph and $v_1, \ldots, v_k \in V(G)$ (not necessarily distinct). For $1 \le i \le k$, $v_i$ is the $i$ th root. Isomorphism of rooted digraphs is defined in the natural way; thus, isomorphic rooted digraphs have the same number of roots, and the isomorphism maps the $i$ th root of one to the $i$ th root of the other, for all $i$. Let $(G, v_1, \ldots, v_k)$, $(H, u_1, \ldots, u_k)$ be rooted digraphs. A *model* of the second in the first is a model $\phi$ of $H$ in $G$ such that $v_i \in V(\phi(u_i))$ $(1 \le i \le k)$, and if there is such a model, we say that $(H, u_1, \ldots, u_k)$ is a minor of $(G, v_1, \ldots, v_k)$. The set of all minors of $(G, v_1, \ldots, v_k)$ is closed under isomorphism, and we call it the *folio* of $(G, v_1, \ldots, v_k)$. Clearly the folio is finite up to isomorphism. If $\delta \ge 0$ is an integer, we say that $(H, u_1, \ldots, u_k)$ has *detail* $\le \delta$ if $|E(H)| \le \delta$ and $|V(H) - \{u_1, \ldots, u_k\}| \le \delta$. The *$\delta$-folio* of $(G, v_1, \ldots, v_k)$ is the set of all minors of $(G, v_1, \ldots, v_k)$ with detail $\le \delta$. Its size (up to isomorphism) is bounded above by a function of $k$ and $\delta$, as is easily seen.

[p. 69] For example, let $G$ be the digraph with four vertices $u, v_1, v_2, v_3$ and three edges $uv_1, uv_2, uv_3$; we shall examine the 0-folio of $(G, v_1, v_2, v_3)$. For $i = 1, 2, 3$ let $N_i$ be the digraph with vertex set $\{1, \ldots, i\}$ and no edges. Then the 0-folio of $(G, v_1, v_2, v_3)$ contains $(N_1, 1, 1, 1)$, $(N_2, 1, 1, 2)$, $(N_2, 1, 2, 1)$, $(N_2, 2, 1, 1)$, $(N_3, 1, 2, 3)$, and all rooted digraphs isomorphic to one of these.

Now the $\delta$-folio of $(G, v_1, \ldots, v_k)$ can be determined from the $\delta$-folio of $(G, u_1, \ldots, u_l)$ if $\{u_1, \ldots, u_l\} = \{v_1, \ldots, v_k\}$, as is easily seen. Thus, if $Z \subseteq V(G)$, by the $\delta$-folio of $G$ relative to $Z$ we mean the $\delta$-folio of $(G, v_1, \ldots, v_k)$ for any choice of $v_1, \ldots, v_k$ with $k = |Z|$ and $Z = \{v_1, \ldots, v_k\}$.

For fixed $\delta, \xi \ge 0$ we shall give an algorithm with running time $O(v^3)$ to solve the following problem.

*Folio.* Instance: A digraph $G$, and a subset $Z \subseteq V(G)$ with $|Z| \le \xi$.

Question: What is the $\delta$-folio of $G$ relative to $Z$?

Let us see how Folio can be used to solve the two problems of the previous section. For Disjoint Connected Subgraphs, with input $G = (V, E)$ and $Z_1, \ldots, Z_t \subseteq V$, we may assume that $Z_1, \ldots, Z_t$ are mutually disjoint; we direct the edges of $G$ arbitrarily; we compute the 0-folio of $G$ relative to $Z_1 \cup \cdots \cup Z_t$, and from this we read off whether the desired connected subgraphs exist. (Compare the example above.) For $H$-Minor Containment, we direct the edges of $G$ arbitrarily and compute the $\delta$-folio of $G$ relative to $\emptyset$, where $\delta = \max(|V(H)|, |E(H)|)$. These both have running time $O(v^3)$.

We shall need some easy observations on folio management~~, the proofs of which we leave to the reader~~:

(2.1) *If $Z' \subseteq Z \subseteq V(G)$ and $\delta \geq 0$, the $\delta$-folio of $G$ relative to $Z'$ is determined by a knowledge of the $\delta$-folio of $G$ relative to $Z$.*

*Proof of (2.1).* Let $k = |Z|$ and $k' = |Z'|$, and let $Z = \{v_1, \ldots, v_k\}$, where $Z' = \{v_1, \ldots, v_{k'}\}$. Let $(H, u_1, \ldots, u_{k'})$ be a rooted digraph with detail $\leq \delta$. We say that a rooted digraph $(H^+, u_1, \ldots, u_k)$ *extends* it if $H$ is a subdigraph of $H^+$ with $E(H^+) = E(H)$, and every vertex of $H^+$ not in $V(H)$ is one of $u_{k'+1}, \ldots, u_k$. Such a rooted digraph has detail $\leq \delta$, because it has the same edges as $H$, and each of its vertices that is not a root is a vertex of $H$ different from $u_1, \ldots, u_{k'}$. We claim that $(H, u_1, \ldots, u_{k'})$ is a minor of $(G, v_1, \ldots, v_{k'})$ if and only if some rooted digraph that extends it is a minor of $(G, v_1, \ldots, v_k)$. Then the $\delta$-folio of $(G, v_1, \ldots, v_{k'})$ consists of the rooted digraphs with detail $\leq \delta$ that are extended by some member of the $\delta$-folio of $(G, v_1, \ldots, v_k)$, and (2.1) follows.

To prove the claim, let $\phi$ be a model in $(G, v_1, \ldots, v_k)$ of a rooted digraph that extends $(H, u_1, \ldots, u_{k'})$. Then the restriction of $\phi$ to $V(H) \cup E(H)$ is a model of $(H, u_1, \ldots, u_{k'})$ in $(G, v_1, \ldots, v_{k'})$. Conversely, let $\phi$ be a model of $(H, u_1, \ldots, u_{k'})$ in $(G, v_1, \ldots, v_{k'})$. For $k' < i \leq k$, if $v_i \in V(\phi(u))$ for some $u \in V(H)$ let $u_i = u$, and otherwise let $u_i$ be a new vertex and let $\phi(u_i)$ be the subgraph of $G$ with vertex set $\{v_i\}$ and no edges. Let $H^+$ be obtained from $H$ by adding the new vertices. Then $(H^+, u_1, \ldots, u_k)$ extends $(H, u_1, \ldots, u_{k'})$, and $\phi$ is a model of it in $(G, v_1, \ldots, v_k)$, because the vertices $v_i$ with $u_i$ new are distinct and belong to no $\phi(u)$ with $u \in V(H)$. This proves (2.1). $\square$[F1]

A separation of a graph or digraph $G$ is a pair $(A, B)$ of subgraphs such that $A \cup B = G$ and $E(A \cap B) = \emptyset$, where for subgraphs $G_1 = (V_1, E_1)$, $G_2 = (V_2, E_2)$ of $G$, $G_1 \cup G_2$ and $G_1 \cap G_2$ are the subgraphs $(V_1 \cup V_2, E_1 \cup E_2)$

[F1] **Missing proof.** The paper leaves the proof of (2.1) to the reader. The argument above is the one used in the formalization.

and $(V_1 \cap V_2, E_1 \cap E_2)$, respectively. The order of the separation $(A, B)$ is $|V(A \cap B)|$.

(2.2) *Let $(A, B)$ be a separation of a digraph $G$, and let $V(A \cap B) \subseteq Z \subseteq V(G)$. For $\delta \geq 0$, the $\delta$-folio of $G$ relative to $Z$ is determined by a knowledge of the $\delta$-folios of $A$ relative to $Z \cap V(A)$ and of $B$ relative to $Z \cap V(B)$.*

*Proof of (2.2).* Let $Z = \{v_1, \ldots, v_k\}$ with $k = |Z|$. Let $I_A$ be the set of all $i$ $(1 \leq i \leq k)$ with $v_i \in V(A)$, and define $I_B$ similarly. Thus every $i$ belongs to $I_A$ or to $I_B$, and $i \in I_A \cap I_B$ if and only if $v_i \in V(A \cap B)$. Let $(H, u_1, \ldots, u_k)$ be a rooted digraph with detail $\leq \delta$. A *split* of it consists of rooted digraphs $(H_A, a_i : i \in I_A)$ and $(H_B, b_i : i \in I_B)$ with roots indexed by $I_A$ and $I_B$, and maps $\pi_A \colon V(H_A) \to V(H)$ and $\pi_B \colon V(H_B) \to V(H)$, such that

(a) $E(H)$ is the disjoint union of $E(H_A)$ and $E(H_B)$, and each $e \in E(H_A)$ with head $a$ and tail $a'$ in $H_A$ has head $\pi_A(a)$ and tail $\pi_A(a')$ in $H$, and similarly for $H_B$

(b) $\pi_A(a_i) = u_i$ for $i \in I_A$, and $\pi_B(b_i) = u_i$ for $i \in I_B$

(c) for each $u \in V(H)$ the set $\pi_A^{-1}(u) \cup \pi_B^{-1}(u)$ is non-empty, and the graph $K_u$ with this vertex set, in which $a_i$ and $b_i$ are adjacent for each $i \in I_A \cap I_B$ with $u_i = u$, is connected.

We claim that $(H, u_1, \ldots, u_k)$ is a minor of $(G, v_1, \ldots, v_k)$ if and only if it has a split in which $(H_A, a_i : i \in I_A)$ and $(H_B, b_i : i \in I_B)$ have detail $\leq \delta$ and are minors of $(A, v_i : i \in I_A)$ and $(B, v_i : i \in I_B)$ respectively. This condition depends only on the $\delta$-folios of $A$ relative to $Z \cap V(A)$ and of $B$ relative to $Z \cap V(B)$, and so (2.2) follows from the claim.

Suppose first that there is such a split, and let $\phi_A, \phi_B$ be models of the two rooted digraphs in $(A, v_i : i \in I_A)$ and $(B, v_i : i \in I_B)$. For $u \in V(H)$ let $\phi(u)$ be the union of the graphs $\phi_A(a)$ $(a \in \pi_A^{-1}(u))$ and $\phi_B(b)$ $(b \in \pi_B^{-1}(u))$, and for $e \in E(H)$ let $\phi(e) = \phi_A(e)$ if $e \in E(H_A)$ and $\phi(e) = \phi_B(e)$ if $e \in E(H_B)$. By (c), $\phi(u)$ is non-null. It is connected, since its parts are connected, $K_u$ is connected, and $v_i \in V(\phi_A(a_i)) \cap V(\phi_B(b_i))$ for $i \in I_A \cap I_B$. Let $u, w \in V(H)$ be distinct, and suppose that some $x$ belongs to $V(\phi(u)) \cap V(\phi(w))$. Since the graphs $\phi_A(a)$ $(a \in V(H_A))$ are mutually disjoint, and so are the graphs $\phi_B(b)$ $(b \in V(H_B))$, we may assume that $x \in V(\phi_A(a)) \cap V(\phi_B(b))$ where $\pi_A(a) = u$ and $\pi_B(b) = w$. Then $x \in V(A \cap B) \subseteq Z$, and so $x = v_i$ for some $i \in I_A \cap I_B$. Since $v_i \in V(\phi_A(a_i))$, it follows that $a = a_i$, and similarly $b = b_i$. So $u = u_i = w$ by (b), a contradiction. Thus the graphs $\phi(u)$ are mutually vertex-disjoint. The edges $\phi(e)$ are distinct, and none of them is an edge of any $\phi(u)$, because $\phi_A$ and $\phi_B$ are models and $E(A \cap B) = \emptyset$. By (a), each $\phi(e)$ has its head

and tail in the right graphs $\phi(u)$, and $v_i \in V(\phi(u_i))$ for $1 \le i \le k$ by (b). Hence $\phi$ is a model of $(H, u_1, \ldots, u_k)$ in $(G, v_1, \ldots, v_k)$.

Conversely, let $\phi$ be a model of $(H, u_1, \ldots, u_k)$ in $(G, v_1, \ldots, v_k)$. For $u \in V(H)$, we call the components of the subgraph $(V(\phi(u)) \cap V(A), E(\phi(u)) \cap E(A))$ the *$A$-pieces* of $u$, and define the $B$-pieces of $u$ similarly. Let $H_A$ be the digraph whose vertices are the $A$-pieces of all vertices of $H$, and whose edges are the edges $e \in E(H)$ with $\phi(e) \in E(A)$, where such an $e$ has as head the $A$-piece containing the head of $\phi(e)$ and as tail the $A$-piece containing the tail of $\phi(e)$. (If $u$ is the head of $e$ in $H$, the head of $\phi(e)$ is in $V(\phi(u)) \cap V(A)$, and so it is in an $A$-piece of $u$.) For $i \in I_A$ let $a_i$ be the $A$-piece of $u_i$ that contains $v_i$, and let $\pi_A$ map each $A$-piece of $u$ to $u$. Define $H_B$, $b_i$ and $\pi_B$ similarly. Mapping each piece to itself and each edge $e$ to $\phi(e)$ gives models of $(H_A, a_i : i \in I_A)$ in $(A, v_i : i \in I_A)$ and of $(H_B, b_i : i \in I_B)$ in $(B, v_i : i \in I_B)$, and (a) and (b) hold by construction. For (c), let $u \in V(H)$. Then $\phi(u)$ is non-null, and so $u$ has a piece. If an $A$-piece and a $B$-piece of $u$ have a common vertex $x$, then $x \in V(A \cap B) \subseteq Z$, and so $x = v_i$ for some $i \in I_A \cap I_B$. Then $u = u_i$, since $v_i \in V(\phi(u_i))$, and the two pieces are $a_i$ and $b_i$, which are adjacent in $K_u$. Since every edge of $\phi(u)$ is an edge of one piece, no two $A$-pieces and no two $B$-pieces of $u$ have a common vertex, and $\phi(u)$ is connected, it follows that $K_u$ is connected. It remains to show that $(H_A, a_i : i \in I_A)$ has detail $\le \delta$ (and similarly for $H_B$). First, $|E(H_A)| \le |E(H)| \le \delta$. Now let $P$ be an $A$-piece of some $u \in V(H)$ that is not a root of $H_A$. Then $V(P) \cap Z = \emptyset$, for if $v_i \in V(P)$ then $i \in I_A$, $u = u_i$ and $P = a_i$. Let $f$ be an edge of $\phi(u)$ with an end $x \in V(P)$. If $f \in E(B)$ then $x \in V(A \cap B) \subseteq Z$, which is impossible. So $f \in E(A)$, and $f \in E(P)$. Since $\phi(u)$ is connected, it follows that $P = \phi(u)$. Moreover $u$ is not a root of $H$, for if $u = u_i$ then $v_i \in V(\phi(u)) = V(P)$. Hence the vertices of $H_A$ that are not roots are graphs $\phi(u)$ for distinct vertices $u$ of $H$ that are not roots, and there are at most $\delta$ of them. This proves the claim, and hence (2.2). $\square$[F2]

For $Z \subseteq V(G)$, a vertex $v$ of $G$ is *irrelevant (to the $\delta$-folio of $G$ relative to $Z$)* if $v \notin Z$ and the $\delta$-folios of $G$ and $G \setminus v$ relative to $Z$ are equal. (We shall use $\setminus$ for deletion.) From (2.1) and (2.2), the following follows easily.

[p. 70] (2.3) *Let $(A, B)$ be a separation of a digraph $G$, and let $Z \subseteq V(A)$. If $v \in V(B)$ is irrelevant to the $\delta$-folio of $B$ relative to $V(A \cap B)$ then it is irrelevant to the $\delta$-folio of $G$ relative to $Z$.*

[F2] **Missing proof.** The paper leaves the proof of (2.2) to the reader. The argument above, which cuts a model along the separation into pieces and glues models of the pieces, is the one used in the formalization.

To see this, let $v$ be irrelevant to the $\delta$-folio of $B$ relative to $V(A \cap B)$. Then $v \notin V(A \cap B)$, and so $v \notin V(A)$, and $v \notin Z$ since $Z \subseteq V(A)$. Let $Z' = Z \cup V(A \cap B)$. Now $(A, B \setminus v)$ is a separation of $G \setminus v$, and $V(A \cap (B \setminus v)) = V(A \cap B) \subseteq Z'$. Since $Z \subseteq V(A)$ and $v \notin Z'$, we have $Z' \cap V(B) = Z' \cap V(B \setminus v) = V(A \cap B)$, and $Z' \cap V(A) = Z'$. By hypothesis the $\delta$-folios of $B$ and $B \setminus v$ relative to $V(A \cap B)$ are equal. Hence by (2.2), applied to the separations $(A, B)$ of $G$ and $(A, B \setminus v)$ of $G \setminus v$, the $\delta$-folios of $G$ and $G \setminus v$ relative to $Z'$ are equal. By (2.1), their $\delta$-folios relative to $Z$ are equal, and since $v \notin Z$, $v$ is irrelevant to the $\delta$-folio of $G$ relative to $Z$.[F3]

The idea of our algorithm for Folio is as follows: If $G$ can be divided into trivial pieces by cutsets of bounded size we can solve Folio directly, using a method of Arnborg and Proskurowski [2]. If not there is an irrelevant vertex, and we give a procedure to find such a vertex. We delete it, and repeat the process as often as necessary, until the graph is reduced so much that the Arnborg-Proskurowski method can be applied. The first part of the paper (Sections 3 and 4) cover detecting and utilizing that $G$ has bounded "branch-width," that is, that $G$ can be divided into trivial pieces by cutsets of bounded size. The remainder of the paper concerns finding an irrelevant vertex. This turns out to be easy if we can find a large clique minor (Sections 5 and 6); the main problem is what should be done if we have not found such a minor. In that case it follows from an earlier result of this series that $G$ has a large grid minor; and the remainder of the paper concerns how to exploit a large grid minor to obtain either a large clique minor or an irrelevant vertex. For this we shall make use of a lemma which is a consequence of the main results of this series and which will be proved in a future paper. (The lemma is an assertion that under certain circumstances, a given vertex is irrelevant.)

## 3. Branch-Width

A *branch-decomposition* of a graph or digraph $G$ is a pair $(T, \tau)$, where
(i) $T$ is a tree and every vertex of $T$ has valency 1 or 3
(ii) $\tau$ is a bijection from $E(G)$ to the set of leaves of $T$.

(The *leaves* of $T$ are the vertices with valency 1.) For each $f \in E(T)$ the edges $e$ of $G$ are divided into two sets, depending on which component of $T \setminus f$ contains $\tau(e)$. The *order* of $f$ is the number of vertices of $G$ incident with an edge in each set. The *width* of $(T, \tau)$ is the maximum order of

[F3] **Missing proof.** The paper says only that (2.3) follows easily from (2.1) and (2.2). The argument above is the one used in the formalization.

edges of $T$, and the *branch-width* of $G$ is the minimum width of all branch-decompositions of $G$. (If there are no branch-decompositions of $G$, then $|E(G)| \leq 1$ and $G$ has branch-width 0 by convention.) Branch-width is closely related to "tree-width"—see [19].

Our first objective is an algorithm which decides either that $G$ has branch-width $\leq 3\omega$, or that $G$ has branch-width $\geq \omega$. For that we need the following lemma.

[p. 71] (3.1) Lemma. *Let $G$ be a graph with branch-width $\leq \omega$, where $\omega \geq 2$, and let $Z \subseteq V(G)$. Then there is a separation $(A, B)$ of $G$ of order $\leq \omega$ with*

$$|Z - V(A)|, |Z - V(B)| \leq \frac{2}{3}|Z| - \frac{1}{2}|Z \cap V(A \cap B)|.$$

*Proof.* Let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G$ of order $\leq \omega$ with

$$|Z - V(A)| + \frac{1}{2}|Z \cap V(A \cap B)| > \frac{2}{3}|Z|.$$

We observe that

(1)(i) *If $(A, B) \in \mathscr{T}$ then $V(A) \neq V(G)$,* (ii) *if $(A_1, B_1), (A_2, B_2) \in \mathscr{T}$ then $B_1$ is not a subgraph of $A_2$* (iii) *if $A_1, A_2, A_3$ are mutually edge-disjoint subgraphs of $G$ with union $G$, then not all of $(A_1, A_2 \cup A_3), (A_2, A_3 \cup A_1), (A_3, A_1 \cup A_2)$ belong to $\mathscr{T}$.*

*Subproof.* For (i), let $(A, B)$ be a separation of $G$ with $V(A) = V(G)$. Then

$$|Z - V(A)| + \frac{1}{2}|Z \cap V(A \cap B)| \leq \frac{1}{2}|Z|,$$

and so $(A, B) \notin \mathscr{T}$. This proves (i).

For (ii), suppose that $(A_1, B_1), (A_2, B_2) \in \mathscr{T}$. Then

$$|Z \cap V(B_1)| = |Z - V(A_1)| + |Z \cap V(A_1 \cap B_1)| > \frac{2}{3}|Z|,$$

but

$$\begin{aligned} |Z| - \frac{1}{2}|Z \cap V(A_2)| &= |Z - V(A_2)| + \frac{1}{2}|Z \cap V(A_2)| \\ &\geq |Z - V(A_2)| + \frac{1}{2}|Z \cap V(A_2 \cap B_2)| > \frac{2}{3}|Z|, \end{aligned}$$

that is, $|Z \cap V(A_2)| < \frac{2}{3}|Z|$. Consequently (ii) holds.

For (iii), suppose that all three of the separations belong to $\mathscr{T}$. For $i = 1, 2, 3$ let $Z_i = Z \cap V(A_i)$. Since $(A_1, A_2 \cup A_3) \in \mathscr{T}$ we deduce that

$$|Z - Z_1| + \frac{1}{2}|Z_1 \cap (Z_2 \cup Z_3)| > \frac{2}{3}|Z|,$$

that is,

$$\frac{1}{3}|Z| > |Z_1| - \frac{1}{2}|Z_1 \cap Z_2| - \frac{1}{2}|Z_1 \cap Z_3| + \frac{1}{2}|Z_1 \cap Z_2 \cap Z_3|.$$

By adding this and the two similar inequalities, we deduce that

$$|Z| > \sum_{1 \le i \le 3} |Z_i| - \sum_{1 \le i < j \le 3} |Z_i \cap Z_j| + \frac{3}{2}|Z_1 \cap Z_2 \cap Z_3|,$$

contrary to the inclusion-exclusion formula, since $Z = Z_1 \cup Z_2 \cup Z_3$. This proves (iii) and so completes the proof of (1).

[p. 72] Suppose that for every separation $(A, B)$ of $G$ of order $\le \omega$, $\mathscr{T}$ contains one of $(A, B)$, $(B, A)$. By (1) and [19, Theorem (4.5)], $\mathscr{T}$ is a "tangle of order $\omega + 1$" in $G$, in the terminology of that paper. But this contradicts [19, Theorem (4.3)], since $G$ has branch-width $\le \omega$ and every edge of $G$ has $\le 2 \le \omega$ ends. It follows that there is a separation $(A, B)$ of $G$ of order $\le \omega$, such that $(A, B), (B, A) \notin \mathscr{T}$; but then $(A, B)$ satisfies the theorem. □

To apply (3.1) we use the following lemma.

(3.2) Lemma. *If $\omega \ge 2$ is an integer and $G$ is a simple graph with branch-width $\le \omega$, then $|E(G)| \le (\frac{3}{2}\omega - 1)|V(G)|$.*

*Proof.* Put $\frac{3}{2}\omega - 1 = \alpha$. We proceed by induction on $|V(G)|$. If $|V(G)| = 0$ the result is trivial, and we assume that $|V(G)| > 0$. By [19, Theorem (5.1)], $G$ has tree-width $\le \alpha$ (see that paper for definitions) and hence has a vertex $v$ of valency $\le \alpha$. From the inductive hypothesis, $G \setminus v$ has $\le \alpha(|V(G)| - 1)$ edges, and so

$$|E(G)| \le \alpha(|V(G)| - 1) + \alpha = \alpha|V(G)|,$$

as required. □

We use (3.1) to show that if we "greedily" attempt to find a branch-decomposition of $G$ of width $\le 3\omega$, and fail, then $G$ has branch-width $\ge \omega$. Actually, we can obtain slightly more than this, as follows. For a graph or digraph $G$, we define $\varepsilon = \varepsilon(G) = |E(G)| + |V(G)| + 1$. Let $\omega \ge 2$ be some integer.

(3.3) Algorithm. *Input:* A graph $G$ with $|E(G)| \geq 2$. *Output:* A subgraph $G'$ of $G$ and a branch-decomposition of $G'$ of width $\leq 3\omega$, such that either $G' = G$ or $G'$ has branch-width $\geq \omega$.

*Running time:* $O(v^2 + \varepsilon)$. *Description:* We construct the simple graph $H_1$ underlying $G$, by deleting all loops and multiple edges. We choose a maximal subgraph $H_2$ of $H_1$ with $|E(H_2)| \leq (\frac{3}{2}\omega - 1)|V(H_1)| + 1$. Thus, either $H_2 = H_1$ or $H_2$ has branch-width $\geq \omega$, by (3.2). We assume that $|E(H_2)| \geq 2$, for otherwise the problem is trivial. We shall construct a subgraph $H_3$ of $H_2$ and a branch-decomposition of $H_3$ of width $\leq 3\omega$, such that either $H_3 = H_2$ or $H_3$ has branch-width $\geq \omega$. Given such an $H_3$, the algorithm is completed as follows. If $H_3 \neq H_2$, we set $G' = H_3$ and stop. If $H_3 = H_2$, we have a branch-decomposition of $H_2$ of width $\leq 3\omega$. If $H_2 \neq H_1$ we set $G' = H_2$ and stop. If $H_2 = H_1$, we convert our branch-decomposition to one for $G$, set $G' = G$, and stop.

[p. 73] Thus it remains to show how to construct $H_3$. We initialize by setting $G_1 = H_2$, we let $T_1$ be a tree with two vertices $t, t'$, and we define $\tau_1 \colon E(G_1) \to V(T_1)$ by setting $\tau_1(e) = t$ for some edge $e$ of $G_1$, and $\tau_1(f) = t'$ for all other edges $f$ of $G_1$. We perform a recursion. At the beginning of the $i$ th iteration we have

(i) a subgraph $G_i$ of $H_2$ such that either $G_i = H_2$ or $G_i$ has branch-width $\geq \omega$

(ii) a tree $T_i$ each vertex of which has valency 1 or 3

(iii) a surjection $\tau_i$ from $E(G_i)$ to the set of leaves of $T_i$, such that each edge of $T_i$ has order $\leq 3\omega$.

(The order of $f \in E(T_i)$ is defined to be the number of vertices of $G_i$ incident with an edge $e_1$ and an edge $e_2$ such that $\tau_i(e_1)$ and $\tau_i(e_2)$ are in different components of $T_i \setminus f$.)

The iteration proceeds as follows:

(1) If $\tau_i$ is a bijection, we set $H_3 = G_i$ and stop. $(T_i, \tau_i)$ is a branch-decomposition of $H_3$ of width $\leq 3\omega$, and either $H_3 = H_2$ or $H_3$ has branch-width $\geq \omega$, as required.

(2) If $\tau_i$ is not a bijection we choose a leaf $v$ of $T_i$ such that $v = \tau_i(e)$ for at least two edges $e$ of $G_i$. Let $S, S'$ be the components of $T_i \setminus f$, where $f \in E(T_i)$ is incident with $v$, and $V(S) = \{v\}$. Let $R$ be the subgraph of $G_i$ formed by all edges $e$ with $\tau_i(e) \in V(S)$ and their ends, and define $R'$ similarly. Then $E(R \cap R') = \emptyset$, $E(R \cup R') = E(G_i)$, and $|V(R \cap R')| \leq 3\omega$. Let $Z = V(R \cap R')$. Let $T'$ be a tree obtained by adding two new vertices $v_1, v_2$ to $T_i$, both adjacent to $v$.

(3) If $Z = \emptyset$ we choose an edge $g$ of $R$, and if $0 < |Z| < 3\omega$ we choose $g \in E(R)$ with an end in $Z$. (This is possible since $E(R) \neq \emptyset$ and every

vertex in $Z$ is incident with an edge of $R$.) We set $T_{i+1} = T'$, $G_{i+1} = G_i$, and define $\tau_{i+1}$ by

$$\begin{aligned} &\tau_{i+1}(e) = \tau_i(e) && (e \in E(G_i), \tau_i(e) \neq v) \\ &\tau_{i+1}(e) = v_1 && (e \in E(G_i) - \{g\}, \tau_i(e) = v) \\ &\tau_{i+1}(g) = v_2. \end{aligned}$$

We return to (1) for the next iteration. It is easy to see that $G_{i+1}$, $T_{i+1}$, $\tau_{i+1}$ satisfy (i), (ii), and (iii). To see this, (i) holds since $G_{i+1} = G_i$, and (ii) holds since $v$ has valency 3 in $T'$ and $v_1, v_2$ are leaves. For (iii), $\tau_{i+1}$ is a surjection, since $\tau_{i+1}(g) = v_2$, and $\tau_{i+1}(e) = v_1$ for some $e \neq g$ because at least two edges $e$ of $G_i$ have $\tau_i(e) = v$. Each edge of $T'$ other than $vv_1$ and $vv_2$ divides $E(G_i)$ in the same way as in $T_i$ (for $f$, the side containing $v$ still contains just the edges of $R$), and so it has the same order. The edge $vv_2$ has order at most $2 \leq 3\omega$, since only the ends of $g$ can be counted. The edges $e$ with $\tau_{i+1}(e) = v_1$ are the edges of $R$ other than $g$, and so each vertex counted in the order of $vv_1$ is incident with an edge of $R$ and with $g$ or an edge of $R'$, that is, it is in $Z$ or it is an end of $g$. If $Z = \emptyset$, the order of $vv_1$ is therefore at most 2. If $0 < |Z| < 3\omega$, then $g$ has an end in $Z$, so at most one end of $g$ is outside $Z$, and the order of $vv_1$ is at most $|Z| + 1 \leq 3\omega$.[F4]

(4) If $|Z| = 3\omega$ we test if there is a separation $(A, B)$ of $R$ of order $\leq \omega$ with

$$|Z - V(A)|, |Z - V(B)| \leq \frac{2}{3}|Z| - \frac{1}{2}|Z \cap V(A \cap B)|.$$

[p. 74] This can be done by enumerating all partitions $(Z_1, Z_2, Z_3)$ of $Z$ with $|Z_1|, |Z_2| \leq 2\omega - \frac{1}{2}|Z_3|$ and then testing for separations $(A, B)$ of order $\leq \omega$ with $V(A) \cap Z = Z_1 \cup Z_3$ and $V(B) \cap Z = Z_2 \cup Z_3$. (Such a separation does not exist if and only if in $R \setminus Z_3$ there are $> \omega - |Z_3|$ paths from $Z_1$ to $Z_2$, mutually vertex-disjoint.)

(5) If $(A, B)$ is a separation as in (4), we set $T_{i+1} = T'$, $G_{i+1} = G_i$, and define $\tau_{i+1}$ by

$$\begin{aligned} &\tau_{i+1}(e) = \tau_i(e) && (e \in E(G_i), \tau_i(e) \neq v) \\ &\tau_{i+1}(e) = v_1 && (e \in E(A)) \\ &\tau_{i+1}(e) = v_2 && (e \in E(B)). \end{aligned}$$

We return to (1) for the next iteration.

[F4] **Missing proof.** The paper says only that this is easy to see. The verification above is the one used in the formalization.

Let us verify that $G_{i+1}$, $T_{i+1}$, $\tau_{i+1}$ satisfy (i), (ii), and (iii). Now (i) is clear, since $G_{i+1} = G_i$, and (ii) is clear from construction. To verify (iii) we observe that $E(A) \neq \emptyset \neq E(B)$; for if $E(A) = \emptyset$, say, then $B = R$ since $R$ has no isolated vertices, and so

$$|Z - V(A)| \leq \frac{2}{3}|Z| - \frac{1}{2}|Z \cap V(A)|,$$

that is, $|Z| \leq \frac{3}{2}|Z \cap V(A)|$, which is impossible since $|Z| = 3\omega$ and

$$|Z \cap V(A)| \leq |V(A)| = |V(A \cap B)| \leq \omega.$$

Thus $E(A), E(B) \neq \emptyset$ and so $\tau_{i+1}$ is a surjection. Every edge of $T_{i+1}$ has the same order as it does in $T_i$ except for the two new edges. The new edges have orders $|(Z \cup V(A)) \cap V(B)|$ and $|(Z \cup V(B)) \cap V(A)|$; and

$$\begin{aligned} |(Z \cup V(A)) \cap V(B)| &= |Z - V(A)| + |V(A) \cap V(B)| \\ &\leq \frac{2}{3}|Z| + |V(A \cap B)| \leq 2\omega + \omega = 3\omega. \end{aligned}$$

Similarly, $|(Z \cup V(B)) \cap V(A)| \leq 3\omega$. This verifies (iii).

(6) If there is no separation $(A, B)$ as in (4), we choose $g \in E(R)$, set $G_{i+1} = G_i \setminus g$, $T_{i+1} = T_i$, and let $\tau_{i+1}$ be the restriction of $\tau_i$ to $E(G_{i+1})$. We return to (1) for the next iteration.

Again, let us verify that $G_{i+1}$, $T_{i+1}$, $\tau_{i+1}$ satisfy (i), (ii), (iii). From (3.1), $G_i$ has branch-width $> \omega$, and so $G_{i+1}$ has branch-width $\geq \omega$ (we leave as an exercise the fact that deleting an edge reduces branch-width by at most one, except when branch-width $\leq 2$ which does not occur here since $\omega \geq 2$). This verifies (i), and (ii) and (iii) are clear. This concludes the description of the algorithm.

We give the details, in the form needed here. Suppose that $G_{i+1}$ has branch-width $\leq \omega - 1$. We shall obtain a contradiction with the fact that $G_i$ has branch-width $> \omega$.

Suppose first that $\omega \geq 3$. Since $G_i$ has branch-width $\geq \omega + 1$ and every edge of $G_i$ has at most $2 \leq \omega + 1$ ends, by [19, Theorem (4.3)] there is a tangle in $G_i$ of order $\geq \omega + 1$. Its members of order $\leq \omega$ form a tangle $\mathscr{T}$ in $G_i$ of order $\omega + 1$. Let $u$ be an end of $g$. By [19, Theorem (8.5)] applied to $\mathscr{T}$ with $W = \{u\}$, there is a tangle of order $\omega$ in $G_i \setminus u$. Since $g$ is incident with $u$, the graph $G_i \setminus u$ is a subgraph of $G_i \setminus g = G_{i+1}$, and so by [19, Theorem (6.1)] there is a tangle of order $\omega$ in $G_{i+1}$. On the other hand, $G_{i+1}$ has an edge, since $G_i$ has branch-width $> 0$ and so has at least two edges. Every edge of $G_{i+1}$ has at most $2 \leq \omega - 1$ ends, and $G_{i+1}$ has

branch-width $\leq \omega - 1$, and so by [19, Theorem (4.3)] there is no tangle in $G_{i+1}$ of order $\geq \omega$. This is a contradiction.

Now suppose that $\omega = 2$, so that $G_{i+1}$ has branch-width $\leq 1$. If $G_{i+1}$ has at most one edge, then $G_i$ has at most two edges, and so it has branch-width $\leq 2$, because a branch-decomposition of $G_i$, if there is one, has a tree with only one edge, and this edge has order at most 2 since all the vertices it counts are ends of one edge of $G_i$. Thus we may assume that $G_{i+1}$ has a branch-decomposition $(T, \tau)$ of width $\leq 1$. Choose $e \in E(G_{i+1})$, sharing an end with $g$ if some edge of $G_{i+1}$ does so. Let $T^+$ be the tree obtained from $T$ by adding two new vertices adjacent to the leaf $\tau(e)$, and let $\tau^+$ map $e$ and $g$ to these two new vertices and agree with $\tau$ on the other edges of $G_i$. Then $(T^+, \tau^+)$ is a branch-decomposition of $G_i$, since $\tau(e)$ now has valency 3. The two new edges of $T^+$ have order at most 2, since each of them separates one edge of $G_i$ from the others. Every other edge $f$ of $T^+$ is an edge of $T$, and it divides $E(G_i)$ as it divides $E(G_{i+1})$ in $T$, with $g$ added to the side that contains $e$. A vertex counted in the order of $f$ in $T^+$ but not in $T$ is therefore an end of $g$ that is incident with no edge of $G_{i+1}$ on the side of $e$. If $e$ shares an end with $g$, this end is not such a vertex, and so the order of $f$ grows by at most one. If no edge of $G_{i+1}$ shares an end with $g$, then no end of $g$ is incident with an edge of $G_{i+1}$, and the order of $f$ does not grow. Hence $(T^+, \tau^+)$ has width $\leq 2$, and so $G_i$ has branch-width $\leq \omega$, a contradiction.[F5]

[p. 75] To find its running time, we observe that at each iteration but the last the quantity $|E(G_i)| - |E(T_i)|$ is reduced by 1; it starts at $O(|V(G)|)$ and finishes at 1, and so there are $O(v)$ iterations. Each one takes a time $O(v)$. The time is spent principally in step 4; there, each partition $(Z_1, Z_2, Z_3)$ can be checked in time $O(|E(G_i)|)$ and, hence, $O(v)$, since $|E(G_i)| \leq O(v)$ and the number of partitions to be checked is bounded.

Thus constructing $H_3$ and its branch-decomposition takes time $O(v^2)$. Converting it to a branch-decomposition of $G$ takes time $O(\varepsilon)$, and so the whole algorithm takes time $O(v^2 + \varepsilon)$. □

Actually, Hans Bodlaender [3] (and see also [12]) has recently found an algorithm that could be used in place of (3.3), with running time $O(v)$ for simple graphs $G$, using tree-width instead of branch-width. Its use could speed up the applications of (3.3); for instance, (7.2) could be made to run in time $O(\varepsilon)$. This would not affect the total running time of our Folio algorithm, and so we omit further details.

[F5] **Missing proof.** The paper leaves this fact as an exercise. The argument above is the one used in the formalization.

## 4. Folio Determination with Bounded Branch-Width

If $G$ is a digraph, $Z \subseteq V(G)$, and $\delta \geq 0$, a modelling of the $\delta$-folio of $G$ relative to $Z$ means some choice of model, for each member of the $\delta$-folio of $G$ relative to $Z$ (up to isomorphism). A method of Arnborg and Proskurowski [2] can be used to solve Folio for graphs of bounded branch-width, as follows. Let $\delta, \xi \geq 0$, $\omega \geq 1$ be integers.

(4.1) Algorithm.

*Input:* A digraph $G$, and a subset $Z \subseteq V(G)$ with $|Z| \leq \xi$, and a branch-decomposition $(T, \tau)$ of $G$ of width $\leq \omega$.

*Output:* The $\delta$-folio of $G$ relative to $Z$, and a modelling of it.

*Running time:* $O(\varepsilon)$.

*Description:* Since the effect of isolated vertices of $G$ on the folio is clear, we may arrange (by deleting them) that $G$ has no isolated vertices. Let $z$ be a leaf of $T$, and number the edges of $T$ as $f_1, \ldots, f_m$, where $m = |E(T)|$, so that the numbers on every path leaving $z$ are decreasing. (Our branch-decomposition in applications usually comes already equipped with such a numbering; but if not, one can be found in time $O(\varepsilon)$.) For $1 \leq i \leq m$, let $S_i, T_i$ be the components of $T \setminus f_i$, with $z \in V(S_i)$. Let $A_i$ be the subgraph of $G$ formed by the edges $e$ with $\tau(e) \in V(S_i)$, and their ends; and define $B_i$ similarly. Since $G$ has no isolated vertices, $(A_i, B_i)$ is a separation of $G$ of order $\leq \omega$. Let $Z_i = (V(A_i) \cup Z) \cap V(B_i)$. We shall compute the $\delta$-folio of $B_i$ relative to $Z_i$ for each $i$, by a recursion as follows. At the start of the $i$ th iteration, the $\delta$-folio of $B_j$ relative to $Z_j$, has been determined for $1 \leq j < i$. [p. 76] (!) If $i \leq m$ and $|V(T_i)| = 1$ then $|E(B_i)| = 1$ and $|V(B_i)| \leq 2$. We determine the folio of $Z_i$ in $B_i$ and return to (1) for the next iteration.

(2) If $i \leq m$ and $|V(T_i)| > 1$, let $f_j, f_k$ be the two edges of $T_i$ with a common end with $f_i$. Then $j, k < i$, and $(B_j, B_k)$ is a separation of $B_i$, and the $\delta$-folios of $B_j$ relative to $Z_j$ and $B_k$ relative to $Z_k$ have already been determined. Since $V(B_j \cap B_k) \subseteq Z_j \cap Z_k$ we can calculate the $\delta$-folio of $B_j \cup B_k$ relative to $Z_j \cup Z_k$ from this information, by (2.2). Since $Z_i \subseteq Z_j \cup Z_k$ and $B_j \cup B_k = B_i$, the $\delta$-folio of $B_i$ relative to $Z_i$ can be calculated by (2.1). (Note that, since $\delta$-folios have size bounded by a constant, these calculations take constant time.) We return to (1) for the next iteration.

(3) If $i = m + 1$, the folio of $B_m$ relative to $Z_m$ has been determined. We determine the $\delta$-folio of $A_m$ relative to $(Z \cup V(B_m)) \cap V(A_m)$ (this is easy, since $|V(A_m)| \leq 2$), and we use (2.1) and (2.2) to determine the $\delta$-folio of $Z$ in $A_m \cup B_m = G$.

(4) For each member of the $\delta$-folio of $Z$ in $G$ (up to isomorphism) we backtrack through this computation to compute a corresponding model, in

the natural way.

Each iteration takes constant time, and there are $|E(T)|+1$ iterations. Thus, steps (1)-(3) take time $O(\varepsilon)$. In step (4), the backtracking to determine how each edge of $G$ should be used to produce the desired member of the $\delta$-folio takes time $O(\varepsilon)$ and converting this information into a model again takes time $O(\varepsilon)$. Thus the total running time is $O(\varepsilon)$. □

To apply (4.1), we use the following theorem, a consequence of Theorems (1.5) of [15] and (5.1) of [19]. See also [25] for a numerical improvement.

(4.2) THEOREM. *For every planar graph $H$ there is a number $\omega$ such that every graph with branch-width $\geq \omega$ has a minor isomorphic to $H$.*

For a first application of (4.1), we have the following, improving a result of [13].

(4.3) *Let $\mathcal{F}$ be a minor-closed class of graphs such that some planar graph is not in $\mathcal{F}$. Then there is an algorithm to determine whether $G \in \mathcal{F}$ with running time $O(v^2)$.*

*Proof.* By Theorem (1.4) of [14], there is a finite list of graphs $H_1, \ldots, H_k$ such that a graph $G$ is in $\mathcal{F}$ if and only if it has no minor isomorphic to any of $H_1, \ldots, H_k$. Since $\mathcal{F}$ does not contain every planar graph, some $H_i$ (say $H_1$) is planar. By (4.2) there is a number $\omega \geq 2$ such that every graph with branch-width $\geq \omega$ has a minor isomorphic to $H_1$. Let $\delta = \max(\varepsilon(H_1), \ldots, \varepsilon(H_k))$. We now describe an algorithm to test membership of $\mathcal{F}$. Let $G$ be the input graph. We may assume that $|E(G)| \geq 2$.

[p. 77] (1) Choose a maximal subgraph $G_1$ of $G$ with at most $\delta$ loops at each vertex and with no $\delta + 12$ edges mutually parallel. Then $G_1 \in \mathcal{F}$ if and only if $G \in \mathcal{F}$, as is easily seen: a model in $G$ of a graph with at most $\delta$ edges uses at most $\delta$ loops at each vertex, and between two distinct vertices of $G$ it uses at most $\delta$ edges that represent edges of the graph and at most one more edge inside a branch set, so its edges can be chosen in $G_1$.[F6]

(2) We apply (3.3) to $G_1$. We obtain a subgraph $G'$ of $G_1$, and a branch-decomposition of $G'$ of width $\leq 3\omega$, such that either $G' = G_1$ or $G'$ has branch-width $\geq \omega$.

(3) If $G' \neq G_1$ then $G'$ has branch-width $\geq \omega$, and so $G'$ has a minor isomorphic to $H_1$ and $G' \notin \mathcal{F}$; thus, $G \notin \mathcal{F}$ and we stop.

(4) If $G' = G_1$, we have a branch-decomposition of $G_1$ of width $\leq 3\omega$.

[F6] **Proof gap.** As printed, $G_1$ keeps too few parallel edges. Let $\mathcal{F}$ be the class of graphs with no minor isomorphic to one vertex with two loops, so that $\delta = 2$. Two vertices joined by three parallel edges are not in $\mathcal{F}$ (contract one edge), but $G_1$, two vertices joined by two edges, is in $\mathcal{F}$. Step (1) of (4.4) keeps $|E(H)|+1$ parallel edges, which is the bound needed.

We direct the edges of $G_1$ arbitrarily, we apply (4.1) to $G_1$ with $Z = \emptyset$ (with $3\omega$ replacing $\omega$) and compute the $\delta$-folio of $G_1$ relative to $\emptyset$. From this we deduce whether $G_1$ has a minor isomorphic to any of $H_1, \ldots, H_k$ and hence determine whether $G_1 \in \mathcal{F}$. This completes the algorithm.

Since at step 1 we arrange that $\varepsilon(G_1) \leq O(v^2)$, the algorithm has running time $O(v^2)$. □

A second, similar, application of (4.1) is the following. Let $H$ be a planar graph. By (4.3) (with $\mathcal{F}$ the class of graphs with no minor isomorphic to $H$) we can test if an input graph has a minor isomorphic to $H$ in time $O(v^2)$. But what if we want to *find* such a minor? The following improves a result of [13].

(4.4) Algorithm.

*Input:* A graph $G$.

*Output:* Decides if there is a model of $H$ in $G$, and outputs such a model if one exists.

*Running time:* $O(v^2)$.

*Description:* Let $\omega$ be as in (4.2), with $\omega \geq 2$.

(1) We choose a maximal subgraph $G_1$ of $G$ with at most $|E(H)|$ loops at every vertex and at most $|E(H)| + 1$ edges mutually parallel. There is a model of $H$ in $G$ if and only if there is one in $G_1$.

(2) We apply (3.3) to $G_1$. We obtain a subgraph $G_2$ of $G_1$, and a branch-decomposition of $G_2$ of width $\leq 3\omega$, such that either $G_2 = G_1$ or $G_2$ has branch-width $\geq \omega$. From the choice of $\omega$, it follows that either $G_2 = G_1$ or there is a model of $H$ in $G_2$. Consequently there is a model of $H$ in $G_1$ if and only if there is one in $G_2$.

(3) We test if $G_2$ has an $H$-minor, and if so find a model of it in $G_2$, by using (4.1) to compute the $\varepsilon(H)$-folio of $G_2$ relative to $\emptyset$ and a modelling of it.

From step (1), $|E(G_1)| \leq O(v^2)$. Step (2) therefore takes time $O(v^2)$, and so does step (3). Thus the algorithm has running time $O(v^2)$ in total. □

## [p. 78] 5. Generic Folios

So far we have discussed how to detect small branch-width and how to solve Folio in digraphs of bounded branch-width. Now we come to the more difficult problem of how to solve Folio in digraphs of large branch-width, and here our strategy is to locate an irrelevant vertex and delete it (and repeat, until we attain small branch-width). The problem of finding such a vertex splits into two subproblems: how to find an irrelevant vertex given

a large clique minor; and how to find either an irrelevant vertex or a large clique minor. We treat the former problem in this section and the next.

The $\delta$-folio of a digraph $G$ relative to $Z \subseteq V(G)$ is *generic* if the $\delta$-folio of the rooted digraph $(G, v_1, \ldots, v_\xi)$ contains every rooted digraph with $\xi$ roots and with detail $\leq \delta$, where $|Z| = \xi$ and $Z = \{v_1, \ldots, v_\xi\}$.

(5.1) *Let $G$ be a digraph with $|V(G)| \geq 2\xi + 3\delta$, and let $Z \subseteq V(G)$ with $|Z| = \xi$. Suppose that $u, v \in Z$, for all distinct $u, v \in V$ that are non-adjacent in $G$. Then the $\delta$-folio of $G$ relative to $Z$ is generic.*

*Proof.* Let $Z = \{v_1, \ldots, v_\xi\}$, and let $(H, u_1, \ldots, u_\xi)$ be a rooted digraph with detail $\leq \delta$. We must show that $(H, u_1, \ldots, u_\xi)$ is a minor of $(G, v_1, \ldots, v_\xi)$. For each $v \in V(H)$, choose $X_v \subseteq V(G)$ such that $|X_v - Z| = 1$, $X_v \cap Z = \{v_i : 1 \leq i \leq \xi, u_i = v\}$, and for distinct $v, v' \in V(H)$, $X_v \cap X_{v'} = \emptyset$. For each edge $e \in E(H)$, choose a non-loop edge $\phi(e)$ of $G$ with head $s_e$ and tail $t_e$, such that $s_e, t_e \notin Z \cup \bigcup(X_v : v \in V(H))$, and for distinct $e, e' \in E(H)$, $s_e, t_e, s_{e'}, t_{e'}$ are all distinct. Such choices are possible since

$$|V(G) - Z| \geq (\xi + \delta) + 2\delta \geq |V(H)| + 2|E(H)|.$$

For each $v \in V(H)$, let $\phi(v)$ be the restriction of $G \setminus \{\phi(e) : e \in E(H)\}$ to

$$X_v \cup \{s_e : e \in E(H) \text{ has head } v\} \cup \{t_e : e \in E(H) \text{ has tail } v\}.$$

Then $G_v$ is connected (since $X_v \not\subseteq Z$) and it is easy to see that $\phi$ is a model of $(H, u_1, \ldots, u_\xi)$ in $(G, v_1, \ldots, v_\xi)$, as required. □

If $G$ is a digraph and $X, Y \subseteq V(G)$ or $X, Y$ are subgraphs of $G$, an edge of $G$ is said to be *between* $X$ and $Y$ if it has one end in $X$ and the other in $Y$.

(5.2) *Let $G$ be a digraph, let $Z \subseteq V(G)$ with $|Z| = \xi$, and let $k = 2\xi + 3\delta$. Suppose that there are $k$ non-null connected subgraphs $G_1, \ldots, G_k$ of $G$, mutually vertex-disjoint, such that*

(i) *$|Z \cap V(G_i)| = 1$ $(1 \leq i \leq \xi)$, and (hence) $Z \cap V(G_i) = \emptyset$ $(\xi + 1 \leq i \leq k)$*

[p. 79] (ii) *for $1 \leq i < j \leq k$, if there is no edge of $G$ between $G_i$ and $G_j$ then $i, j \leq \xi$.*

*Then the $\delta$-folio of $G$ relative to $Z$ is generic.*

*Proof.* We delete every vertex of $G$ not in $V(G_1) \cup \cdots \cup V(G_k)$, and then contract every edge of each $G_i$. We obtain a digraph $G'$ with $|V(G')| = k$. Let $Z' \subseteq V(G')$ correspond to $Z$. By (5.1), the $\delta$-folio of $G'$ relative to $Z'$ is generic, and so the $\delta$-folio of $G$ relative to $Z$ is generic, as required. □

We shall use (5.2) to prove the following lemma.

(5.3) *Let $G$ be a graph and let $Z \subseteq V(G)$ with $|Z| = \xi$. Let $k \geq \lfloor \frac{3}{2}\xi \rfloor$ and let $G_1, \ldots, G_k$ be subgraphs of $G$, mutually vertex-disjoint, such that*

(i) *for $1 \leq i \leq k$, either $G_i$ is connected or every component of $G_i$ has a vertex in $Z$,*

(ii) *for $1 \leq i < j \leq k$, either $V(G_i)$ and $V(G_j)$ both meet $Z$ or there is an edge of $G$ between $G_i$ and $G_j$, and*

(iii) *for $1 \leq i \leq k$, there is no separation $(A, B)$ of $G$ of order $< \xi$ with $Z \subseteq V(A)$ and with $A \cap G_i$ null.*

*Then for some $\mu$ with $0 \leq \mu \leq \xi$ there are $\kappa = k - \lfloor \frac{1}{2}(\xi - \mu) \rfloor$ non-null connected subgraphs $H_1, \ldots, H_\kappa$ of $G$, mutually vertex-disjoint, such that*

(a) *$|V(H_i) \cap Z| = 1$ ($1 \leq i \leq \xi$) and $V(H_i) \cap Z = \emptyset$ ($\xi + 1 \leq i \leq \kappa$), and*

(b) *for $1 \leq i < j \leq \kappa$, if there is no edge of $G$ between $H_i$ and $H_j$ then $i, j \leq \mu$.*

*In particular, for any $\delta \geq 0$, if $k \geq \lfloor \frac{5}{2}\xi \rfloor + 3\delta$ and $G$ is a digraph then the $\delta$-folio of $G$ relative to $Z$ is generic.*

*Proof.* The second assertion follows from the first and (5.2). For if $\mu, \kappa, H_1, \ldots, H_\kappa$ satisfy the first assertion of the theorem, and $k \geq \lfloor \frac{5}{2}\xi \rfloor + 3\delta$, then

$$\kappa \geq k - \lfloor \frac{1}{2}\xi \rfloor \geq 2\xi + 3\delta$$

and so the second assertion follows from (5.2).

We prove the first assertion by contradiction. Suppose it is false and choose a counterexample $G$, $Z$ with $|V(G)| + |E(G)|$ minimum. Clearly (using hypothesis (iii) for vertices in $Z$), $G$ has no loops or multiple edges, or isolated vertices. We shall show that

[p. 80] (1) *There is no separation $(A, B)$ of $G$ of order $\leq \xi$, such that $Z \subseteq V(A)$, $A \cap G_i$ is null for some $i$ ($1 \leq i \leq k$), and $B \neq G$.*

*Subproof.* Suppose that $(A, B)$ is such a separation, with $A \cap G_k$ null, say. By hypothesis (iii), $(A, B)$ has order $\xi$. Let $Z' = V(A \cap B)$, and for $1 \leq i \leq k$ let $G'_i = G_i \cap B$. Then $G'_1, \ldots, G'_k$ are mutually vertex-disjoint subgraphs of $B$. Moreover,

(i)' *For $1 \leq i \leq k$, either $G'_i$ is connected or every component of $G'_i$ has a vertex in $Z'$.* For if $G'_i$ has more than one component and one of them does not meet $Z'$, then it is also a component of $G_i$ and does not meet $Z$ and does not equal $G_i$, contrary to hypothesis.

(ii)' *For $1 \leq i < j \leq k$, either $V(G'_i)$ and $V(G'_j)$ both meet $Z'$ or there is an edge of $B$ between $V(G'_i)$ and $V(G'_j)$.* For if $V(G'_i) \cap Z' = \emptyset$ then $G'_i = G_i$ and $V(G_i) \cap Z = \emptyset$. Thus there is an edge of $G$ between $V(G_i)$ and $V(G_j)$.

Since $V(G'_i) \cap Z' = \emptyset$, this edge is an edge of $B$, and hence its other end is in $V(G'_j)$, as required.

(iii)' *For $1 \le i \le k,$, there is no separation $(A', B')$ of $B$ of order $< \xi$ such that $Z' \subseteq V(A')$ and $G'_i \cap A'$ is null.* For if $(A', B')$ is such a separation, then $(A \cup A', B')$ is a separation of $G$ of order $< \xi$, with $Z \subseteq V(A \cup A')$ and with $G'_i \cap (A \cup A')$ null, a contradiction since $G'_i = G_i$.

From (i)', (ii)', (iii)' and the minimality of $G$, we deduce, since $B$ is smaller than $G$, that for some $\mu$ with $0 \le \mu \le \xi$ there are $\kappa = k - \lfloor \frac{1}{2}(\xi - \mu) \rfloor$ non-null connected subgraphs $H'_1, \ldots, H'_\kappa$ of $B$, mutually vertex-disjoint, such that

(a) $|V(H'_i) \cap Z'| = 1$ $(1 \le i \le \xi)$, $V(H'_i) \cap Z' = \emptyset$ $(\xi + 1 \le i \le \kappa)$, and

(b) for $1 \le i < j \le \kappa$, if there is no edge of $B$ between $V(H'_i)$ and $V(H'_j)$ then $i, j \le \mu$.

By Menger's theorem and hypothesis (iii) there are $\xi$ paths $P_1, \ldots, P_\xi$ of $A$ from $Z$ to $Z'$, mutually vertex-disjoint. Let us choose the numbering so that for $1 \le i \le \xi$ the end of $P_i$ in $Z'$ is in $V(H'_i)$. Let

$$\begin{aligned} H_i &= P_i \cup H'_i && (1 \le i \le \xi) \\ H_i &= H'_i && (\xi + 1 \le i \le \kappa). \end{aligned}$$

Then it is easy to see that $H_1, \ldots, H_\kappa$ satisfy the theorem, a contradiction. To see this, we may assume that each $P_i$ has no vertex in $Z \cup Z'$ except its ends (replacing it by a subpath if necessary). Since $|Z| = |Z'| = \xi$, each vertex of $Z$ and each vertex of $Z'$ is an end of exactly one of $P_1, \ldots, P_\xi$. Since $Z \subseteq V(A)$, we have $Z \cap V(B) \subseteq Z'$. For $1 \le i \le \xi$, $H_i$ is connected, since $P_i$ and $H'_i$ are connected and both contain the end $z_i$ of $P_i$ in $Z'$. The graphs $H_1, \ldots, H_\kappa$ are mutually vertex-disjoint, since the $P_i$ are disjoint subgraphs of $A$, the $H'_j$ are disjoint subgraphs of $B$, and a vertex of $P_i$ in $V(B)$ is in $V(A \cap B) = Z'$ and so equals $z_i$, which belongs to no $H'_j$ with $j \ne i$. For (a), if $1 \le i \le \xi$ then $V(H'_i) \cap Z \subseteq V(H'_i) \cap Z' = \{z_i\} \subseteq V(P_i)$, and so $V(H_i) \cap Z = V(P_i) \cap Z$, which has exactly one member. If $\xi + 1 \le i \le \kappa$, then $V(H_i) \cap Z \subseteq V(H'_i) \cap Z' = \emptyset$. For (b), every edge of $B$ is an edge of $G$, so if there is no edge of $G$ between $H_i$ and $H_j$ then there is none of $B$ between $H'_i$ and $H'_j$, and so $i, j \le \mu$.[F7] This proves (1).

(2) *Every edge of $G$ is between $V(G_i)$ and $V(G_j)$ for some choice of $i, j$ with $1 \le i < j \le k$.*

*Subproof.* If $e \in E(G)$ is not between any two distinct $V(G_i)$'s, contracting $e$ produces a smaller counterexample; for by (1) $e$ does not join [p. 81]

[F7] **Missing proof.** The paper says only that this is easy to see. The verification above is the one used in the formalization.

two members of $Z$, and by (1) again, condition (iii) holds after contraction of $e$. This proves (2).

Let $X = V(G) - Z$, and $Y = \bigcup(V(G_i) : 1 \le i \le k, |V(G_i)| \ge 2)$.

(3) *$Y \subseteq Z$; every vertex $x \in X$ is adjacent to every member of $(X - \{x\}) \cup (Z - Y)$; and $|X| - |Y| \ge k - \lfloor \frac{1}{2}|Y| \rfloor - \xi \ge 0$.*

*Subproof.* From (2) and hypothesis (i), it follows that for $1 \le i \le k$, either $|V(G_i)| = 1$ or $V(G_i) \subseteq Z$; and so $Y \subseteq Z$. Since no vertex is isolated, it follows from (2) that

$$V(G) = \bigcup(V(G_i) : 1 \le i \le k).$$

Hence the second claim holds and, moreover,

$$X \cup (Z - Y) = \bigcup(V(G_i) : 1 \le i \le k, |V(G_i)| = 1).$$

Consequently,

$$|X| + |Z - Y| = |\{1 \le i \le k, |V(G_i)| = 1\}|;$$

but

$$\lfloor \frac{1}{2}|Y| \rfloor \le |\{i : 1 \le i \le k, |V(G_i)| \ge 2\}|$$

and, adding, we deduce the first inequality. The second holds since $k \ge \lfloor \frac{3}{2}\xi \rfloor$ and $|Y| \le \xi$. This proves (3).

(4) *There is a matching $M \subseteq E(G)$ with $|M| = |Y|$, such that every member of $M$ has one end in $Y$ and the other in $X$.*

*Subproof.* Let $Y' \subseteq Y$ and let $X' \subseteq X$ be the set of vertices of $X$ with a neighbour in $Y'$. Suppose that $|X'| < |Y'|$. Then $X' \ne X$, since $|X| \ge |Y| \ge |Y'|$ by (3); choose $x \in X - X'$. Choose $i$ with $1 \le i \le k$ and $V(G_i) = \{x\}$. Let $(A, B)$ be a separation of $G$ with $V(A) = Z \cup X'$, $V(B) = V(G) - Y'$. Then $A \cap G_i$ is null, and yet $(A, B)$ has order $|Z - Y'| + |X'| < \xi$, contrary to (iii). Thus $|X'| \ge |Y'|$, and (4) follows from Hall's theorem.

Let $\mu = |Z - Y|$, let $\kappa = k - \lfloor \frac{1}{2}|Y| \rfloor$, and let $M$ be as in (4). For $1 \le i \le \kappa$ we define $H_i$ as follows, mutually vertex-disjoint:

(i) for $1 \le i \le \mu$, $H_i$ consists of a vertex of $Z - Y$

(ii) for $\mu + 1 \le i \le \xi$, $H_i$ consists of an edge of $M$ and its ends

(iii) for $\xi + 1 \le i \le \kappa$, $H_i$ consists of a vertex of $X$ incident with no member of $M$.

[p. 82] These choices are possible by (3). Since every vertex of $X$ and every (different) vertex of $X \cup (Z - Y)$ are adjacent, $H_1, \ldots, H_\kappa$ satisfy the theorem, a contradiction, as required. □

As a consequence of (5.3) we have the following, which we shall need in later papers.

(5.4) *Let $G$ be a graph and let $Z \subseteq V(G)$ with $|Z| = \xi$. Let $k \geq \lfloor \frac{3}{2}\xi \rfloor$ and let $G_1, \ldots, G_k$ be connected subgraphs of $G$, mutually vertex-disjoint, such that for $1 \leq i < j \leq k$ there is an edge of $G$ between $G_i$ and $G_j$. Suppose that there is no separation $(A, B)$ of $G$ of order $< |Z|$ with $Z \subseteq V(A)$ and $A \cap G_i$ null for some $i$ $(1 \leq i \leq k)$. Then for every partition $(Z_1, \ldots, Z_n)$ of $Z$ into non-empty subsets, there are $n$ connected subgraphs $T_1, \ldots, T_n$ of $G$, mutually disjoint and with $V(T_t) \cap Z = Z_t$ $(1 \leq t \leq n)$.*

*Proof.* By (5.3) we may choose $\mu$ with $0 \leq \mu \leq \xi$ and $\kappa = k - \lfloor \frac{1}{2}(\xi - \mu) \rfloor$, and $H_1, \ldots, H_\kappa$ as in (5.3). Let $Z = \{z_1, \ldots, z_\xi\}$, where $z_i \in V(H_i)$ $(1 \leq i \leq \xi)$. For $1 \leq t \leq n$, let $S_t$ be the subgraph of $G$ induced on the set

$$\bigcup \{V(H_i) : 1 \leq i \leq \xi, z_i \in Z_t\}.$$

Then $S_1, \ldots, S_n$ are mutually vertex-disjoint subgraphs of $G$, and $S_t \cap Z = Z_t$ for each $t$, but they may not be connected. If $S_t$ is connected, define $T_t = S_t$. Let $N \subseteq \{1, \ldots, n\}$ be

$$N = \{t : 1 \leq t \leq n, S_t \text{ is not connected}\}.$$

(1) *If $t \in N$ then $|Z_t| \geq 2$ and $Z_t \subseteq \{z_1, \ldots, z_\mu\}$.*

For if $|Z_t| = 1$ then $S_t$ is connected since each $H_i$ is connected. If $z_i \in Z_t$ for some $i$ with $\mu + 1 \leq i \leq \xi$, then since there is an edge between $H_i$ and every other $H_j$ again it follows that $S_t$ is connected.

From (1) it follows that $|N| \leq \lfloor \frac{1}{2}\mu \rfloor$. Since $k \geq \lfloor \frac{3}{2}\xi \rfloor$ it follows that

$$\kappa - \xi = k - \lfloor \frac{1}{2}(\xi - \mu) \rfloor - \xi \geq \lfloor \frac{3}{2}\xi \rfloor - \lfloor \frac{1}{2}(\xi - \mu) \rfloor - \xi \geq \lfloor \frac{1}{2}\mu \rfloor \geq |N|.$$

Let $f$ be an injection from $N$ into $\{1, \ldots, \kappa\}$ such that $V(H_{f(t)}) \cap Z = \emptyset$ for all $t \in N$. (Such an injection $f$ exists since there are $\kappa - \xi \geq |N|$ values of $i \in \{1, \ldots, \kappa\}$ with $V(H_i) \cap Z = \emptyset$). For each $t \in N$ let $T_t$ be the subgraph of $G$ induced on $V(S_t) \cup V(H_{f(t)})$. Then $T_t$ is connected, and $V(T_t) \cap Z = V(S_t) \cap Z = Z_t$ for each $t \in N$; and so $T_1, \ldots, T_n$ satisfy the theorem. □

(5.4) implies a theorem of Larman and Mani [10]. The latter asserts that if $s_1, t_1, \ldots, s_n, t_n$ are distinct vertices of a $2n$-connected graph $G$ which has a subgraph that is a subdivision of $K_{3n}$, then there are $n$ mutually disjoint paths joining $s_i$, $t_i$ $(1 \leq i \leq n)$. It can be derived from (5.4) by setting [p. 83]

$\xi = 2n$, $k = 3n$, $Z = \{s_1, \ldots, s_n, t_1, \ldots, t_n\}$, $Z_1 = \{s_1, t_1\}, \ldots, Z_n = \{s_n, t_n\}$ and by choosing the $G_i$'s suitably from the subdivided $K_{3n}$.

Let us digress for a moment to get a corollary of (5.4) of interest in "pure" graph theory. The following is due to Thomason [28].[F8]

(5.5) *Let $p \geq 1$ be an integer, and let $G$ be a non-null simple graph with*

$$|E(G)| \geq 2.68p(\log_2(p))^{1/2}|V(G)|.$$

*Then $G$ has a $K_p$-minor.*

A graph $G$ is said to be *$p$-linked*, where $p \geq 0$ is an integer, if for all $s_1, \ldots, s_p$, $t_1, \ldots, t_p \in V(G)$ with $s_i, t_i \neq s_j, t_j$ for $1 \leq i < j \leq p$, there are $p$ mutually vertex-disjoint paths of $G$ linking $s_i$ and $t_i$ for $1 \leq i \leq p$. It is known that for any $p$, every sufficiently connected graph is $p$-linked, but it is an open problem to determine the minimum $f(p)$ such that every $f(p)$-connected graph is $p$-linked. It seems that, up to now, the only known bounds on $f(p)$ were exponential in $p$. But from (5.4) and (5.5) we obtain that $f(p) \leq 16.08p(\log_2(3p))^{1/2}$. More precisely,

(5.6) *For every integer $p \geq 0$, every $2p$-connected simple graph $G$ with $|E(G)| \geq 8.04p(\log_2 3p)^{1/2}|V(G)|$ is $p$-linked.*

*Proof.* By (5.5) (since we may assume that $p \geq 1$ and $V(G) \neq \emptyset$) $G$ has a $K_{3p}$-minor. Hence there are $3p$ connected subgraphs $G_1, \ldots, G_{3p}$ of $G$, mutually vertex-disjoint, such that for $1 \leq i < j \leq 3p$ there is an edge of $G$ between $G_i$ and $G_j$. Let $Z \subseteq V(G)$ with $|Z| \leq 2p$. There is no separation $(A, B)$ of $G$ of order $< |Z|$ with $Z \subseteq V(A)$ and $A \cap G_i$ null for some $i$, since $G$ is $2p$-connected. From (5.4), setting $Z = \{s_1, \ldots, s_p, t_1, \ldots, t_p\}$, it follows that $G$ is $p$-linked, as required. □

## 6. What To Do with a Clique

In this section we use (5.3) to show that, in solving Folio, if we find a large clique minor we can find an irrelevant vertex. We need the following.

(6.1) *Let $G$ be a digraph and let $Z \subseteq V(G)$ with $|Z| \leq \xi$. Let $k = \lfloor \frac{5}{2}\xi \rfloor + 3\delta + 1$ and let $G_1, \ldots, G_k$ be mutually vertex-disjoint connected subgraphs of $G$ such that for $1 \leq i < j \leq k$ there is an edge of $G$ between $G_i$ and $G_j$. Let $(A, B)$ be a separation of $G$ such that*

*(i) $A \cap G_i$ is null for some $i$ $(1 \leq i \leq k)$*

[F8] **Proof gap.** Thomason's paper [28] proves the constant 2.68 only for all sufficiently large $p$. The bound for every $p \geq 1$ is proved in the formalization, and a written proof for $p \geq 51$ is in the folder docs/thomason of the formalization.

*(ii)* $Z \subseteq V(A)$

[p. 84] *(iii) subject to (i) and (ii),* $(A, B)$ *has minimum order*

*(iv) subject to (i), (ii), and (iii),* $A$ *is maximal.*

*Let* $v \in V(B) - V(A)$*. Then* $v$ *is irrelevant to the* $\delta$*-folio of* $G$ *relative to* $Z$*.*

*Proof.* By (2.3), it suffices to show that $v$ is irrelevant to the $\delta$-folio of $B$ relative to $V(A \cap B)$. To prove this, it suffices to show that the $\delta$-folio of $B \setminus v$ relative to $V(A \cap B)$ is generic. Let $V(A \cap B) = Z'$, and $|Z'| = \xi' \leq \xi$. Let $G_1, \ldots, G_k$ be numbered so that $v \notin V(G_1) \cup \cdots \cup V(G_{k-1})$, and let $G'_i = G_i \cap B$ $(1 \leq i \leq k-1)$. Let $G' = B \setminus v$. We shall apply (5.3) to $G', Z', G'_1, \ldots, G'_{k-1}$. Let us verify its hypotheses. We observe that $k - 1 \geq \lfloor \frac{5}{2}\xi' \rfloor + 3\delta$, and $G'_1, \ldots, G'_{k-1}$ are subgraphs of $G'$, mutually vertex-disjoint. Moreover, for $1 \leq i \leq k-1$, either $G'_i$ is connected or every component of $G'_i$ has a vertex in $Z'$; for if some component of $G'_i$ has no vertex in $Z'$, then it is a component of $G_i$, and so equals $G_i$ since $G_i$ is connected; and so $G'_i$ is connected. Furthermore, for $1 \leq i < j \leq k-1$, either $V(G'_i)$ and $V(G'_j)$ both meet $Z$ or there is an edge of $G'$ between $V(G'_i)$ and $V(G'_j)$; for if say $V(G'_i) \cap Z = \emptyset$ then $G'_i = G_i$, and since there is an edge G between $V(G_i)$ and $V(G_j)$, this edge is therefore an edge of $G'$ between $V(G'_i)$ and $V(G'_j)$. Finally, we claim

(1) *For* $1 \leq i \leq k-1$*, there is no separation* $(A', B')$ *of* $G'$ *of order* $< \xi'$ *with* $Z' \subseteq V(A')$ *and with* $A' \cap G'_i$ *null.*

*Subproof.* If $(A', B')$ is such a separation, there is a separation $(A'', B'')$ of $B$ of order $\leq \xi'$ with $v \in V(A'' \cap B'')$ and $A'' \setminus v = A'$, $B'' \setminus v = B'$. Then $(A \cup A'', B'')$ is a separation of $G$ of order $\leq \xi'$. Now $Z \subseteq V(A \cup A'')$, and $(A \cup A'') \cap G_i$ is null, and $A \cup A'' \neq A$ since $v \in V(A'')$. This contradicts hypothesis (iv), and proves (1).

From (1) and (5.3) we deduce that the $\delta$-folio of $G'$ relative to $Z'$ is generic, as required. □

We use (6.1) for the following algorithm. Let $\xi, \delta \geq 0$ be integers, and let $p = \lfloor \frac{5}{2}\xi \rfloor + 3\delta + 1$.

(6.2) Algorithm.

*Input:* A digraph $G$, a subset $Z \subseteq V(G)$ with $|Z| \leq \xi$, and $p$ mutually vertex-disjoint connected subgraphs $G_1, \ldots, G_p$ of $G$ such that for $1 \leq i < j \leq p$ there is an edge of $G$ between $G_i$ and $G_j$.

*Output:* A vertex of $G$ irrelevant to the $\delta$-folio of $G$ relative to $Z$.

*Running time:* $O(\varepsilon)$.

*Description:* For each $i$ with $1 \leq i \leq p$ such that $Z \cap V(G_i) = \emptyset$ we find a separation $(A_i, B_i)$ of $G$ with $Z \subseteq V(A_i)$ and $A_i \cap G_i$ null, of minimum

order and subject to that with $A_i$ maximal. (The usual max-flow min-cut algorithm provides a "min-cut" which yields a separation $(A, B)$ with $A$ [p. 85] automatically maximal, if we run the augmenting paths starting from $V(G_i)$. Thus this is easy.) We choose $\xi'$ minimum such that some $(A_i, B_i)$ has order $\xi'$, and we choose $i$ with $A_i$ maximal such that $(A_i, B_i)$ has order $\xi'$. We choose $v \in V(G_i)$. Then $v$ is the desired irrelevant vertex, by (6.1), since $(A_i, B_i)$ satisfies the conditions for $(A, B)$ in (6.1).

The algorithm involves computing $(A_i, B_i)$ for $\leq p$ values of $i$; and each computation can be performed with a standard network flow algorithm in time $O(\varepsilon)$. Thus the algorithm has running time $O(\varepsilon)$. □

(6.2) will form one of the key steps of our algorithm for Folio, but there is also a peripheral application for it, to reduce our general problem to the case when $|E(G)| \leq O(\nu)$ and, thus, to improve the overall running time of the main algorithm. In the remainder of this section we explain this.

If $G$ is a graph and $X \subseteq V(G)$, we denote by $N(X)$ the set of vertices in $V(G) - X$ with a neighbour in $X$. First, we need the following.

(6.3) Algorithm.

*Input:* An integer $k > 0$ and a non-null simple graph $G$ with $|E(G)| \geq k|V(G)|$.

*Output:* A subset $X \subseteq V(G)$ such that

(i) the restriction of $G$ to $X$ is non-null and connected

(ii) $N(X)$ is non-null, and

(iii) every vertex of $N(X)$ has $\geq k$ neighbours in $N(X)$.

*Running time:* $O(\nu\varepsilon)$.

*Description:* (1) Choose a component $H$ of $G$ with $|E(H)| \geq k|V(H)|$, and choose $v_1 \in V(H)$.

(2) We perform a recursion. Set $X_1 = \{v_1\}$. At the start of the $i$ th step of the recursion we have a subset $X_i \subseteq V(H)$ with $|X_i| = i$ such that $X_i$ meets at most $|N(X_i)| + k(i-1)$ edges, and such that the restriction of $G$ to $X_i$ is connected.

(3) If there exists $v \in N(X_i)$ with $< k$ neighbours in $N(X_i)$, we set $X_{i+1} = X_i \cup \{v\}$ and return for the next iteration. To check the inequality, let $v$ have $d_1$ neighbours in $N(X_i)$ and $d_2$ in $V(H) - (X_i \cup N(X_i))$. Then

$$|N(X_{i+1})| = |N(X_i)| - 1 + d_2$$

and $X_{i+1}$ meets $d_1 + d_2 \leq k - 1 + d_2$ edges not met by $X_i$. It follows that $X_{i+1}$ meets at most

$$|N(X_i)| + k(i-1) + k - 1 + d_2 = |N(X_{i+1})| + ki$$

edges in total, and so $X_{i+1}$ has the properties claimed.

[p. 86] (4) If there is no such vertex $v$, we set $X = X_i$ and stop. We claim that $X$ has the desired properties for the output. For certainly (i) and (iii) hold; suppose that $N(X)$ is null. Then $X_i = V(H)$ and so $i = |V(H)|$. But $X_i$ meets at most $|N(X_i)| + k(i-1)$ edges, and $N(X_i) = \emptyset$, and so

$$|E(H)| \leq k(|V(H)| - 1),$$

contrary to our choice of $H$. Thus $N(X)$ is non-null, as required.

There are at most $v(G)$ iterations of the recursion, and each takes time $O(\varepsilon)$. Thus the total running time is $O(v\varepsilon)$. □

We use (6.3) for the following, which is related to (5.5) and to a theorem of Mader [11]. Let $p \geq 1$ be an integer.

(6.4) Algorithm.

*Input:* A non-null simple graph $G$ with $|E(G)| \geq 2^{p-3}|V(G)|$.

*Output:* A model of $K_p$ in $G$.

*Running time:* $O(v^2)$.

*Description:* We may assume that $p \geq 3$, for otherwise the problem is trivial.

(1) Choose a subgraph $H \subseteq G$ with $V(H) = V(G)$ and $|E(H)| = 2^{p-3}|V(H)|$.

(2) We perform a recursion. Let $Y_1 = V(H)$. At the start of the $i$ th iteration we have $i-1$ disjoint connected subgraphs $G_1, \ldots, G_{i-1} \subseteq H$ and a non-null subset $Y_i \subseteq V(H) - V(G_1 \cup \cdots \cup G_{i-1})$ such that

(i) there are at least $2^{p-2-i}|Y_i|$ edges of $H$ with both ends in $Y_i$,

(ii) every vertex in $Y_i$ has a neighbour in each of $V(G_1), \ldots, V(G_{i-1})$,

and (iii) for $1 \leq j < j' \leq i-1$, each vertex of $V(G_{j'})$ has a neighbour in $V(G_j)$. The $i$ th iteration proceeds as follows.

(3) We apply (6.3) to the restriction of $H$ to $Y_i$, taking $k = 2^{p-2-i}$. We obtain $X \subseteq Y_i$ such that the restriction of $H$ to $X$ is non-null and connected, $N(X) \cap Y_i$ is non-null, and every vertex of $N(X) \cap Y_i$ has at least $2^{p-2-i}$ neighbours in $N(X) \cap Y_i$. We set $G_i$ to be the restriction of $H$ to $X$, and $Y_{i+1} = N(X) \cap Y_i$. If $i = p-2$ the recursion terminates, and otherwise we return for the next iteration.

(4) Let $G_{p-1}, G_p$ be one-vertex graphs with $V(G_{p-1}) = \{u\}$, $V(G_p) = \{v\}$, for some adjacent $u, v \in Y_{p-1}$. (These exist since $Y_{p-1} \neq \emptyset$ and every [p. 87] $u \in Y_{p\ 1}$ has a neighbour in $Y_{p-1}$.) Then $G_1, \ldots, G_p$ are disjoint, non-null, connected subgraphs of $G$, and for $1 \leq i < j \leq p$ every vertex of $G_j$ has a neighbour in $G_i$. From this we obtain a model of $K_p$.

There are $p-2$ iterations of the recursion, and each takes time $O(v(H)\varepsilon(H))$. Since $|E(H)| = 2^{p-3}|V(G)|$, the running time is $O(v(G)^2)$. □

Next, let $\xi, \delta \geq 0$ be integers.

(6.5) Algorithm.

*Input:* A digraph $G$ and a subset $Z \subseteq V(G)$ with $|Z| \leq \xi$.

*Output:* A subdigraph $G'$ of $G$ with $Z \subseteq V(G')$, such that $G$ and $G'$ have the same $\delta$-folio relative to $Z$, and such that $|E(G')| \leq C|V(G')|$, where $C = \delta + 2(\delta+1)2^{p-3}$ and where $p = \lfloor \frac{5}{2}\xi \rfloor + 3\delta + 1$.

*Running time:* $O(v^3)$.

*Description:*

(1) Let $H$ be a maximal subdigraph of $G$ with at most $\delta$ loops at each vertex and at most $\delta+1$ edges with head $u$ and tail $v$, for all distinct $u, v \in V(G)$. Certainly $G$ and $H$ have the same $\delta$-folios relative to $Z$.

(2) We set $G_1 = H$ and perform a recursion. At the start of the $i$ th iteration we have a subdigraph $G_i$ of $H$ with $Z \subseteq V(G_i)$ and $|V(G_i)| = |V(H)| - i + 1$ such that $G_i$ and $H$ (and hence $G$) have the same $\delta$-folios relative to $Z$. If $|E(G_i)| \leq C|V(G_i)|$ we set $G' = G_i$ and stop. Otherwise, the $i$ th iteration proceeds as follows.

(3) Let $J_i$ be the simple (undirected) graph underlying $G_i$. Then

$$\begin{aligned}(\delta + 2(\delta+1)2^{p-3})|V(J_i)| = C|V(G_i)| &\leq |E(G_i)| \\ &\leq \delta|V(J_i)| + 2(\delta+1)|E(J_i)|\end{aligned}$$

and so $2^{p-3}|V(J_i)| \leq |E(J_i)|$. We apply (6.4) to $J_i$ and obtain a model of $K_p$ in $J_i$ (and hence in $G_i$).

(4) We apply (6.2) to $G_i$ and the model of $K_p$ and find a vertex $v$ of $G_i$ irrelevant to the $\delta$-folio of $G_i$ relative to $Z$. We set $G_{i+1} = G_i \setminus v$ and return for the next iteration.

There are at most $v(G)$ iterations, and each iteration involves an application of (6.4) to $J_i$, which takes time $O(v^2)$, and an application of (6.2) to $G_i$, which takes time $O(\varepsilon(G_i)) \leq O(v^2)$. Thus the total running time for constant $\delta, \xi$ is $O(v^3)$.

(6.3)-(6.5) can all be implemented more quickly than stated, but for us it makes no difference.

## [p. 88] 7. Walls

Now we come to the third part of the paper, which concerns what to do to solve Folio when the branch-width of our graph is too high to apply the methods of Section 4 and yet we have not found a large clique minor. In

such a graph there is a large grid minor, by (4.2), and, roughly, our method is to find a section of such a minor that does not stand out among the other sections of the grid close to it in any significant way (with regard to how the rest of $G$ is attached to it.) From a theorem which we postpone to a later paper, any vertex in the middle of such an "ordinary" section is irrelevant. Here we give an algorithm which, given a grid minor, either finds a large clique minor, or finds an "ordinary" section of the grid, or moves to a better grid minor, when we start again.

Although grids are natural objects to use from some viewpoints, grid minors introduce a complication (due to the fact that their vertices are mostly 4-valent) which is unnecessary and can be avoided by using elementary walls instead (which are parts of the hexagonal lattice instead of the square lattice). Figure 1 shows an elementary wall of height 8. It has four distinguished vertices, called its *corners*, shaded in the figure. The *elementary wall of height* $h$ (where $h \geq 2$ is even) is similar; there are $h$ rows of "bricks," and $h$ "bricks" in each row, and four "corners." Thus there are $h+1$ "horizontal" paths and its "vertical" edges divide into $2h+2$ "collinear" sets of $\frac{1}{2}h$ edges each, and the corners are the ends of the first and the last horizontal paths. The *perimeter* is the circuit which bounds the infinite region when drawn as in Fig. 1.

A *wall of height* $h$ (where $h \geq 2$ is even) is a subdivision of an elementary wall of height $h$. It too therefore has four distinguished vertices, called its *corners*. A wall is thus a pair $(H, P)$, where $H$ is a graph and $P \subseteq V(H)$ is

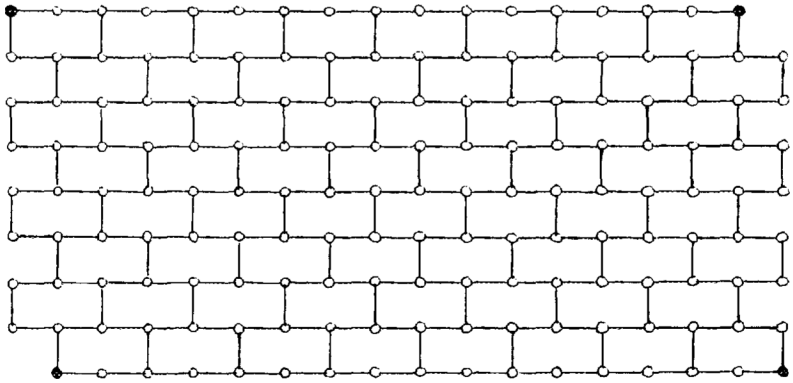


FIG. 1. Elementary 8-wall.

[p. 89] the set of corners, but for convenience we speak of a wall as a graph and leave the corners implicit. The *perimeter* of a wall is defined as before.

Now since every vertex of an elementary wall $H$ of height $h$ has valency

$\leq 3$, if a graph has a minor isomorphic to $H$ then it has a wall of height $h$ as a subgraph (and algorithmically, it is easy to see how to construct the wall in time $O(\varepsilon)$, given a model of $H$). To see both statements, let $\phi$ be a model of $H$ in the graph. Let $u \in V(H)$, and let $t_1, \ldots, t_k$ be the ends in $V(\phi(u))$ of the edges $\phi(e)$, one for each edge $e$ of $H$ incident with $u$. Thus $k \leq 3$, and $t_1, \ldots, t_k$ need not be distinct. Take a breadth-first search tree of $\phi(u)$ from $t_1$, and let $Q_2, Q_3$ be the paths of this tree from $t_2, t_3$ to $t_1$ (if $k = 2$, let $t_3 = t_1$, so that $Q_3$ is the one-vertex path $t_1$). Let $b_u$ be the first vertex of $Q_3$ that belongs to $Q_2$. Since both paths run to the root of the tree, $Q_3$ follows $Q_2$ from $b_u$ on, and so $\phi(u)$ contains three paths from $b_u$ to $t_1, t_2, t_3$, mutually disjoint except for $b_u$. For each edge $e = uw$ of $H$, the path from $b_u$ to the end of $\phi(e)$ in $\phi(u)$, the edge $\phi(e)$, and the path from the end of $\phi(e)$ in $\phi(w)$ to $b_w$ form a path $P_e$ between $b_u$ and $b_w$. Since the graphs $\phi(u)$ are mutually disjoint, the paths $P_e$ ($e \in E(H)$) are mutually disjoint except for their ends, and so their union is a subdivision of $H$, that is, a wall of height $h$ whose corners are the vertices $b_u$ for the corners $u$ of $H$. Each search takes time linear in $|V(\phi(u))| + |E(\phi(u))|$, and the graphs $\phi(u)$ are disjoint, so the wall is found in time $O(\varepsilon)$.[F9]

(7.1) *For $h \geq 2$, even, if every graph with branch-width $\geq \omega/3$ has a minor isomorphic to the elementary wall of height $h$, then there is an algorithm as in (7.2), with running time $O(v^2 + \varepsilon)$.*

(7.2) Algorithm.

*Input:* A graph $G$.

*Output:* Either a branch-decomposition of $G$ of width $\leq \omega$, or a wall of height $h$ which is a subgraph of $G$.

*Description:* Given input $G$, we apply (4.4) to $G$ to determine if there is a model of the elementary wall of height $H$ in $G$ and to find one if one exists. If we find one, we construct from it a wall of height $h$ and stop. If there is none, then $G$ has branch-width $< \omega/3$. We apply (3.3) to $G$ and obtain a subgraph $G'$ of $G$ and a branch-decomposition of $G'$ of width $\leq \omega$, such that either $G' = G$ or $G'$ has branch-width $\geq \omega/3$. Thus $G' = G$, and we have a branch-decomposition of $G$ of width $\leq \omega$. We stop.

This algorithm involves one application of (4.4) and one of (3.3); thus, it has running time $O(v^2 + \varepsilon)$. □

If $H$ is a wall, drawn in the plane as in Fig. 1, its *distance function* $d$ is defined as follows. For $s, t$ in the plane, $d(s, t) = 0$ if $s = t$, and otherwise

[F9] **Missing proof.** The paper says only that this is easy to see. The construction above, by breadth-first search in each branch set, is the one used in the formalization.

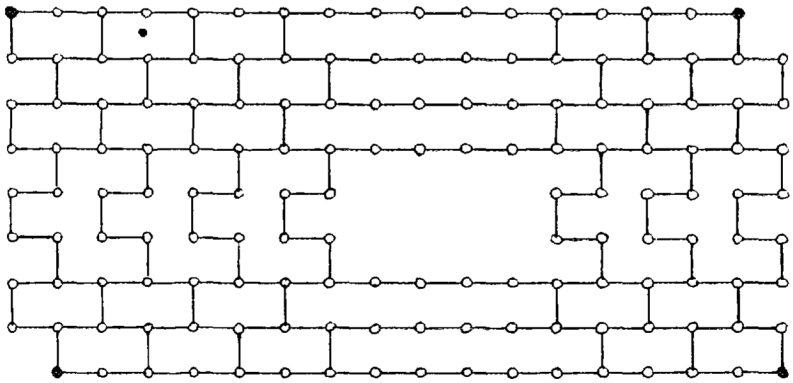


FIG. 2. A hole in a wall.

[p. 90] $d(s,t)$ is the minimum of the number of points of $F$ in the drawing, taken over all subsets $F$ in the plane homeomorphic to $[0,1]$ with ends $s$ and $t$. We shall speak of the distance function of $H$ without reference to any drawing, but we assume implicitly that there is given some fixed drawing of $H$. If $u,v \in V(H)$, we define $d(u,v) = d(s,t)$, where $s,t$ are the points of the drawing representing $u,v$. If $H$ and $H'$ are both walls and $H'$ is a subgraph of $H$, let the corresponding distance functions (using the drawing of $H'$ obtained from the given drawing of $H$) be $d, d'$. If $d'(s,t) \geq d(s,t) - k$ for all points $s,t$ of the plane, we write $d' \geq d - k$.

Given a wall $H$ of height $h$, we can obtain another wall of height $h-2$ as indicated in Fig. 2, omitting half the horizontal edges of two consecutive rows and all the vertical edges in four consecutive columns and the four single-edge components that result. By generalizing this construction, we deduce

(7.3) *Let $H$ be a wall of height $h$ with distance function $d$ and let $v \in V(H)$. Let $r \geq 0$ be an integer with $h \geq 2r+4$. Then there is a subgraph $H'$ of $H$, which is a wall of height $h-2r-2$, such that*

*(i) $d' \geq d - 4r - 4$, where $d'$ is the distance function in $H'$*

*(ii) there is no vertex $u$ of $H'$ with $d(u,v) \leq r$.*

~~The proof is left to the reader.~~ (The most difficult case is when $v$ is close to the infinite region of $H$, when it is necessary to delete a band of $H$ around the outside.)

We prove (7.3). Let $k = r+1$, so that $h \geq 2k+2$. We need a subgraph $H'$ of $H$ which is a wall of height $h-2k$, such that $d' \geq d-4k$ and $d(u,v) \geq k$ for every vertex $u$ of $H'$. The distances $d$ and $d'$ depend only on the drawings of $H$ and $H'$ as subsets of the plane. A homeomorphism of the plane maps subsets homeomorphic to $[0,1]$ onto such subsets, and so applying it to the

drawing of $H$ changes neither the inequality $d' \geq d - 4k$ nor the numbers $d(u, v)$. Hence we may assume that $H$ is drawn exactly as in Fig. 1, in the coordinates below.

We proceed as follows. In (1) and (2) we express $d$ through the regions of $H$. In (6) we give a test which shows $d' \geq d - 2c$ for a subwall $H'$, by comparing the regions of $H'$ with those of $H$. Then we build three kinds of subwalls and check the test for each: a copy of a smaller wall moved inside $H$, used when $v$ is near a side of $H$, a wall with a small hole, used when $r = 0$, and a wall with a large hole around $v$, used when $r \geq 1$ and $v$ is far from the sides.

The elementary wall of height $h$ has as vertices the integer points $(x, y)$ with $0 \leq x \leq 2h + 1$ and $0 \leq y \leq h$, other than $(0, 0)$ and $(2h + 1, h)$. Its *horizontal* edges join $(x, y)$ and $(x + 1, y)$, and its *vertical* edges join $(x, y)$ and $(x, y+1)$ when $x+y$ is odd. Each edge is drawn as the straight segment between its ends, and this is the drawing of Fig. 1. We say that $(x, y)$ is in *row* $y$ and *column* $x$, and that the horizontal edge from $(x, y)$ to $(x + 1, y)$ and the vertical edge from $(x, y)$ to $(x, y + 1)$ are *at* $(x, y)$. The vertices and the edges of the elementary wall are its *elements*, where an edge, as an element, is the open segment without its ends. The *midpoint* of an element is the vertex itself or the middle of the segment. The wall $H$ is drawn as the elementary wall, with the vertices of the subdivision placed on the segments, and so every point of the drawing of $H$ lies in exactly one element. Each wall $H' \subseteq H$ that we construct has its branch vertices at vertices of the elementary wall, and its drawing is a union of elements. We say that $H'$ *keeps* the elements in its drawing and *deletes* the others.

For integers $x, y$ the *square* $(x, y)$ is the open unit square $(x, x+1)\times(y, y+1)$, and the *strip* $y$ is the set of points whose second coordinate lies strictly between $y$ and $y + 1$. A *position* is a pair $(X, Y)$ of integers with $X + Y$ even. The position of the square $(x, y)$ is $(x, y)$ if $x+y$ is even, and $(x+1, y)$ if $x + y$ is odd. A position $(X, Y)$ is a *brick of height* $h$ if $0 \leq Y \leq h-1$ and $1 \leq X \leq 2h$. In this case there is no vertical edge at $(X, Y)$, since $X + Y$ is even, and so the open rectangle $(X - 1, X + 1) \times (Y, Y + 1)$ is a region of $H$. These regions are the bricks of Fig. 1, and we call this region the brick $(X, Y)$ as well. It consists of the two squares with position $(X, Y)$, namely $(X - 1, Y)$ and $(X, Y)$, and the open segment between them. Hence the regions of $H$ are the bricks of height $h$ and the infinite region, and each square lies in the brick of its position if this position is a brick of height $h$, and in the infinite region otherwise. The perimeter of $H$ consists of the points of the drawing with first coordinate at most 1 or at least $2h$, or with second coordinate 0 or $h$.

A region is *at* a point $s$ if $s$ is in its closure. The regions at a point of an element are the regions containing the squares whose closures contain the midpoint of the element. We call these squares the squares *at* the element, and these regions the regions at the element. If the midpoint is $(p, q)$, the squares at the element are the squares $(u, w)$ with $\lceil p \rceil - 1 \le u \le \lfloor p \rfloor$ and $\lceil q \rceil - 1 \le w \le \lfloor q \rfloor$. So their positions $(X, Y)$ satisfy $|X - p| \le 1$ and $q - 1 \le Y \le q$. Moreover, some square at the element has a position with $X \le p - \frac{1}{2}$, some has $X \ge p + \frac{1}{2}$, some has $Y \le q - \frac{1}{2}$ and some has $Y \ge q - \frac{1}{2}$. For the last two take $w = \lceil q \rceil - 1$ and $w = \lfloor q \rfloor$. For the first two, at a vertex $(x, y)$ one of the squares $(x - 1, y - 1)$ and $(x - 1, y)$ has position $x - 1$, and one of $(x, y - 1)$ and $(x, y)$ has position $x + 1$. At the horizontal edge at $(x, y)$ the squares $(x, y - 1)$ and $(x, y)$ have positions $x$ and $x + 1$ in some order, and at the vertical edge at $(x, y)$ the squares $(x - 1, y)$ and $(x, y)$ have positions $x - 1$ and $x + 1$. For every element at $(x, y)$, the square $(x, y)$ is at it.

For a pair $w = (w_1, w_2)$ of integers let $\|w\| = \max(|w_2|, \frac{1}{2}(|w_1| + |w_2|))$. Each of the two terms of this maximum is unchanged when $w$ is replaced by $-w$, and its value at $w + w'$ is at most the sum of its values at $w$ and at $w'$. So $\|-w\| = \|w\|$ and $\|w + w'\| \le \|w\| + \|w'\|$. If $w$ is the difference of two positions then $w_1 + w_2$ is even, and so $\|w\|$ is an integer. For a position $F = (X, Y)$ let

$$\sigma_h(F) = \min\left(Y + 1,\ h - Y,\ \lfloor (X + 1)/2 \rfloor,\ \lfloor (2h + 2 - X)/2 \rfloor\right).$$

We call the four numbers in this minimum the *bottom*, *top*, *left* and *right terms* of $F$, for height $h$. By (2) below, $\|F - G\|$ is the number of steps between touching bricks from $F$ to $G$, and $\sigma_h(F)$ is the number of steps from $F$ to the infinite region. If $F$ is a brick of height $h$, each term is at least 1, and $\sigma_h(F) = 1$ exactly when $Y \in \{0, h - 1\}$ or $X \in \{1, 2, 2h - 1, 2h\}$, that is, exactly when the closure of $F$ meets the perimeter. If $F$ is not a brick of height $h$, then one of its terms is at most 0, and so $\sigma_h(F) \le 0$. We use two simple facts. First, for integers $A \ge B$ we have $\lfloor A/2 \rfloor - \lfloor B/2 \rfloor \le \lceil (A - B)/2 \rceil$, since $\lfloor A/2 \rfloor \le A/2$ and $\lfloor B/2 \rfloor \ge (B - 1)/2$. Second, to show that $\sigma_h(F) \le \sigma_{h'}(F') + c$ for positions $F, F'$ and heights $h, h'$, it is enough to show that each term of $F'$ for height $h'$, plus $c$, is at least some term of $F$ for height $h$.

Let $J$ be a graph drawn in the plane. Two regions of $J$ *touch* if their closures have a common point. For regions $R, R'$ let $D_J(R, R')$ be the least $n \ge 0$ such that there are regions $R = R_0, R_1, \ldots, R_n = R'$ with $R_{i-1}$ touching $R_i$ for $1 \le i \le n$. Thus $D_J$ satisfies the triangle inequality. We

define $d_J$ from the drawing of $J$ in the same way as $d$ is defined from the drawing of $H$. The *depth* of a region $R$ is $D_J(R, R_\infty)$, where $R_\infty$ is the infinite region. For a point $s$ let $[s] = 1$ if $s$ is a point of the drawing, and $[s] = 0$ otherwise.

(1) *For distinct points $s, t$ we have $d_J(s,t) = [s] + [t] + \min D_J(R, R')$, where the minimum is over all regions $R$ at $s$ and $R'$ at $t$.*

Let $F$ be as in the definition of $d_J$, with the fewest points in the drawing. Since there is such a set with finitely many points in the drawing, $F$ has finitely many. Between two consecutive ones, $F$ runs inside one region, and the first and last of these regions are at $s$ and at $t$. At each point of the drawing other than $s$ and $t$, $F$ passes from one region into a region that touches it (or into the same region). So the regions met by $F$, in order, give a sequence of touching regions from a region at $s$ to a region at $t$, with at most as many steps as $F$ has points in the drawing other than $s$ and $t$. This proves "$\geq$". Conversely, choose a region $R$ at $s$, a region $R'$ at $t$, and regions $R = R_0, \ldots, R_n = R'$ as above, with $n$ as small as possible, and for $1 \leq i \leq n$ choose a point $x_i$ of the drawing in the closures of $R_{i-1}$ and $R_i$. The regions $R_i$ are distinct, and the points $s, x_1, \ldots, x_n, t$ are distinct, since otherwise the sequence could be shortened (if $x_i = x_j$ with $i < j$, then $R_{i-1}$ touches $R_j$, and if $x_1 = s$ then $R_1$ is at $s$). Join each two consecutive points of $s, x_1, \ldots, x_n, t$ by an arc inside the corresponding region $R_i$, apart from its ends. The union of these arcs is a set $F$ as in the definition, and it meets the drawing in $[s] + [t] + n$ points. This proves "$\leq$", and so proves (1).

(2) *Two distinct bricks $F, G$ of $H$ touch if and only if $\|F - G\| = 1$, and a brick $F$ touches the infinite region if and only if $\sigma_h(F) = 1$. The depth of a brick $F$ is $\sigma_h(F)$, and $D_H(F,G) = \min(\|F - G\|, \sigma_h(F) + \sigma_h(G))$ for all bricks $F, G$.*

Let $F = (X, Y)$ and $G = (X', Y')$. Their closures are the rectangles $[X-1, X+1] \times [Y, Y+1]$ and $[X'-1, X'+1] \times [Y', Y'+1]$, which meet if and only if $|X - X'| \leq 2$ and $|Y - Y'| \leq 1$. Since $X - X'$ and $Y - Y'$ have the same parity and $F \neq G$, this holds exactly when $(|X - X'|, |Y - Y'|)$ is $(2,0)$ or $(1,1)$, that is, when $\|F - G\| = 1$. The closure of the infinite region is the infinite region together with the perimeter, and so $F$ touches the infinite region exactly when the closure of $F$ meets the perimeter, that is, when $\sigma_h(F) = 1$.

Let $F = R_0, R_1, \ldots, R_n = R_\infty$ be touching regions with $n$ equal to the depth of $F$. Then $R_1, \ldots, R_{n-1}$ are bricks. Between touching bricks each term changes by at most 1, since the second coordinate changes by at most 1 and the first by at most 2, and so $\sigma_h$ changes by at most 1. Since $R_{n-1}$ touches the infinite region, $\sigma_h(R_{n-1}) = 1$, and so $\sigma_h(F) \leq n$. Conversely,

a brick $(X, Y)$ with $Y \geq 1$ touches the brick $(X-1, Y-1)$ or the brick $(X+1, Y-1)$, one of which is a brick of height $h$ since $1 \leq X \leq 2h$. So $Y$ such steps lead from $F$ to a brick in row 0, which touches the infinite region, and the depth of $F$ is at most $Y+1$. In the same way, going up, the depth is at most $h-Y$. Going left or right by steps that change the first coordinate by 2, until it is in $\{1,2\}$ or in $\{2h-1, 2h\}$, shows that the depth is at most $\lfloor (X+1)/2 \rfloor$ and at most $\lfloor (2h+2-X)/2 \rfloor$. Hence the depth of $F$ is $\sigma_h(F)$.

Now let $F = R_0, \ldots, R_n = G$ be touching regions with $n = D_H(F, G)$. If one of them is the infinite region, then $n \geq \sigma_h(F) + \sigma_h(G)$, by the depths. Otherwise they are all bricks, and $n \geq \|F - G\|$ by the triangle inequality, since $\|R_i - R_{i-1}\| = 1$. Conversely, $D_H(F, G) \leq \sigma_h(F) + \sigma_h(G)$, by passing through the infinite region. Moreover $D_H(F, G) \leq \|F - G\|$. For let $w = G - F$. If $|w_1| \geq |w_2|$, take $|w_2|$ steps that add $(\pm 1, \pm 1)$ and then $\frac{1}{2}(|w_1| - |w_2|)$ steps that add $(\pm 2, 0)$, with all signs towards $G$. If $|w_1| < |w_2|$, take $|w_2|$ steps that add $(\pm 1, \pm 1)$, all with the second sign towards $G$. The first $|w_1|$ of them have the first sign towards $G$, and the remaining ones (an even number) alternately move the first coordinate away from $X'$ and back, to the side of $X'$ on which it stays between 1 and $2h$. In both cases all the positions passed are bricks, since their coordinates stay in the ranges of bricks, and the number of steps is $\|F - G\|$. This proves (2).

We shall use (2) in the following form. If $R$ and $R'$ are the regions of $H$ that contain squares with positions $F$ and $G$, then $R$ has depth $\max(\sigma_h(F), 0)$, and

$$D_H(R, R') \geq \min(\|F - G\|, \mathrm{depth}(R) + \mathrm{depth}(R')).$$

If $R$ and $R'$ are bricks this is (2). If one of them is the infinite region, then $D_H(R, R')$ is the sum of the depths, and the infinite region has depth 0 while $\sigma_h(F) \leq 0$ if $F$ is not a brick.

We use the following form of a theorem of Whitney (H. Whitney, Congruent graphs and the connectivity of graphs, *Amer. J. Math.* **54** (1932), 150–168): if two drawings in a sphere are subdivisions of the same simple 3-connected graph, then some homeomorphism of the sphere maps the first onto the second, each branch vertex to the corresponding branch vertex, and each subdivided edge onto the corresponding subdivided edge. For a wall $W$ of height $h'$, the *side vertices* of $W$ are its branch vertices that correspond to the vertices $(x, y)$ of the elementary wall with $y \in \{0, h'\}$ or $x \in \{0, 2h'+1\}$.

(3) *Let $W$ be a wall of height $h' \geq 2$ drawn in the plane, and suppose that for each side vertex $w$ there is a half-line starting at $w$ that meets the drawing of $W$ only in $w$, such that these half-lines are mutually disjoint.*

*Then some homeomorphism of the plane maps the drawing of the elementary wall of height $h'$ as in Fig. 1 onto the drawing of $W$, each vertex to the corresponding branch vertex, and each edge onto the corresponding path.*

Let $W^+$ be the graph obtained from the elementary wall of height $h'$ by adding a vertex $z$ adjacent to all side vertices. Add a point $\infty$ to the plane, to obtain a sphere. With $z$ drawn at $\infty$ and its edges drawn along the given half-lines, the drawing of $W$ becomes a drawing in the sphere of a subdivision of $W^+$. The drawing of Fig. 1 gives another one, with half-lines going down from the vertices in row 0, up from the vertices in row $h'$, left from the vertices $(0, y)$ and right from the vertices $(2h' + 1, y)$ with $0 < y < h'$. These half-lines meet the drawing only in their starts, and they are mutually disjoint. We show below that $W^+$ is 3-connected. It is simple, and so by Whitney's theorem some homeomorphism of the sphere maps the second drawing onto the first, with each branch vertex and each subdivided edge going to the corresponding one. It maps $\infty$ to $\infty$, since $\infty$ is the point of $z$ in both drawings, and so it restricts to the required homeomorphism of the plane.

It remains to show that $W^+ - X$ is connected for every set $X$ of at most two vertices of $W^+$. If $z \in X$, then $W^+ - X$ is the elementary wall minus at most one vertex, which is connected since the elementary wall is 2-connected. (It is the union of the circuits around its bricks, and these can be ordered so that each shares an edge with an earlier one, row by row from the bottom and from left to right in each row. Adding a circuit that shares at least two vertices with a 2-connected graph gives a 2-connected graph.) If $z \notin X$, it is enough to show that every vertex $w \notin X$ of the elementary wall is joined by a path of $W^+ - X$ to a side vertex, since the side vertices are adjacent to $z$. If $w$ is a side vertex this holds trivially. Otherwise $w = (x, y)$ with $0 < x < 2h' + 1$ and $0 < y < h'$, and three paths join $w$ to side vertices and meet only in $w$: the path along row $y$ to $(0, y)$, the path along row $y$ to $(2h' + 1, y)$, and a staircase. If $x + y$ is odd the staircase is $(x, y), (x, y+1), (x+1, y+1), (x+1, y+2), \ldots$, and if $x+y$ is even it is $(x, y), (x, y - 1), (x - 1, y - 1), (x - 1, y - 2), \ldots$, in both cases stopped at the first side vertex. These are paths of the wall, since each vertical step starts at a vertex with odd coordinate sum (going up) or ends at one (going down), and before the first side vertex all the vertices passed are inside the wall. The staircase leaves row $y$ at its first step. At least one of the three paths avoids $X$, since $|X| \leq 2$ and $w \notin X$. This proves (3).

If (3) applies to a wall $H' \subseteq H$ of height $h'$, we name the regions of $H'$ by the regions of the elementary wall of height $h'$ drawn as in Fig. 1, using the homeomorphism. Thus the regions of $H'$ are named by the bricks of height

$h'$ and by the infinite region. The region named by the brick $B'$ is the inside of the circuit $C(B')$ of $H'$ that corresponds to the circuit around $B'$, and the infinite region of $H'$ is named by the infinite region. A homeomorphism preserves touching, and so (2) for height $h'$ gives $D_{H'}$ and the depths in $H'$. Moreover every point of the drawing of $H'$ is at two different regions of $H'$, since this is true for the drawing of Fig. 1. Each region $F$ of $H$ is connected and disjoint from the drawing of $H'$, and so it lies in a region $\pi(F)$ of $H'$. In particular $\pi(R_\infty)$ is the infinite region of $H'$, since $R_\infty$ is unbounded.

All our walls $H' \subseteq H$ of height $h'$ have the following form. There are integers $R(0) < R(1) < \cdots < R(h')$ such that the branch vertex of $H'$ corresponding to $(x, y)$ is a vertex of $H$ in row $R(y)$, whose column increases with $x$. Each horizontal edge of $H'$ is the path of $H$ along a row between its ends. The vertical edges of $H'$ are mutually disjoint paths of $H$. The vertical edge of $H'$ at $(x, y)$ meets rows $R(y)$ and $R(y + 1)$ of $H$ only in its ends, and it contains exactly one vertical edge of $H$ in each strip $t$ with $R(y) \leq t < R(y + 1)$, and no other vertical edges of $H$. We write $c_t(x)$ for the column of this vertical edge of $H$ in strip $t$. Since the rows $R(y)$ are distinct, the columns of the branch vertices increase along each row, and each vertical edge of $H'$ lies between the heights of the two rows it joins and meets these rows only in its ends, these paths of $H$ meet only in common ends, and no branch vertex is an inner vertex of one of them. So $H'$ is a wall of height $h'$. Moreover (3) applies to $H'$, with half-lines going down from its branch vertices in row 0, up from those in row $h'$, left from the first and right from the last branch vertex of every other row. For the drawing of $H'$ lies between the horizontal lines at heights $R(0)$ and $R(h')$, and its points at height $R(y)$ are the points of the path of $H$ along row $R(y)$ between the first and the last branch vertex of row $y$.

Finally, in each of our walls there are an integer $c_0$, and for every strip $t$ integers $y_t$ and $a_t \leq a'_t$ with $a_t \equiv a'_t \equiv t - y_t \pmod 2$, such that the following holds. If $t < R(0)$ then $y_t < 0$, and if $t \geq R(h')$ then $y_t \geq h'$. If $R(0) \leq t < R(h')$, then $R(y_t) \leq t < R(y_t+1)$, and $c_t(x) = x+a_t$ if $x \leq c_0-1$ and $c_t(x) = x + a'_t$ if $x \geq c_0$, for every $x$ such that $H'$ has a vertical edge at $(x, y_t)$.

(4) *Let $H'$ be a wall of this form, and let $F = (X, Y)$ be a brick of $H$. Let $z = X - a_Y$ if $X - a_Y \leq c_0 - 1$, let $z = X - a'_Y$ if $X - a'_Y \geq c_0$, and let $z = c_0 - 1$ otherwise. Let $P(F)$ be the position of the square $(z, y_Y)$. Then $\pi(F)$ is the region of $H'$ named by $P(F)$ if $P(F)$ is a brick of height $h'$, and the infinite region of $H'$ otherwise. Moreover $F - P(F) = (\Delta X, Y - y_Y)$, where $\Delta X = a_Y$ in the first case, $\Delta X = a'_Y$ in the second case, and $\Delta X \in \{X - c_0, X - c_0 + 1\}$ in the third case. In all cases $a_Y \leq \Delta X \leq a'_Y$.*

The first two cases cannot both hold, since $a_Y \le a'_Y$. Let $s$ be the square $(X, Y)$, so that $s \subseteq F$ and $\pi(F)$ is the region of $H'$ that contains $s$. Let $y = y_Y$, and let $x$ be the integer with $x + y$ odd and $x + 1 \in \{z, z + 1\}$. Thus $P(F) = (x + 1, y)$. We first prove the last two statements. Since $X + Y$ and $x + 1 + y$ are even, $\Delta X = X - x - 1 \equiv Y - y \equiv a_Y \equiv a'_Y \pmod 2$. In the first case $X - z = a_Y$ and $\Delta X \in \{a_Y, a_Y - 1\}$, so $\Delta X = a_Y$ by parity. In the same way $\Delta X = a'_Y$ in the second case. In the third case $\Delta X \in \{X - c_0, X - c_0 + 1\}$ since $z = c_0 - 1$, and $X - a_Y \ge c_0$ and $X - a'_Y \le c_0 - 1$, and so $a_Y \le \Delta X \le a'_Y$.

If $Y < R(0)$ or $Y \ge R(h')$, then the horizontal line through the centre of $s$ is disjoint from the drawing of $H'$. This line and $s$ form an unbounded connected set disjoint from the drawing of $H'$, and so $s$ lies in the infinite region of $H'$. Moreover $y < 0$ or $y \ge h'$, and so $P(F)$ is not a brick of height $h'$.

Now let $R(0) \le Y < R(h')$. Then $R(y) \le Y < R(y + 1)$, and so $0 \le y \le h' - 1$. Let $\ell$ be the half-line going left from the centre $(X + \frac{1}{2}, Y + \frac{1}{2})$ of $s$. For a brick $B' = (x' + 1, y')$ of height $h'$, the circuit $C(B')$ consists of the vertical edges of $H'$ at $(x', y')$ and $(x' + 2, y')$ and of parts of rows $R(y')$ and $R(y' + 1)$ of $H$. The half-line $\ell$ meets $C(B')$ only in vertical edges of $H$ in strip $Y$, and it crosses each of them whose column is at most $X$. If $y' \ne y$, then $C(B')$ contains no vertical edge of $H$ in strip $Y$. If $y' = y$, then the vertical edges of $H$ in strip $Y$ in $C(B')$ are in columns $c_Y(x') < c_Y(x' + 2)$. A point not on a closed polygon is inside it if and only if a half-line from the point that meets the polygon only in crossings of its segments crosses it an odd number of times. Hence $s$ lies in the region named by $(x' + 1, y)$ if $c_Y(x') \le X < c_Y(x' + 2)$, and if there is no such $x'$, then $s$ lies in the infinite region of $H'$.

For every integer $w$ let $g(w) = w + a_Y$ if $w \le c_0 - 1$, and $g(w) = w + a'_Y$ if $w \ge c_0$. Then $g$ is strictly increasing, since $a_Y \le a'_Y$, and $c_Y(w) = g(w)$ whenever $H'$ has a vertical edge at $(w, y)$, that is, when $0 \le w \le 2h' + 1$ and $w + y$ is odd. We claim that $g(x) \le X < g(x + 2)$. In the first case $x \le z \le c_0 - 1$, so $g(x) = x + a_Y \le z + a_Y = X$, and $g(x+2) \ge x+2+a_Y > z+a_Y = X$. In the second case $x + 2 > z \ge c_0$, so $g(x + 2) = x + 2 + a'_Y > z + a'_Y = X$, and $g(x) \le x + a'_Y \le z + a'_Y = X$. In the third case $c_0 - 2 \le x \le c_0 - 1$, so $g(x) = x + a_Y < c_0 + a_Y \le X$ and $g(x + 2) = x + 2 + a'_Y \ge c_0 + a'_Y > X$. This proves the claim.

If $x \ge 0$ and $x + 2 \le 2h' + 1$, then $c_Y(x) = g(x)$ and $c_Y(x + 2) = g(x + 2)$, since $x + y$ is odd. So $s$ lies in the region named by $(x + 1, y) = P(F)$, which is a brick of height $h'$. If $x < 0$, then every $w$ such that $H'$ has a vertical edge at $(w, y)$ satisfies $w \ge x + 2$, since $w \ge 0$ and $w \equiv x \pmod 2$. So

$c_Y(w) = g(w) \geq g(x+2) > X$, and $s$ lies in the infinite region of $H'$. In this case $x+1 \leq 0$, and $P(F)$ is not a brick of height $h'$. If $x+2 > 2h'+1$, then every such $w$ satisfies $w \leq x$, since $w \leq 2h'+1 \leq x+1$ and $w \equiv x \pmod 2$. So $c_Y(w) = g(w) \leq g(x) \leq X$, and again $s$ lies in the infinite region. In this case $x+1 \geq 2h'+1$, and $P(F)$ is not a brick of height $h'$. This proves (4).

For a brick $F$ of $H$ we call $F - P(F)$ the *move* of $F$. By (4), for each brick $F$ the depth of $\pi(F)$ is at least $\sigma_{h'}(P(F))$. For if $P(F)$ is a brick of height $h'$, then $\pi(F)$ is named by it and (2) for height $h'$ applies, and otherwise $\sigma_{h'}(P(F)) \leq 0$.

(5) *Let $F$ be a brick of $H$ whose move $(\Delta X, \Delta Y)$ satisfies $0 \leq \Delta Y \leq h - h'$ and $0 \leq \Delta X \leq 2(h-h')$. Then the bottom, top, left and right terms of $F$ for height $h$ exceed those of $P(F)$ for height $h'$ by at most $\Delta Y$, $h - h' - \Delta Y$, $\lceil \Delta X/2 \rceil$ and $\lceil (2(h-h') - \Delta X)/2 \rceil$, respectively.*

Let $F = (X, Y)$, so $P(F) = (X - \Delta X, Y - \Delta Y)$. The bottom terms are $Y+1$ and $Y - \Delta Y + 1$, and the top terms are $h - Y$ and $h' - Y + \Delta Y$, which gives the first two bounds. The left terms are $\lfloor A/2 \rfloor$ and $\lfloor B/2 \rfloor$ with $A = X+1$ and $B = A - \Delta X$, and the right terms are $\lfloor A/2 \rfloor$ and $\lfloor B/2 \rfloor$ with $A = 2h+2-X$ and $B = 2h'+2-X+\Delta X = A - (2(h-h') - \Delta X)$. In both cases $A \geq B$, and the first simple fact gives the last two bounds. This proves (5).

We call the four bounds in (5) the *drops* of the four terms. The following is the test mentioned above.

(6) *Let $H' \subseteq H$ be a wall of height $h'$ to which (3) applies, and let $P$ map the bricks of $H$ to positions, such that $\pi(F)$ is the region named by $P(F)$ if $P(F)$ is a brick of height $h'$, and the infinite region otherwise. Let $c \geq 1$ be an integer such that*

(a) *$\|F - G\| \leq \|P(F) - P(G)\| + 2c - 2$ for all bricks $F, G$ of $H$,*

(b) *$\sigma_h(F) \leq \sigma_{h'}(P(F)) + c$ for every brick $F$ of $H$, and*

(c) *every element deleted by $H'$ is at a region $F$ of $H$ whose depth is at most the depth of $\pi(F)$ plus $c-1$.*

*Then $d' \geq d - 2c$.*

As above, the depth of $\pi(F)$ is at least $\sigma_{h'}(P(F))$ for every brick $F$. So by (b) every region $F$ of $H$ has depth at most the depth of $\pi(F)$ plus $c$. For the infinite region this holds since both depths are 0. Let $s \neq t$ be points (for $s = t$ both distances are 0), write $[s]'$ for $[s]$ taken in $H'$, and choose regions $F', G'$ of $H'$ at $s, t$ such that $d'(s,t) = [s]' + [t]' + D_{H'}(F', G')$, by (1) for $H'$. Since $F'$ is open and the drawing of $H$ is a finite union of segments, $F'$ contains points arbitrarily close to $s$ that are not in the drawing of $H$. There are finitely many regions of $H$, and so one region $F$ of $H$ contains such points arbitrarily close to $s$. Then $F$ is at $s$, and $F \subseteq F'$ since $F$ is

connected and disjoint from the drawing of $H'$. If $s$ is a point of $H$ but not of $H'$, then a small disc around $s$ meets no point of $H'$, since the drawing of $H'$ is closed. This disc lies in $F'$, and so all the regions of $H$ at $s$ lie in $F'$. In this case $s$ lies in an element deleted by $H'$, and by (c) we may choose $F$ with depth at most that of $F'$ plus $c-1$. In every case $[s]-[s]' \leq 1$, and $[s]-[s]'$ plus the depth of $F$ is at most the depth of $F'$ plus $c$. If $s$ is a point of $H$ but not of $H'$ this is the choice just made, and otherwise $[s]=[s]'$ and it follows from the second sentence of this proof. Choose $G$ in the same way for $t$. By (1) for $H$, $d(s,t) \leq [s]+[t]+D_H(F,G)$. By (2) for $H'$, either $D_{H'}(F',G')$ is the sum of the depths of $F'$ and $G'$, or $F',G'$ are named by bricks $B_1, B_2$ of height $h'$ and $D_{H'}(F',G') = \|B_1 - B_2\|$. In the first case,

$$\begin{aligned} d(s,t) &\leq [s]+[t]+\mathrm{depth}(F)+\mathrm{depth}(G) \\ &\leq [s]'+[t]'+\mathrm{depth}(F')+\mathrm{depth}(G')+2c = d'(s,t)+2c. \end{aligned}$$

In the second case $F, G$ are bricks, since $\pi$ maps the infinite region of $H$ to that of $H'$. Moreover $\pi(F)=F'$ and $\pi(G)=G'$, so $P(F)=B_1$ and $P(G)=B_2$. By (2) for $H$ and by (a),

$$\begin{aligned} d(s,t) &\leq [s]+[t]+\|F-G\| \\ &\leq ([s]'+1)+([t]'+1)+\|B_1-B_2\|+2c-2 = d'(s,t)+2c. \end{aligned}$$

This proves (6).

By (4), the map $P$ defined there satisfies the hypothesis of (6) for every wall of our form. We check (a), (b), (c) as follows. If all the moves of bricks of $H$ satisfy the hypothesis of (5), and all the drops are at most $c$, then (b) holds, by the second simple fact. For (c) and an element $m$ deleted by $H'$, it is enough to find a square at $m$ whose position $F$ is not a brick of height $h$, or satisfies $\sigma_h(F) \leq c-1$, or satisfies $\sigma_h(F) \leq \sigma_{h'}(P(F))+c-1$. This is because the region of $H$ containing the square lies at $m$. It is the infinite region if $F$ is not a brick of height $h$, and otherwise it is the brick $F$, of depth $\sigma_h(F)$, while $\pi(F)$ has depth at least $\max(\sigma_{h'}(P(F)),0)$. For the last inequality it is enough that each term of $P(F)$, plus $c-1$, is at least some term of $F$.

For (ii) we use (1) as follows. Let $m_v$ be the element of $H$ that contains $v$. The regions of $H$ at $v$ are the regions at $m_v$, and similarly for every point of the drawing. Let $u$ be a vertex of $H'$ with $u \neq v$, such that every region of $H$ at $u$ has distance $D_H$ at least $k-2$ from every region of $H$ at $v$. Then $d(u,v) \geq 1+1+(k-2) = k$ by (1). If $H'$ deletes $m_v$, then $u \neq v$ for every vertex $u$ of $H'$, since $u$ lies in an element kept by $H'$.

We use the rotation $\rho$ of the plane given by $\rho(x,y) = (2h+1-x, h-y)$. Since $h$ is even, $\rho$ maps each element of $H$ onto an element: the vertex $(x,y)$ to the vertex $(2h+1-x, h-y)$, the horizontal edge at $(x,y)$ to the horizontal edge at $(2h-x, h-y)$, and the vertical edge at $(x,y)$ to the vertical edge at $(2h+1-x, h-1-y)$, which exists because $(2h+1-x)+(h-1-y) = 3h-x-y$ is odd. It maps the square $(x,y)$ to the square $(2h-x, h-1-y)$, and the brick $(X,Y)$ to the brick $(2h+1-X, h-1-Y)$. We let $\rho$ act on all positions by this formula. It exchanges the bottom and top terms of positions, and the left and right terms, and so it preserves $\sigma_h$, and also $\|F-G\|$ for positions $F, G$. Since $\rho$ maps the drawing of $H$ onto itself, it maps regions of $H$ to regions of $H$ and preserves touching and depths. Let $\rho'$ be defined in the same way with $h'$ in place of $h$. Let $H' \subseteq H$ and $P$ be as in (6). Then the subgraph $\rho(H')$ of $H$, whose drawing is the image under $\rho$ of the drawing of $H'$, is a wall of height $h'$ whose branch vertex corresponding to $(x,y)$ is the image under $\rho$ of the branch vertex of $H'$ corresponding to $\rho'(x,y)$. Composing the homeomorphism of (3) for $H'$ with $\rho$ and $\rho'$ shows that (3) applies to $\rho(H')$, and that the map $F \mapsto \rho'(P(\rho(F)))$ has the property of $P$ in (6) for $\rho(H')$. Moreover $\rho(H')$ deletes an element $m$ if and only if $H'$ deletes $\rho(m)$. Hence if $H'$ satisfies (a), (b), (c) of (6) for some $c$, then so does $\rho(H')$. We call $\rho(H')$ the *turned* $H'$.

We come to the subwalls. Let $h' = h - 2k$. Let $1 \le i \le 4k-1$ and $1 \le j \le 2k-1$ with $i+j$ even. The *moved wall* $T(i,j)$ is the subgraph of $H$ whose drawing is the drawing of the elementary wall of height $h'$, as in Fig. 1, translated by $(i,j)$. Each vertex $(x,y)$ of the elementary wall of height $h'$ is translated to the vertex $(x+i, y+j)$ of $H$, since $1 \le x+i \le 2h'+1+4k-1 = 2h$ and $1 \le y+j \le h'+2k-1 = h-1$. Each vertical edge is translated to a vertical edge of $H$, since $i+j$ is even. So $T(i,j)$ is a wall of our form, with $R(y) = y+j$ and $c_t(x) = x+i$, and we take $c_0 = 0$ and $a_t = a'_t = i$ and $y_t = t-j$ for every strip $t$. By (4), $P(F) = F - (i,j)$ for every brick $F$ of $H$. The wall $T(i,j)$ keeps exactly the elements whose midpoint minus $(i,j)$ is the midpoint of an element of the elementary wall of height $h'$. In particular, the midpoint $(p,q)$ of every element kept by $T(i,j)$ satisfies $i \le p \le i+2h'+1$ and $j \le q \le j+h'$.

(7) *The moved wall $T(i,j)$ satisfies (a), (b), (c) of (6) with $c = 2k$, and so $d' \ge d - 4k$ for $H' = T(i,j)$.*

Since all moves equal $(i,j)$, we have $\|F-G\| = \|P(F)-P(G)\|$, which gives (a). By (5) with $h-h' = 2k$, the drops are $j$, $2k-j$, $\lceil i/2 \rceil$ and $\lceil (4k-i)/2 \rceil$, all at most $2k$. This gives (b). For (c), let $m$ be an element deleted by $T(i,j)$, with midpoint $(p,q)$. Let the *infinite wall* have all integer points as vertices, with horizontal and vertical edges defined as for the elementary

wall. Translating by $-(i,j)$ maps $m$ to a vertex or edge of the infinite wall, since $i+j$ is even. A vertex or edge of the infinite wall whose midpoint $(p'',q'')$ satisfies $0 \le p'' \le 2h'+1$ and $0 < q'' < h'$ is an element of the elementary wall of height $h'$, since the vertices and edges of the infinite wall in this box that are missing from the elementary wall have $q'' \in \{0,h'\}$. Hence $q \le j$, or $q \ge j+h'$, or $p \le i-\frac{1}{2}$, or $p \ge i+2h'+\frac{3}{2}$, since $p$ is a multiple of $\frac{1}{2}$. In the first case choose a square at $m$ whose position $F=(X,Y)$ has $Y \le q-\frac{1}{2}$, so that $Y \le j-1$. The bottom term of $F$ is $Y+1 \le j \le 2k-1$, and the second coordinate of $P(F)$ is $Y-j<0$. In the second case choose a square with $Y \ge q-\frac{1}{2}$, so that $Y \ge j+h'$. The top term of $F$ is $h-Y \le 2k-j \le 2k-1$, and the second coordinate of $P(F)$ is at least $h'$. In the third case choose a square with $X \le p-\frac{1}{2} \le i-1$. The left term of $F$ is at most $\lfloor i/2 \rfloor \le 2k-1$, and the first coordinate of $P(F)$ is negative. In the fourth case choose a square with $X \ge p+\frac{1}{2} \ge i+2h'+2$. The right term of $F$ is at most $\lfloor (2h-2h'-i)/2 \rfloor = \lfloor (4k-i)/2 \rfloor \le 2k-1$, and the first coordinate of $P(F)$ is at least $2h'+2$. In each case $\sigma_h(F) \le 2k-1$, which gives (c). By (6), $d' \ge d-4k$. This proves (7).

Since the drawing of the elementary wall of height $h'$ is mapped onto itself by $\rho'$, the rotation $\rho$ maps the drawing of $T(i,j)$ onto the drawing of $T(4k-i,2k-j)$. Thus $\rho(T(i,j)) = T(4k-i,2k-j)$.

Next, let $k=1$, so that $h'=h-2$. We use a variant of the construction of Fig. 2, turned upside down. Let $Y_0, c_0$ be integers with $1 \le Y_0 \le h'$ and $1 \le c_0 \le 2h'+1$. The wall $S(Y_0,c_0)$ is the subgraph of $H$ defined as follows. Its branch vertex corresponding to $(x,y)$ is the vertex $(x,y)$ of $H$ if $y \le Y_0-1$ and $x \le c_0-1$, the vertex $(x+4,y)$ if $y \le Y_0-1$ and $x \ge c_0$, and the vertex $(x+2,y+2)$ if $y \ge Y_0$. Its horizontal edges are the paths of $H$ along rows between their ends. Its vertical edge at $(x,y)$ is the vertical edge of $H$ at the vertex of $H$ corresponding to $(x,y)$, if $y \ne Y_0-1$. If $y=Y_0-1$, then $x \equiv Y_0 \pmod 2$, and the vertical edge at $(x,y)$ is a staircase made of five edges of the elementary wall. If $x \le c_0-1$, it goes from $(x,Y_0-1)$ up to $(x,Y_0)$, right to $(x+1,Y_0)$, up to $(x+1,Y_0+1)$, right to $(x+2,Y_0+1)$, and up to $(x+2,Y_0+2)$. If $x \ge c_0$, it goes from $(x+4,Y_0-1)$ up to $(x+4,Y_0)$, left to $(x+3,Y_0)$, up to $(x+3,Y_0+1)$, left to $(x+2,Y_0+1)$, and up to $(x+2,Y_0+2)$. These vertical steps are edges, since they start at vertices with odd coordinate sum, as $x \equiv Y_0$.

(8) *The wall $S(Y_0,c_0)$ is a wall of height $h'$ of our form, and every move is one of $(0,0)$, $(2,0)$, $(4,0)$, $(1,1)$, $(3,1)$, $(2,2)$. It deletes the vertical edges at $(x,y)$ with $y \le Y_0-1$ and $c_0 \le x \le c_0+3$. If $c_0 \equiv Y_0 \pmod 2$, it also deletes the vertices $(c_0,Y_0)$, $(c_0+1,Y_0)$, $(c_0+2,Y_0)$ and $(c_0+1,Y_0+1)$, the horizontal edges at $(c_0-1,Y_0)$, $(c_0,Y_0)$, $(c_0,Y_0+1)$ and $(c_0+1,Y_0+1)$, and*

*the vertical edge at* $(c_0+1, Y_0)$*. Every element* $m$ *it deletes, with midpoint* $(p,q)$*, is of one of the following kinds:*

(i) *a vertical edge at* $(x,y)$ *with* $y \le Y_0 - 1$ *and* $c_0 \le x \le c_0+3$*,*

(ii) *an element with* $Y_0 \le q \le Y_0+1$*,*

(iii) *an element with* $p < 2$*, or* $p > 2h-1$*, or* $q = h$*,*

(iv) *if* $c_0 = 1$*, a vertex* $(x,0)$ *or a horizontal edge at* $(x,0)$ *with* $1 \le x \le 4$*.*

The branch vertices are vertices of $H$, since $x+4 \le 2h'+5 = 2h+1$ and $y+2 \le h'+2 = h$, and the staircases use only the rows $Y_0, Y_0+1 \le h-1$. We check the form with $R(y) = y$ for $y \le Y_0 - 1$ and $R(y) = y+2$ for $y \ge Y_0$. The columns of the branch vertices increase along each row. Each staircase meets rows $Y_0 - 1$ and $Y_0 + 2$ only in its ends and has one vertical edge of $H$ in each of the strips $Y_0 - 1$, $Y_0$, $Y_0 + 1$. In rows $Y_0$ and $Y_0 + 1$ the staircase for $x \le c_0 - 1$ uses the columns $x, x+1$ and $x+1, x+2$, and the staircase for $x \ge c_0$ uses the columns $x+3, x+4$ and $x+2, x+3$. Since staircases on the same side differ by 2 in $x$, and since the columns used on the left are at most $c_0 + 1$ and those on the right at least $c_0 + 2$, the staircases are mutually disjoint. So $S(Y_0, c_0)$ has our form. We take $c_0$ as given, and for every strip $t$ we take $y_t = t$ and $(a_t, a'_t) = (0,4)$ if $t \le Y_0 - 1$, $y_t = Y_0 - 1$ and $(a_t, a'_t) = (1,3)$ if $t = Y_0$, $y_t = Y_0 - 1$ and $(a_t, a'_t) = (2,2)$ if $t = Y_0+1$, and $y_t = t-2$ and $(a_t, a'_t) = (2,2)$ if $t \ge Y_0+2$. These satisfy the conditions of our form, by the definition of the vertical edges, and $R(0) = 0$ and $R(h') = h$. By (4), the moves in strips $t \le Y_0 - 1$ are $(\Delta X, 0)$ with $0 \le \Delta X \le 4$ and $\Delta X$ even, those in strip $Y_0$ are $(1,1)$ and $(3,1)$, and those in higher strips are $(2,2)$.

By the form, the vertical edges of $H$ that $S(Y_0, c_0)$ keeps in strips $t \le Y_0 - 1$ are in columns $c_t(x) = x \le c_0 - 1$ or $c_t(x) = x+4 \ge c_0 + 4$, which shows that the elements of kind (i) are deleted. Now let $c_0 \equiv Y_0 \pmod 2$. The vertices and edges of $H$ in rows $Y_0$ and $Y_0 + 1$ and in strip $Y_0$ that $S(Y_0, c_0)$ keeps are those of the staircases, since the other rows and vertical edges of $S(Y_0, c_0)$ lie in rows $y \le Y_0 - 1$ or $y \ge Y_0 + 2$ and in strips other than $Y_0$. The staircases are at the points $(x, Y_0 - 1)$ with $x \equiv Y_0 \pmod 2$, so that $x \equiv c_0 \pmod 2$, and hence $x \le c_0 - 2$ or $x \ge c_0$. The staircase for $x \le c_0 - 2$ meets these rows and this strip in the vertices $(x, Y_0)$, $(x+1, Y_0)$, $(x+1, Y_0+1)$, $(x+2, Y_0+1)$, the horizontal edges at $(x, Y_0)$ and $(x+1, Y_0+1)$, and the vertical edge at $(x+1, Y_0)$. The staircase for $x \ge c_0$ meets them in the vertices $(x+4, Y_0)$, $(x+3, Y_0)$, $(x+3, Y_0+1)$, $(x+2, Y_0+1)$, the horizontal edges at $(x+3, Y_0)$ and $(x+2, Y_0+1)$, and the vertical edge at $(x+3, Y_0)$. Hence a vertex $(x', Y_0)$ or a vertical edge at $(x', Y_0)$ that is kept has $x' \le c_0 - 1$ or $x' \ge c_0 + 3$, a horizontal edge at $(x', Y_0)$ that is kept has

$x' \leq c_0 - 2$ or $x' \geq c_0 + 3$, a vertex $(x', Y_0 + 1)$ that is kept has $x' \leq c_0$ or $x' \geq c_0 + 2$, and a horizontal edge at $(x', Y_0 + 1)$ that is kept has $x' \leq c_0 - 1$ or $x' \geq c_0 + 2$. This shows that the other listed elements are deleted.

Now let $m$ be deleted and not of kinds (i) and (ii). Suppose first that $q \leq Y_0 - \frac{1}{2}$. Then $m$ is a vertex or a horizontal edge in a row $y \leq Y_0 - 1$, since the vertical edges of $H$ in strips $t \leq Y_0 - 1$ in other columns than $c_0, \ldots, c_0 + 3$ are kept. The path of row $y$ of $S(Y_0, c_0)$ runs along row $y$ of $H$ from column 0, or from column 1 if $y = 0$ and $c_0 \geq 2$, or from column 5 if $y = 0$ and $c_0 = 1$, to column $2h' + 5 = 2h + 1$. So it is all of row $y$ of $H$ unless $y = 0$ and $c_0 = 1$, and $m$ is of kind (iv). Suppose next that $q \geq Y_0 + \frac{3}{2}$. The part of the drawing of $S(Y_0, c_0)$ above height $Y_0 + 1$ is the part of the drawing of the elementary wall of height $h'$ above height $Y_0 - 1$, translated by $(2, 2)$. Indeed, the rows $y \geq Y_0$ and the vertical edges at $(x, y)$ with $y \geq Y_0$ are translated by $(2, 2)$, and so is the last step of each staircase. So $m$ translated by $-(2, 2)$ is a vertex or edge of the infinite wall which is not an element of the elementary wall of height $h'$, while $0 < q - 2 \leq h'$. As in the proof of (7), it follows that $p - 2 < 0$, or $p - 2 > 2h' + 1$, or $q - 2 = h'$, and so $m$ is of kind (iii). This proves (8).

We call $S(Y_0, c_0)$ *good* if $c_0 \geq 2Y_0$ when $Y_0$ is odd, and $c_0 + 2Y_0 \leq 2h' + 2$ when $Y_0$ is even.

(9) *If $S(Y_0, c_0)$ is good, then it satisfies (a), (b), (c) of (6) with $c = 2$.*

Each of the six moves $\mu$ of (8) satisfies $|\mu_1 - 2| + |\mu_2| \leq 2$, and so any two moves $\mu, \mu'$ satisfy $|\mu_1 - \mu'_1| + |\mu_2 - \mu'_2| \leq 4$ and $|\mu_2 - \mu'_2| \leq 2$. Hence $\|\mu - \mu'\| \leq 2$, and so $\|F - G\| \leq \|P(F) - P(G)\| + 2$ for all bricks $F, G$, since $F - G = (P(F) - P(G)) + (\mu - \mu')$ where $\mu, \mu'$ are the moves of $F, G$. This gives (a). By (5) with $h - h' = 2$, all the drops are at most 2, since $0 \leq \Delta Y \leq 2$ and $0 \leq \Delta X \leq 4$. This gives (b).

For (c) let $m$ be a deleted element, of one of the kinds of (8). Let $(x, y)$ be such that $m$ is the vertex $(x, y)$ or an edge at $(x, y)$. If $m$ is of kind (iii) or (iv), let $F = (X, Y)$ be the position of the square $(x, y)$, which is at $m$. If $p < 2$ then $x \leq 1$ and $X \leq 2$, so the left term of $F$ is at most 1. If $p > 2h - 1$ then $x \geq 2h - 1$ and $X \geq 2h - 1$, so the right term of $F$ is at most 1. If $q = h$, then $Y = h$ and $F$ is not a brick of height $h$. If $m$ is of kind (iv), the bottom term of $F$ is 1. So in these cases $F$ is as required.

Let $m$ be of kind (ii). Then $m$ has a square in strip $Y_0$ at it: the square $(x, Y_0)$ if $y = Y_0$, and the square $(x, Y_0)$ below $m$ if $m$ is a vertex or horizontal edge in row $Y_0 + 1$. Let $F = (X, Y_0)$ be its position. If $F$ is not a brick of height $h$ we are done, so we may assume that it is. Its move is $(1, 1)$ or $(3, 1)$. By (4), the move is $(1, 1)$ if $X \leq c_0$, it is $(3, 1)$ if $X \geq c_0 + 3$, and otherwise it is $(X - c_0, 1)$ or $(X - c_0 + 1, 1)$. So if the move is $(1, 1)$ then

$X \le c_0 + 1$, and if it is $(3,1)$ then $X \ge c_0 + 2$. By (5), the bottom and top terms drop by 1. Suppose that $Y_0$ is odd, so that $X$ is odd. If the move is $(1,1)$, the left term drops by $\lceil 1/2 \rceil = 1$, and the right term drops from $(2h+1-X)/2$ to $\lfloor (2h-1-X)/2 \rfloor = (2h-1-X)/2$, that is, by 1. If the move is $(3,1)$, the right term drops by $\lceil 1/2 \rceil = 1$. The left term of $P(F)$ is $\lfloor (X-2)/2 \rfloor = (X-3)/2$, and since $X$ is odd and $X \ge c_0 + 2 \ge 2Y_0 + 2$ because $S(Y_0, c_0)$ is good, we have $X \ge 2Y_0 + 3$. So this left term plus 1 is at least $Y_0 + 1$, the bottom term of $F$. Now suppose that $Y_0$ is even, so that $X$ is even. If the move is $(3,1)$, the left term drops from $X/2$ to $\lfloor (X-2)/2 \rfloor = X/2 - 1$, and the right term by 1. If the move is $(1,1)$, the left term drops by 1. The right term of $P(F)$ is $\lfloor (2h'+3-X)/2 \rfloor$, and since $X \le c_0 + 1 \le 2h' + 3 - 2Y_0$ because $S(Y_0, c_0)$ is good, it is at least $Y_0$. So this right term plus 1 is at least $Y_0 + 1$, the bottom term of $F$. In all cases each term of $P(F)$ plus 1 is at least some term of $F$, as required.

Let $m$ be the vertical edge at $(x,y)$ of kind (i), so that $y \le Y_0 - 1$ and $c_0 \le x \le c_0 + 3$. Suppose first that $Y_0$ is odd, and let $F = (X,y)$ be the position of the square $(x,y)$, so that $X = x+1$ and $c_0 + 1 \le X \le c_0 + 4$. In the definition of $P(F)$ in (4), with $a_y = 0$ and $a'_y = 4$, the first case does not occur. So $z = c_0 - 1$ if $X \le c_0 + 3$, and $z = c_0$ if $X = c_0 + 4$. In both cases $P(F) = (X', y)$ with $X' \in \{c_0 - 1, c_0\}$, and the move is $(X - X', 0)$ with $X - X' \in \{2, 4\}$, since it is positive and even. By (5), the bottom term does not drop, and the right term drops by at most $\lceil (4-2)/2 \rceil = 1$. The left term of $P(F)$ is $\lfloor (X'+1)/2 \rfloor \ge \lfloor c_0/2 \rfloor \ge Y_0 \ge y+1$, using $c_0 \ge 2Y_0$. So it is at least the bottom term of $P(F)$, which equals that of $F$. The right term of $P(F)$ is $\lfloor (2h'+2-X')/2 \rfloor \le \lfloor (2h'+3-2Y_0)/2 \rfloor = h' + 1 - Y_0 \le h' - y$, which is the top term of $P(F)$. So the top term of $P(F)$ plus 1 is at least its right term plus 1, which is at least the right term of $F$. Now suppose that $Y_0$ is even, and let $F = (X,y)$ be the position of the square $(x-1, y)$, so that $X = x - 1$ and $c_0 - 1 \le X \le c_0 + 2$. If $X = c_0 - 1$, we are in the first case of (4) and $P(F) = F$. Otherwise we are in the third case, $P(F) = (X', y)$ with $X' \in \{c_0 - 1, c_0\}$, and the move $(X - X', 0)$ satisfies $X - X' \in \{0, 2\}$. So the bottom term does not drop, and the left term drops by at most 1. The left term of $P(F)$ is $\lfloor (X'+1)/2 \rfloor \le \lfloor (c_0+1)/2 \rfloor \le h' + 1 - Y_0 \le h' - y$, using $c_0 \le 2h' + 2 - 2Y_0$, and this is the top term of $P(F)$. The right term of $P(F)$ is $\lfloor (2h'+2-X')/2 \rfloor \ge \lfloor (2h'+2-c_0)/2 \rfloor \ge Y_0 \ge y+1$, the bottom term of $F$. In both cases each term of $P(F)$ plus 1 is at least some term of $F$. This proves (9).

(10) *If $k = 1$, there is a wall $H' \subseteq H$ of height $h - 2$ which satisfies (a), (b), (c) of (6) with $c = 2$ and deletes $m_v$.*

If $T(1,1)$ or $T(3,1)$ deletes $m_v$, we take it as $H'$, by (7). So we may

assume that both keep $m_v$. Let $(x, y)$ be such that $m_v$ is the vertex $(x, y)$ or an edge at $(x, y)$. Since $T(1, 1)$ and $T(3, 1)$ keep $m_v$, the elementary wall of height $h'$ has an element of the same kind at $(x-1, y-1)$ and at $(x-3, y-1)$. Hence $3 \le x \le 2h' + 2$ and $1 \le y \le h' + 1$. Moreover $x \le 2h' + 1$ if $m_v$ is a horizontal edge, and $y \le h'$ if $m_v$ is a vertical edge. Since $h = h' + 2$, the rotation $\rho$ maps the vertex $(x, y)$ to the vertex $(2h' + 5 - x, h' + 2 - y)$, the horizontal edge at $(x, y)$ to the horizontal edge at $(2h' + 4 - x, h' + 2 - y)$, and the vertical edge at $(x, y)$ to the vertical edge at $(2h' + 5 - x, h' + 1 - y)$.

We take as $H'$ the wall $S(Y_0, c_0)$ or the turned $S(Y_0, c_0)$, for integers $Y_0, c_0$ with $c_0 \equiv Y_0 \pmod 2$ such that $S(Y_0, c_0)$ is defined and good, that is, such that $1 \le Y_0 \le h'$ and $1 \le c_0 \le 2h' + 1$, and $c_0 \ge 2Y_0$ if $Y_0$ is odd and $c_0 + 2Y_0 \le 2h' + 2$ if $Y_0$ is even. Then $H'$ satisfies (a), (b), (c) of (6) with $c = 2$, by (9) and the remark about turned walls. By (8), $S(Y_0, c_0)$ deletes $m_v$ if $m_v$ is one of the elements listed there, and the turned $S(Y_0, c_0)$ deletes $m_v$ if $\rho(m_v)$ is one of them, since it deletes $m_v$ if and only if $S(Y_0, c_0)$ deletes $\rho(m_v)$.

For each kind of $m_v$ and each parity of $x$ and $y$, we give below two or three candidates for $H'$, in order. In every candidate $c_0 \equiv Y_0 \pmod 2$. For each candidate we name the element listed in (8) that $m_v$ is, or that $\rho(m_v)$ is for a turned candidate, and we give a condition that holds if and only if $S(Y_0, c_0)$ is defined and good. We take as $H'$ the first candidate whose condition holds, and we show that the condition of the last candidate holds if the earlier ones fail. Each condition comes from putting the values of $Y_0$ and $c_0$ into the inequalities above. In this, $1 \le c_0 \le 2h' + 1$ always follows from the bounds on $x$ and the parity of $x$, and $1 \le Y_0 \le h'$ follows from the bounds on $y$ and the parity of $y$, except that $Y_0 = h' + 2 - y$ with $y$ odd needs $y \ge 3$, $Y_0 = y$ with $y$ odd needs $y \le h'$, and $Y_0 = y + 1$ needs $y \le h' - 1$. These are part of the conditions below, or follow from them and the bounds.

Let $m_v$ be a vertex. If $x$ and $y$ are even, the candidates are $S(y-1, x-1)$, where $m_v$ is the vertex $(c_0 + 1, Y_0 + 1)$, with the condition $x \ge 2y - 1$; then $S(y, x - 2)$, where $m_v$ is the vertex $(c_0 + 2, Y_0)$, with the condition $x + 2y \le 2h' + 4$; and then the turned $S(h' + 2 - y, 2h' + 4 - x)$, where $\rho(m_v)$ is the vertex $(c_0 + 1, Y_0)$, with the condition $x + 2y \ge 2h' + 6$. If the second condition fails, then $x + 2y \ge 2h' + 5$, and so the third holds, since $x + 2y$ is even.

If $x$ and $y$ are odd, the candidates are the turned $S(h' + 2 - y, 2h' + 4 - x)$, where $\rho(m_v)$ is the vertex $(c_0 + 1, Y_0)$, with the condition that $y \ge 3$ and $x \le 2y$; and then $S(y, x)$, where $m_v$ is the vertex $(c_0, Y_0)$, with the condition that $y \le h'$ and $x \ge 2y$. If the first condition fails, then $y = 1$ or $x > 2y$. If

$y = 1$, then $y \leq h'$ and $x \geq 3 > 2y$. If $x > 2y$, then $2y < x \leq 2h' + 2$, and so $y \leq h'$. In both cases the second condition holds.

If $x$ is odd and $y$ is even, the candidates are the turned $S(h' + 1 - y, 2h' + 4 - x)$, where $\rho(m_v)$ is the vertex $(c_0 + 1, Y_0 + 1)$, with the condition $x \leq 2y + 2$; then $S(y, x - 1)$, where $m_v$ is the vertex $(c_0 + 1, Y_0)$, with the condition $x + 2y \leq 2h' + 3$; and then the turned $S(h' + 2 - y, 2h' + 3 - x)$, where $\rho(m_v)$ is the vertex $(c_0 + 2, Y_0)$, with the condition $x + 2y \geq 2h' + 5$. If the second condition fails, then $x + 2y \geq 2h' + 4$, and so the third holds, since $x + 2y$ is odd.

If $x$ is even and $y$ is odd, the candidates are $S(y, x - 1)$, where $m_v$ is the vertex $(c_0 + 1, Y_0)$, with the condition $x \geq 2y + 1$, which gives $y \leq h'$ since $x \leq 2h' + 2$; and then the turned $S(h' + 2 - y, 2h' + 5 - x)$, where $\rho(m_v)$ is the vertex $(c_0, Y_0)$, with the condition that $y \geq 3$ and $x \leq 2y + 1$. If the first condition fails, then $x \leq 2y$, and so $y \geq 3$, since $x \geq 3$ and $y$ is odd. So the second condition holds.

Let $m_v$ be a horizontal edge. If $x$ and $y$ are even, the candidates are $S(y - 1, x - 1)$, where $m_v$ is the horizontal edge at $(c_0 + 1, Y_0 + 1)$, with the condition $x \geq 2y - 1$; and then the turned $S(h' + 1 - y, 2h' + 3 - x)$, where $\rho(m_v)$ is the horizontal edge at $(c_0 + 1, Y_0 + 1)$, with the condition $x \leq 2y + 1$, which holds if the first fails.

If $x$ and $y$ are odd, the candidates are $S(y, x)$, where $m_v$ is the horizontal edge at $(c_0, Y_0)$, with the condition $x \geq 2y$, which gives $y \leq h'$ since $x \leq 2h' + 1$; and then the turned $S(h' + 2 - y, 2h' + 4 - x)$, where $\rho(m_v)$ is the horizontal edge at $(c_0, Y_0)$, with the condition that $y \geq 3$ and $x \leq 2y$. If the first condition fails, then $x < 2y$, and so $y \geq 3$, since $x \geq 3$ and $y$ is odd. So the second condition holds.

If $x$ is odd and $y$ is even, the candidates are $S(y - 1, x)$, where $m_v$ is the horizontal edge at $(c_0, Y_0 + 1)$, with the condition $x \geq 2y - 2$; and then the turned $S(h' + 1 - y, 2h' + 4 - x)$, where $\rho(m_v)$ is the horizontal edge at $(c_0, Y_0 + 1)$, with the condition $x \leq 2y + 2$, which holds if the first fails.

If $x$ is even and $y$ is odd, the candidates are $S(y, x + 1)$, where $m_v$ is the horizontal edge at $(c_0 - 1, Y_0)$, with the condition $x + 1 \geq 2y$, which gives $y \leq h'$ since $x \leq 2h'$ as $x$ is even; and then the turned $S(h' + 2 - y, 2h' + 5 - x)$, where $\rho(m_v)$ is the horizontal edge at $(c_0 - 1, Y_0)$, with the condition that $y \geq 3$ and $x \leq 2y + 1$. If the first condition fails, then $x \leq 2y - 2$, and so $y \geq 3$, since $x \geq 3$. So the second condition holds.

Let $m_v$ be a vertical edge, so that $x + y$ is odd. If $y$ is even, the candidates are the turned $S(h' + 1 - y, 2h' + 4 - x)$, where $\rho(m_v)$ is the vertical edge at $(c_0 + 1, Y_0)$, with the condition $x \leq 2y + 2$; and then $S(y + 1, x)$, where $m_v$ is the vertical edge at $(c_0, Y_0 - 1)$, with the condition that $y \leq h' - 1$ and

$x \geq 2y+2$. If the first condition fails, then $x \geq 2y+3$, and so $2y+3 \leq 2h'+2$, which gives $y \leq h'-1$. So the second condition holds.

If $y$ is odd, the candidates are $S(y, x-1)$, where $m_v$ is the vertical edge at $(c_0+1, Y_0)$, with the condition $x \geq 2y+1$; and then the turned $S(h'+2-y, 2h'+5-x)$, where $\rho(m_v)$ is the vertical edge at $(c_0, Y_0-1)$, with the condition that $y \geq 3$ and $x \leq 2y+1$. If the first condition fails, then $x \leq 2y$, and so $y \geq 3$, since $x \geq 3$ and $y$ is odd. So the second condition holds.

In every case the chosen wall $H'$ deletes $m_v$ and satisfies (a), (b), (c) of (6) with $c = 2$. This proves (10).

Now let $k \geq 2$. Let $(p, q)$ be the midpoint of $m_v$, and let $\delta$ be the least of the numbers $\lfloor q \rfloor$, $h - \lceil q \rceil$, $\lfloor p/2 \rfloor$ and $\lfloor (2h+1-p)/2 \rfloor$. These are the distances of $v$ from the bottom, top, left and right sides of $H$, counted in bricks. Every square at $m_v$ has a position $G = (X, Y)$ with $\sigma_h(G) \geq \delta$. For $\lceil q \rceil - 1 \leq Y \leq \lfloor q \rfloor$ gives $Y+1 \geq \lceil q \rceil \geq \lfloor q \rfloor$ and $h - Y \geq h - \lfloor q \rfloor \geq h - \lceil q \rceil$, and $|X - p| \leq 1$ gives $X+1 \geq p$ and $2h+2-X \geq 2h+1-p$. Hence every region of $H$ at $m_v$ has depth at least $\delta$.

(11) *If $k \geq 2$ and $\delta \leq k$, then there are $i, j$ as in (7) such that $T(i, j)$ deletes $m_v$, and every region of $H$ at an element kept by $T(i, j)$ has distance $D_H$ at least $k-2$ from every region of $H$ at $m_v$.*

We choose $i, j$ by a side of $H$ at distance $\delta$ from $v$, so that $T(i, j)$ leaves a wide margin of $H$ on that side. In the *bottom* case, where $\delta = \lfloor q \rfloor$, let $j = \min(2k-1, k+1+\delta)$ and $i = 2k + (j \bmod 2)$. Otherwise, in the *top* case, where $\delta = h - \lceil q \rceil$, let $j = \max(1, k-1-\delta)$ and $i = 2k + (j \bmod 2)$. Otherwise, in the *left* case, where $\delta = \lfloor p/2 \rfloor$, let $i = \min(4k-1, 2k+2+2\delta)$ and $j = k + ((i+k) \bmod 2)$. Otherwise we are in the *right* case, where $\delta = \lfloor (2h+1-p)/2 \rfloor$, and we let $i = \max(1, 2k-2-2\delta)$ and $j = k + ((i+k) \bmod 2)$. These satisfy the conditions of (7), using $k \geq 2$ and $\delta \leq k$. In the first two cases $i+j$ is even, $k+1 \leq j \leq 2k-1$ or $1 \leq j \leq k-1$, and $i \in \{2k, 2k+1\}$. In the last two cases $i+j$ is even, $j \in \{k, k+1\}$, and $2k+2 \leq i \leq 4k-1$ or $1 \leq i \leq 2k-2$.

Let $m$ be an element kept by $T(i, j)$, let $s$ be a square at $m$ with position $F = (X, Y)$, and let $s'$ be a square at $m_v$ with position $G = (X', Y')$. By the remark after (2), it is enough to show that the regions $R, R'$ of $H$ containing $s, s'$ satisfy $\mathrm{depth}(R) + \mathrm{depth}(R') \geq k-2$ and $\|F - G\| \geq k-2$. The midpoint of $m$ lies in $[i, i+2h'+1] \times [j, j+h']$, and so $i-1 \leq X \leq i+2h'+2$ and $j-1 \leq Y \leq j+h'$. Hence the terms of $F$ are at least $j$, $2k-j$, $\lfloor i/2 \rfloor$ and $\lfloor (2h-i-2h')/2 \rfloor = \lfloor (4k-i)/2 \rfloor$, and so $\mathrm{depth}(R) \geq \min(j, 2k-j, \lfloor i/2 \rfloor, \lfloor (4k-i)/2 \rfloor)$. In each of the four cases this minimum is at least $k-1-\delta$. For in the first two cases $\lfloor i/2 \rfloor = k$ and $\lfloor (4k-i)/2 \rfloor \geq k-1$,

and the smaller of $j$ and $2k-j$ is at least $\min(k-1, 2k-(k+1+\delta))$ in the first case and at least $\min(k-1-\delta, k+1)$ in the second. In the last two cases $j \geq k$ and $2k-j \geq k-1$, and the smaller of $\lfloor i/2 \rfloor$ and $\lfloor (4k-i)/2 \rfloor$ is at least $\min(k-1, \lfloor (2k-2-2\delta)/2 \rfloor)$. Since $\mathrm{depth}(R') \geq \delta$, we get $\mathrm{depth}(R) + \mathrm{depth}(R') \geq k-1$.

Now we bound $\|F-G\|$. In the bottom case $Y' \leq \lfloor q \rfloor = \delta$, so $\|F-G\| \geq Y - Y' \geq j-1-\delta$, which is $k$ if $j = k+1+\delta$, and $2k-2-\delta \geq k-2$ if $j = 2k-1$. In the top case $Y' \geq \lceil q \rceil - 1 = h-\delta-1$ and $Y \leq j+h' = h-2k+j$, so $\|F-G\| \geq Y' - Y \geq 2k-1-\delta-j$, which is $k$ if $j = k-1-\delta$, and $2k-2-\delta \geq k-2$ if $j = 1$. In the left case $p < 2\delta+2$, so $X' \leq p+1$ gives $X' \leq 2\delta+2$, and $\|F-G\| \geq (X-X')/2 \geq (i-3-2\delta)/2$. If $i = 2k+2+2\delta$ this is $k-\frac{1}{2}$, and so $\|F-G\| \geq k$ since it is an integer. If $i = 4k-1$ this is $2k-2-\delta \geq k-2$. In the right case $p > 2h-1-2\delta$, so $X' \geq p-1$ gives $X' \geq 2h-1-2\delta$, while $X \leq i+2h'+2 = i+2h-4k+2$. So $\|F-G\| \geq (X'-X)/2 \geq (4k-3-2\delta-i)/2$, which is $k-\frac{1}{2}$ if $i = 2k-2-2\delta$ and $2k-2-\delta$ if $i = 1$. In all cases $\|F-G\| \geq k-2$.

Finally $T(i,j)$ deletes $m_v$, by the bounds on the midpoints of kept elements. In the bottom case $q < \delta+1 \leq j$, since $2k-1 \geq k+1 \geq \delta+1$. In the top case $q > h-\delta-1 \geq h-2k+j = j+h'$, since $j \leq 2k-1-\delta$. In the left case $p < 2\delta+2 \leq i$, since $4k-1 \geq 2k+2 \geq 2\delta+2$. In the right case $p > 2h-1-2\delta \geq i+2h'+1$, since $i \leq 4k-2-2\delta$. This proves (11).

For $v$ far from the sides we use a wall with a large hole. Let $k \geq 2$ and $b = k-1 \geq 1$, so that $h' = h-2b-2$, and let $e$ be the even one of $b+1$ and $b+2$. Let $Y_0, c_0$ be integers with $1 \leq Y_0 \leq h'$ and $1 \leq c_0 \leq 2h'+1$, and let $Y_1 = Y_0 + b - 1$. For every integer $t$ let $w_t = \max(0, b-|t-Y_1|)$. Thus $w_t = 0$ unless $Y_0 \leq t \leq Y_0+2b-2$, consecutive values of $w_t$ differ by 1 for $Y_0 - 1 \leq t \leq Y_0+2b-1$, and $w_{Y_1} = b$. The wall $L(Y_0, c_0)$ is the subgraph of $H$ defined as follows. Let $R(y) = y$ if $y \leq Y_0 - 1$ and $R(y) = y+2b$ if $y \geq Y_0$. The branch vertex corresponding to $(x,y)$ is the vertex of $H$ in row $R(y)$ and column $x+e$ if $x \leq c_0-1$, and column $x+e+2b$ if $x \geq c_0$. The horizontal edges are the paths of $H$ along rows between their ends. The vertical edge at $(x,y)$ is the vertical edge of $H$ at the vertex corresponding to $(x,y)$ if $y \neq Y_0-1$. If $y = Y_0-1$, the vertical edge at $(x,y)$ is the staircase which has, in each strip $t$ with $Y_0-1 \leq t \leq Y_0+2b-1$, the vertical edge of $H$ in column $x+e-w_t$ if $x \leq c_0-1$, and in column $x+e+2b+w_t$ if $x \geq c_0$, together with the horizontal edges of $H$ joining the consecutive ones. So the band of rows $Y_0, \ldots, Y_0+2b-1$ of $H$ is not used as rows of $L(Y_0, c_0)$, and the staircases on the left of $c_0$ move one step to the left in each of the first $b$ of these rows, and back to the right in the others. The staircases on the right of $c_0$ do the mirror image. Between them lies a hole shaped like a

hexagon, which is the reason for this construction.

(12) *The wall $L(Y_0, c_0)$ is a wall of height $h'$ of our form, with $c_0$ as given, $y_t = t$ for $t \le Y_0 - 1$, $y_t = Y_0 - 1$ for $Y_0 \le t \le Y_0 + 2b - 1$, $y_t = t - 2b$ for $t \ge Y_0 + 2b$, and $a_t = e - w_t$, $a'_t = e + 2b + w_t$ for every $t$. Let $x_L$ be the largest integer with $x_L \le c_0 - 1$ and $x_L \equiv Y_0 \pmod 2$, and let $B_0 = (x_L + 1 + e + b, Y_1)$. Then all bricks $F$ of $H$ with $\|F - B_0\| \le b$ lie in the same region of $L(Y_0, c_0)$. If moreover $2Y_0 \le h$, then $L(Y_0, c_0)$ satisfies (a), (b), (c) of (6) with $c = 2k$.*

The staircases are paths of $H$. For since $|t + 1 - Y_1| - |t - Y_1| = \pm 1$, consecutive vertical edges of a staircase are in adjacent columns. Moreover $w_t \equiv b + t - Y_1 \equiv t + Y_0 + 1 \pmod 2$, so the column $x + e - w_t$ plus $t$ is odd, since $x \equiv Y_0$ and $e$ is even. The same holds on the right. Since $w_{Y_0 - 1} = w_{Y_0 + 2b - 1} = 0$, each staircase starts and ends at the branch vertices it should join, and it meets rows $Y_0 - 1$ and $Y_0 + 2b$ only in its ends. The columns used lie between $e - b \ge 1$ and $2h' + 1 + e + 3b \le 2h + 1$, and the rows used are at most $h' + 2b = h - 2$. In each row of the band, a staircase uses two adjacent columns, and the staircases on the same side differ by 2 in $x$. The left staircases use columns at most $c_0 - 1 + e$, and the right ones columns at least $c_0 + e + 2b$. So the staircases are mutually disjoint, and $L(Y_0, c_0)$ has our form with the values given. Here $w_t = 0$ outside the band, and $t - y_t$ is 0, $t - Y_0 + 1$ or $2b$, which has the parity of $a_t$ and $a'_t$ in each case.

By (4), the move $(\Delta X, \Delta Y)$ of a brick $F = (X, Y)$ satisfies $e - w_Y \le \Delta X \le e + 2b + w_Y$, and $\Delta Y = 0$ if $Y \le Y_0 - 1$, $\Delta Y = Y - Y_0 + 1$ if $Y_0 \le Y \le Y_0 + 2b - 1$, and $\Delta Y = 2b$ if $Y \ge Y_0 + 2b$. In the band, $\Delta Y - b = Y - Y_1$ and $|Y - Y_1| = b - w_Y$. Outside the band $w_Y = 0$ and $|\Delta Y - b| = b$. Hence in all cases

$$|\Delta X - e - b| + |\Delta Y - b| \le (b + w_Y) + (b - w_Y) = 2b,$$

and $0 \le \Delta Y \le 2b$, with $\Delta Y = 0$ only if $Y \le Y_0 - 1$. Moreover $\Delta X \ge e - w_Y \ge b + 1 - w_Y \ge 1$, and $\Delta X \ge 2$ unless $w_Y = b$, that is, unless $Y = Y_1$.

Let $F = (X, Y)$ be a brick with $\|F - B_0\| \le b$. Then $|Y - Y_1| \le b$, so $Y$ is in the band or $Y = Y_0 - 1$, and $y_Y = Y_0 - 1$. Moreover $|X - (x_L + 1 + e + b)| \le 2b - |Y - Y_1| = b + w_Y$. Hence $X - a_Y = X - e + w_Y \ge x_L + 1 \ge c_0 - 1$ and $X - a'_Y = X - e - 2b - w_Y \le x_L + 1 \le c_0$. In the definition of $P(F)$ in (4), this gives $z = c_0 - 1$ in the first and third cases, and $z = c_0 = x_L + 1$ in the second case. In each case the position of $(z, Y_0 - 1)$ is $(x_L + 1, Y_0 - 1)$, since $x_L + 1 \in \{c_0 - 1, c_0\}$ and $x_L + 1 \equiv Y_0 - 1$. So $P(F) = (x_L + 1, Y_0 - 1)$ for all these bricks, and by (4) they lie in the same region of $L(Y_0, c_0)$.

Now let $2Y_0 \le h$. Any two moves $\mu, \mu'$ satisfy $|\mu_1 - \mu'_1| + |\mu_2 - \mu'_2| \le 4b$ and $|\mu_2 - \mu'_2| \le 2b$, so $\|\mu - \mu'\| \le 2b \le 2c-2$, and (a) follows as in the proof of (9). Since $h - h' = 2b+2$, $0 \le \Delta Y \le 2b$ and $1 \le \Delta X \le e+3b \le 4b+2$, (5) applies, and the drops are at most $2b$, $2b+2$, $2b+1$ and $\lceil (4b+3)/2 \rceil = 2b+2$. This gives (b) with $c = 2b+2 = 2k$. Moreover the top term drops by at most $2b+1$ unless $\Delta Y = 0$, and the right term drops by at most $2b+1$ if $\Delta X \ge 2$.

For (c) let $m$ be a deleted element, and let $(x, y)$ be such that $m$ is the vertex $(x, y)$ or an edge at $(x, y)$. Let $s$ be the square $(x, y-1)$ if $m$ is a vertex or a horizontal edge with $y = Y_1$, and the square $(x, y)$ otherwise. In both cases $s$ is at $m$. Let $F = (X, Y)$ be its position, and suppose that $F$ is a brick of height $h$. We show that each term of $P(F)$ plus $2b+1$ is at least some term of $F$. The bottom and left terms drop by at most $2b+1$. If $\Delta Y = 0$, then $Y \le Y_0 - 1$, and the top term of $P(F)$ plus $2b+1$ is $h' - Y + 2b + 1 = h - 1 - Y \ge Y + 1$, since $2Y + 2 \le 2Y_0 \le h$. So it is at least the bottom term of $F$. It remains to consider the right term when $\Delta X = 1$, which happens only if $Y = Y_1$. Then $m$ is the vertical edge at $(x, Y_1)$, by the choice of $s$, and $X = x+1$ since $x + Y_1$ is odd. Let $x'' = x - e + b$. Then $x'' \equiv Y_0 \pmod 2$, since $x \equiv Y_1 + 1 = Y_0 + b$. If $0 \le x'' \le c_0 - 1$, then $x = x'' + e - w_{Y_1}$ is the column of the vertical edge in strip $Y_1$ of the staircase at $(x'', Y_0 - 1)$, and so $m$ is kept, which is false. If $x'' < 0$, then $X = x'' + e - b + 1 \le e - b \le 2$, and so $\sigma_h(F) \le 1$. If $x'' \ge c_0$, then $X - a_{Y_1} = x'' + 1 \ge c_0 + 1$, so we are in the second or third case of (4). There $\Delta X = e + 3b$ or $\Delta X \ge X - c_0 = x'' + 1 + e - b - c_0 \ge 2$. So $\Delta X \ge 2$ in all cases where $\sigma_h(F) > 1$. This gives (c), and proves (12).

(13) *If $k \ge 2$ and $\delta \ge k+1$, there is a wall $H' \subseteq H$ of height $h - 2k$ which satisfies (a), (b), (c) of (6) with $c = 2k$, and a brick $B$ of $H$ at $m_v$, such that all bricks $F$ of $H$ with $\|F - B\| \le k-1$ lie in the same region of $H'$.*

Let $b = k-1$ and $e$ be as above. The square $(\lceil p \rceil - 1, \lceil q \rceil - 1)$ is at $m_v$. Let $B = (X_0, Y_1)$ be its position, so $Y_1 = \lceil q \rceil - 1$ and $\lceil p \rceil - 1 \le X_0 \le \lceil p \rceil$. Let $Y_0 = Y_1 - b + 1$ and $c_0 = X_0 - e - b$. Then $c_0 - 1 \equiv Y_0 \pmod 2$, since $X_0 \equiv Y_1$ and $e$ is even, and so $x_L = c_0 - 1$ in (12), and $B = B_0$. Since $\delta \ge b+2$, the four side distances of $m_v$ give $\lceil q \rceil \ge b+2$, $\lceil q \rceil \le h - b - 2$, $p \ge 2b+4$ and $p \le 2h - 2b - 3$. Hence $Y_0 \ge 2$ and $Y_0 = \lceil q \rceil - b \le h - 2b - 2 = h'$. Moreover $c_0 \ge (2b+3) - (b+2) - b = 1$, using $X_0 \ge p - 1$ and $e \le b+2$, and $c_0 \le (2h - 2b - 3) - (b+1) - b = 2h'$, using $X_0 \le \lceil p \rceil$ and $e \ge b+1$. If $2Y_0 \le h$, then (12) shows that $H' = L(Y_0, c_0)$ and $B$ are as required. Otherwise $2(\lceil q \rceil - b) > h$. Now $\rho(m_v)$ has midpoint $(2h + 1 - p, h - q)$, its four side distances are those of $m_v$ in another order, and so the same

construction applies to it, with $Y_0$ replaced by $\lceil h-q\rceil - b = h - \lfloor q\rfloor - b$. Since $\lfloor q\rfloor \geq \lceil q\rceil - 1$, twice this number is less than $2h - h - 2b + 2 - 2b \leq h$. So the construction gives a wall $L$ for $\rho(m_v)$ with (a), (b), (c), and a brick $B'$ at $\rho(m_v)$ such that all bricks within $\|\cdot\|$-distance $b$ of $B'$ lie in one region of $L$. We take $H' = \rho(L)$, which satisfies (a), (b), (c) by the remark about turned walls, and $B = \rho(B')$. Then $B$ is a brick at $m_v$, since $\rho$ maps the squares at $\rho(m_v)$ to the squares at $m_v$. All bricks within $\|\cdot\|$-distance $b$ of $B$ lie in one region of $\rho(L)$, since $\rho$ preserves $\|\cdot\|$ and maps regions of $L$ to regions of $\rho(L)$. This proves (13).

Now we put the cases together. Let $m_v$ be the element of $H$ that contains $v$. If $r = 0$, then $k = 1$ and (10) gives a wall $H'$ of height $h-2$ with $d' \geq d-4$ by (6), which deletes $m_v$. Every vertex $u$ of $H'$ satisfies $u \neq v$, and so $d(u,v) \geq 2 \geq 1$ by (1). So let $r \geq 1$, that is, $k \geq 2$, and let $\delta$ be as above. If $\delta \leq k$, let $H' = T(i,j)$ for $i,j$ as in (11). Then $d' \geq d - 4k$ by (7), and every vertex $u$ of $H'$ satisfies $u \neq v$ and lies in an element kept by $H'$. So by (11), every region at $u$ has distance $D_H$ at least $k-2$ from every region at $v$, and $d(u,v) \geq k$. If $\delta \geq k+1$, let $H'$ and $B$ be as in (13), and let $O$ be the region of $H'$ that contains the bricks $F$ with $\|F - B\| \leq k-1$. Then $d' \geq d - 4k$ by (6). Let $u$ be a vertex of $H'$. It is at two different regions of $H'$, so at a region $E' \neq O$ of $H'$. As in the proof of (6), some region $E$ of $H$ at $u$ lies in $E'$. So $E$ is not a brick with $\|E - B\| \leq k-1$. Since $B$ is at $m_v$, $\sigma_h(B) \geq \delta \geq k+1$. If $E$ is the infinite region, then $D_H(E,B) = \sigma_h(B) \geq k+1$. Otherwise, by (2), $D_H(E,B) = \min(\|E-B\|, \sigma_h(E) + \sigma_h(B)) \geq k$. In both cases $D_H(E,B) \geq k$. Every region of $H$ at $u$ touches $E$ or equals it, and every region of $H$ at $v$ touches $B$ or equals it, since $B$ is at $v$. So every region at $u$ has distance $D_H$ at least $k-2$ from every region at $v$. Moreover $u \neq v$, since otherwise $E$ would be at $v$ and $D_H(E,B) \leq 1$. Hence $d(u,v) \geq k$. In each case $H'$ is a wall of height $h-2k = h-2r-2$ in $H$ with $d' \geq d-4k = d-4r-4$, and no vertex $u$ of $H'$ has $d(u,v) \leq r$. This proves (7.3).[F10]

Let $G$ be a graph. A *wall in* $G$ is a subgraph of $G$ which is a wall. Let $H$ be a subgraph of $G$. An *$H$-path in $G$* is a path with distinct ends, both in $V(H)$, and with no edge or internal vertex in $H$. We shall need the following lemma.

(7.4) *For any $p \geq 2$ there exists $r \geq 0$ such that the following holds. Let $H$ be a wall in a graph $G$. Let $P_1, \ldots, P_{\binom{p}{2}}$ be mutually vertex-disjoint $H$-paths in $G$, with their $p(p-1)$ ends mutually at distance $\geq r$. Then $G$ has a minor isomorphic to $K_p$. Moreover, there is an algorithm, with*

[F10] **Missing proof.** The paper leaves the proof of (7.3) to the reader. The proof above is the one used in the formalization.

*running time* $\leq C\varepsilon(G)$*, where* $C$ *depends only on* $p$*, which, given* $G$*,* $H$*, and* $P_1, \ldots, P_{\binom{p}{2}}$ *as above, finds a model of* $K_p$ *in* $G$*.*

This follows from the main theorem of [16], or more directly from Theorem (4.4) of [20], taking $\mathcal{T}$ to be the set of all separations $(A, B)$ of $H$ of order $< r$ such that $A$ contains no horizontal path of $H$, taking $\Sigma$ to be a sphere, and taking $\Gamma = H$.

Let $H$ be a wall in $G$. For $l \geq 3$, $m \geq 0$ an $(l, m)$-*star over* $H$ in $G$ is a subgraph $P_1 \cup \cdots \cup P_l$ of $G$ such that for some vertex $v$,

[p. 91] (i) $P_1, \ldots, P_l$ are paths of $G$ and all have one end $v$, and are otherwise mutually vertex-disjoint

(ii) for $1 \leq i \leq l$, let the ends of $P_i$ be $v, u_i$; then $u_i \neq v$, and $u_i \in V(H)$, and no edge or internal vertex of $P_i$ is in $H$

(iii) for $1 \leq i < j \leq l$, $u_i$ and $u_j$ are at distance $\geq m$ in $H$.

We say that $v$ is the *center* of the star. (It is unique.) The star is *external* if $v \notin V(H)$.

(7.5) *For any* $p \geq 2$ *there are integers* $l \geq 3$*,* $m \geq 0$ *with the following property. Let* $G$ *be a graph and let* $H$ *be a wall in* $G$*. Suppose that there are* $\binom{p}{2}$ *external* $(l, m)$*-stars over* $H$ *with distinct centers (but not necessarily disjoint). Then* $G$ *has a* $K_p$*-minor.*

*Proof.* We shall prove the following for $k, m \geq 0$ by induction on $k$.

*Let* $l = 2k^2 - k + 3$*. For any* $k$ *external* $(l, m)$*-stars* $S_1, \ldots, S_k$ *over* $H$ *with distinct centers there are* $k$ *mutually vertex-disjoint* $H$*-paths of* $G$*, each a subgraph of* $S_1 \cup \cdots \cup S_k$*, with their* $2k$ *ends mutually at distance* $\geq \frac{1}{2}m$*.*

The claim is true for $k = 0$, and we assume that $k > 0$. For $1 \leq i \leq k-1$, let $S_i'$ be an $(l-1, m)$-star with the same center as $S_i$, with $S_i' \subseteq S_i$, and with $v_k \notin V(S_i')$, where $v_k$ is the center of $S_k$. Since $l - 1 \geq 2(k-1)^2 - (k-1) + 3$, there are (from our inductive hypothesis) $k - 1$ mutually vertex-disjoint $H$-paths $Q_1, \ldots, Q_{k-1}$ of $G$, with

$$Q_1 \cup \cdots \cup Q_{k-1} \subseteq S_1' \cup \cdots \cup S_{k-1}'$$

and with their $2(k-1)$ ends mutually at distance $\geq \frac{1}{2}m$.

Let $S_k = P_1 \cup \cdots \cup P_l$, with the $P_i$'s as in the definition of a star. Suppose that for some $j$ with $1 \leq j \leq k-1$ there are $\geq 2k$ values of $i$ with $1 \leq i \leq l$ such that $P_i \cap Q_j$ is non-null. For each such $i$, let $u_i$ be the first vertex of $P_i$ which belongs to $V(Q_j)$ (first as $P_i$ is traversed starting from its end in $V(H)$). Choose distinct $i_1, i_2, \ldots, i_{2k}$ such that $P_{i_t} \cap Q_j$ is non-null for each $t$, numbered so that $u_{i_1}, \ldots, u_{i_{2k}}$ occur in this order on $Q_j$. Then we may construct $k$ $H$-paths $R_1, \ldots, R_k$ with their ends mutually at distance $\geq m \geq \frac{1}{2}m$, where each $R_t$ is formed from subpaths of $P_{i_{2t-1}}$ and $P_{i_{2t}}$ and

the path of $Q_j$ between $u_{i_{2t-1}}, u_{i_{2t}}$. We may assume therefore that there is no such $j$.

Hence the total number of pairs $i, j$ with $P_i \cap Q_j$ non-null is at most $(k-1)(2k-1)$; and so there are at least

$$l - (k-1)(2k-1) \geq 2k$$

[p. 92] values of $i$ such that $P_i \cap Q_j$ is null for all $j$ with $1 \leq j \leq k-1$. For $1 \leq i \leq l$, let $u_i, v_k$ be the ends of $P_i$, and for $1 \leq j \leq k-1$ let $s_j, t_j$ be the ends of $Q_j$. Now for any $x \in \{s_1, t_1, \ldots, s_{k-1}, t_{k-1}\}$, $d(u_i, x) < \frac{1}{2}m$ for at most one value of $i$ $(1 \leq i \leq l)$, since $d(u_i, u_{i'}) \geq m$ for $i \neq i'$. Thus there are at least two values of $i$ such that

(i) $P_i \cap Q_j$ is null for $1 \leq j \leq k-1$, and

(ii) $d(u_i, x) \geq \frac{1}{2}m$ for all $x \in \{s_1, t_1, \ldots, s_{k-1}, t_{k-1}\}$.

The union of the corresponding two $P_i$'s yields an $H$-path $Q_k$, and $Q_1, \ldots, Q_k$ are the desired $H$-paths. This completes the induction.

In particular, the statement holds if $k = \binom{p}{2}$. Let $l = 2\binom{p}{2}^2 - \binom{p}{2} + 3$, choose $r$ as in (7.4) and let $m = 2r$. Then $l, m$ satisfy the theorem, by (7.4). □

(7.6) *Let $H$ be a wall in a graph $G$. Suppose there are $k$ $(l, m+4k)$-stars in $G$ over $H$ with distinct centers $v_1, \ldots, v_k$, where $m \geq 3$. Then there is a wall $H' \subseteq H$ with $k$ external $(l, m)$-stars over $H'$, with centers $v_1, \ldots, v_k$.*

*Proof.* Let $v_1, \ldots, v_k \in V(G)$ be distinct, and each the center of an $(l, m+4k)$-star over $H$. Let $H_0 = H$. By $k$ applications of (7.3), with $r = 0$ and setting $v = v_1, \ldots, v_k$ in turn, we deduce that there are walls $H_1, \ldots, H_k$ (this is possible, since $H$ has height $h \geq 2k+2$: two ends of legs of a star over $H$ have distance $\geq m + 4k \geq 4k+3$, and any two vertices of a wall of height $h$ have distance $\leq 2h+2$)[F11] such that for $1 \leq i \leq k$

(i) $H_i$ is a subgraph of $H_{i-1}$, and $v_i \notin V(H_i)$

(ii) $d_i \geq d_{i-1} - 4$, where $d_0, \ldots, d_k$ are the corresponding distance functions.

(1) *If $v \in V(G)$ is the center of an $(l, m+4(k-i+1))$-star over $H_{i-1}$, then $v$ is the center of an $(l, m+4(k-i))$-star over $H_i$, for $1 \leq i \leq k$.*

*Subproof.* Let $P_1 \cup \cdots \cup P_l$ be an $(l, m+4(k-i+1))$-star over $H_{i-1}$, with center $v$. For $1 \leq j \leq l$, let $P_j$ have ends $v, u_j$; and let $Q_j$ be a minimal path of $H_{i-1}$ between $u_j$ and $V(H_i)$, with ends $u_j, w_j$ say. Now for distinct

[F11] **Proof gap.** The proof does not check the height. Each application of (7.3) with $r = 0$ needs height $\geq 4$ and lowers the height by 2, so the $k$ applications need $h \geq 2k+2$. The inserted reason shows that this follows from the hypotheses.

$j, j'$,

$$d_i(u_j, u_{j'}) \geq d_{i-1}(u_j, u_{j'}) - 4 \geq m + 4(k-i+1) - 4 = m + 4(k-i) \geq 3.$$

Moreover $d_i(w_j, w_{j'}) \geq d_i(u_j, u_{j'})$, since a curve from $w_j$ to $w_{j'}$ can be extended along the drawings of $Q_j$ and $Q_{j'}$, which meet the drawing of $H_i$ only in $w_j$ and $w_{j'}$.[F12] Consequently, $u_j$ and $u_{j'}$ do not belong to the closure of the same region of $H_i$, and hence $Q_j$ and $Q_{j'}$ are vertex-disjoint. If $v \in V(Q_j)$, replace $P_j \cup Q_j$ below by the subpath of $Q_j$ between $v$ and $w_j$.[F13] It follows that

$$(P_1 \cup Q_1) \cup (P_2 \cup Q_2) \cup \cdots \cup (P_l \cup Q_l)$$

is an $(l, m + 4(k-i))$-star over $H_i$, as required.

From ~~(1)~~the proof of (1), used once with $H_0, H_k$ in place of $H_{i-1}, H_i$ and with $d_k \geq d_0 - 4k$ (by (ii)) in place of $d_i \geq d_{i-1} - 4$, we deduce that $v_1, \ldots, v_k$ are all centers of $(l, m)$-stars over $H_k$;[F14] and these stars are all external since $v_1, \ldots, v_k \notin V(H_k)$. The result follows. □

[p. 93] From (7.5) and (7.6) we deduce

(7.7) *For any $p \geq 2$ there are integers $l \geq 3$, $m \geq 0$ with the following property. Let $G$ be a graph and let $H$ be a wall in $G$, so that there are $\binom{p}{2}$ $(l, m)$-stars over $H$, with distinct centers; then $G$ has a $K_p$-minor. Moreover, there is an algorithm, with running time $\leq C\varepsilon(G)$, where $C$ depends only on $p$, which, given $\binom{p}{2}$ $(l, m)$-stars with distinct centers over a wall $H$ in a graph $G$, finds a model of $K_p$ in $G$.*

The first assertion is clear from (7.5) and (7.6). The second follows since the proofs of (7.5) and of (7.6) may easily be converted to algorithms.

We give the details for the steps that need a method. We assume that a wall is given with its identification with a subdivision of an elementary wall, so that the position of each of its vertices in the drawing of Fig. 1 is known: a vertex of the elementary wall, or a point of the line that represents one of

[F12] **Proof gap.** The ends of the legs of the new star are $w_j$ and $w_{j'}$, not $u_j$ and $u_{j'}$, so the bound on $d_i(u_j, u_{j'})$ is not enough by itself. The inserted sentence gives the missing step.

[F13] **Proof gap.** The stars of (7.6) need not be external, so the center $v$ may be a vertex of $H_{i-1}$, and then it may lie on $Q_j$ (for instance, $v = v_i$). Then $P_j \cup Q_j$ is not a path. The inserted sentence deals with this case.

[F14] **Proof gap.** Claim (1) is applied at each step $i$ to all of $v_1, \ldots, v_k$, but $v_{i+1}, \ldots, v_k$ may be vertices of $H_i$. If such a center $v$ lies on $Q_j$, then $v = w_j$, since $w_j$ is the only vertex of $Q_j$ in $H_i$. Then $P_j \cup Q_j$ is a circuit, and the sentence added to the proof of (1) for this case gives a leg with only one vertex. Used once, from $H_0$ to $H_k$, the proof of (1) works, because no center is in $V(H_k)$, so every new leg ends at a vertex other than its center. With these corrections, (7.6) holds as printed.

its edges. A wall that is a subgraph of it gets its positions from these. The numbers $k = \binom{p}{2}$ and $l$ depend only on $p$, and so do the numbers of stars, legs and ends handled below.

For (7.6), each application of (7.3) with $r = 0$ chooses $H_i$ among a bounded number of explicit subwalls of $H_{i-1}$, according to the position of $v_i$ in $H_{i-1}$ (see the proof of (7.3)), and this takes time $O(\varepsilon)$. Then we extend every leg of every star as in (1). Let $P_j$ be a leg with ends $v, u_j$, and let $q$ be the first vertex of the bottom row of $H_i$. In the elementary wall of $H_{i-1}$ there is a path from the position of $u_j$ to that of $q$ that runs along the row of $u_j$ to a vertical edge, along this edge to the next row, and so on until it reaches the row of $q$, and then along this row to $q$ (if $u_j$ is inside an edge of the elementary wall, the path first runs along this edge). We follow the corresponding path of $H_{i-1}$ from $u_j$ and stop at its first vertex in $V(H_i)$. This gives a minimal path $Q_j$ of $H_{i-1}$ between $u_j$ and $V(H_i)$, in time $O(\varepsilon)$. If $v \in V(Q_j)$, the new leg is the subpath of $Q_j$ between $v$ and $w_j$, as in (1), and otherwise it is $P_j \cup Q_j$. In this way a new leg may consist of $v$ alone, when $v = w_j$ (this is why the proof of (7.6) uses (1) only once). We allow such legs before the last step. The argument of (1) applies to them unchanged and gives the bound $m + 4(k - i)$ on the distances between the ends of the legs, since it uses only these ends. After step $k$ no center is in $V(H_k)$, and so every leg has at least two vertices, and we have $k$ external $(l, m)$-stars over $H_k$.

For (7.5) we follow the induction on $k$. Given $S_1, \dots, S_k$, we obtain $S'_i$ by deleting from $S_i$ the leg that contains $v_k$, or any one leg if there is none. There is at most one such leg, since $v_k$ is not the center of $S_i$ and the legs of $S_i$ meet only in its center. We apply the algorithm to $S'_1, \dots, S'_{k-1}$ and obtain $Q_1, \dots, Q_{k-1}$. Next we label each vertex of $Q_j$ with $j$ $(1 \le j \le k-1)$, number the vertices of each $Q_j$ in order, and traverse each leg $P_i$ of $S_k$ from its end in $V(H)$, recording for each $j$ the first vertex of $P_i$ with label $j$. This takes time $O(\varepsilon)$, since the legs of $S_k$ meet only in $v_k$, and it decides whether some $j$ has $\ge 2k$ values of $i$ with $P_i \cap Q_j$ non-null. If so, it gives the vertices $u_{i_1}, \dots, u_{i_{2k}}$ in order on $Q_j$, and the paths $R_1, \dots, R_k$ are formed in time $O(\varepsilon)$. Otherwise we need the distances $d(u_i, x)$ for the $2(k-1)$ ends $x$ of $Q_1, \dots, Q_{k-1}$, and we compute them from the positions of $u_i$ and $x$. By the formula for $d$ in the proof of (7.3), for distinct vertices $s, t$ of $H$, $d(s, t)$ is 2 plus the minimum of $D_H(R, R')$ over the regions $R$ at $s$ and $R'$ at $t$. Moreover $D_H$ of a brick and the infinite region is the depth of the brick, and $D_H$ of two bricks is the smaller of their lattice distance and the sum of their depths. Let $H$ have height $h$. We may take the drawing of the elementary wall to have as its vertices the points $(x, y)$ with integers $0 \le x \le 2h + 1$

and $0 \le y \le h$, other than $(0,0)$ and $(2h+1,h)$, with the rows horizontal, and with $(x,y)$ joined to $(x,y+1)$ when $x+y$ is odd. We name each brick $(X,y)$, where the brick lies between the rows $y$ and $y+1$ ($0 \le y < h$) and $X$ is the abscissa of its middle. Then the lattice distance of the bricks $(X,y)$ and $(X',y')$ is

$$\max\bigl(|y-y'|, \tfrac{1}{2}(|X-X'|+|y-y'|)\bigr),$$

and the depth of the brick $(X,y)$ is

$$\min\bigl(y+1,\ h-y,\ \lfloor (X+1)/2\rfloor,\ \lfloor (2h+2-X)/2\rfloor\bigr).$$

The regions at a vertex are read off from its position, and so each distance $d(u_i,x)$ is computed in constant time. Thus each of the $\binom{p}{2}$ steps of the induction takes time $O(\varepsilon)$. Together with the algorithm of (7.4), this gives a model of $K_p$ in time $\le C\varepsilon(G)$.[F15]

## 8. The 2-Paths Problem

Let $G$ be a graph and let $\Omega$ be a cyclic permutation of a subset (denoted by $\bar{\Omega}$) of $V(G)$. We call $(G,\Omega)$ a *society*. Actually, for our application in this paper we could assume that $|\bar{\Omega}| = 4$ throughout, but we have chosen to give the results in general. A *cross* in a society $(G,\Omega)$ is a subgraph $P \cup Q$ of $G$, where $P,Q$ are vertex-disjoint paths of $G$ with ends $p_1,p_2$ and $q_1,q_2$, respectively, and $p_1,q_1,p_2,q_2 \in \bar{\Omega}$ and occur in this order or its reverse under $\Omega$. We need the following.

(8.1) Algorithm.
*Input:* A society $(G,\Omega)$.
*Output:* A cross in $(G,\Omega)$ if one exists.
*Running time:* $O(v\varepsilon)$.
*Description:* We assume that $|\bar{\Omega}| \ge 4$, for otherwise no cross exists. Let $H$ be the cubic graph formed by two vertex-disjoint circuits $C_1,C_2$, both of length $|\bar{\Omega}|$, and for $1 \le i \le |\bar{\Omega}|$ an edge joining the $i$ th vertices of $C_1$ and $C_2$. Identify $V(C_2)$ with $\bar{\Omega}$, in order, and let $s_1,s_2,t_1,t_2$ be distinct vertices of $C_1$, in order. We apply an algorithm of Shiloach [27][F16] to find two paths

[F15] **Missing proof.** The paper says only that the proofs of (7.5) and (7.6) may easily be converted to algorithms. The paragraphs above give the method and the running time of each step that is not read off directly from these proofs. The method is the one used in the formalization.

[F16] **Missing proof.** Shiloach's paper [27] gives the algorithm in outline. On page 454 it says "We have not written the solution as an explicit, long, and tedious algorithm". A complete algorithm for finding a cross, with a proof, is in the note *Extracting a crossing in time $O(nm)$*, in the folder docs/crossing of the formalization.

of $G \cup H$, vertex-disjoint, joining $s_1, t_1$ and $s_2, t_2$, respectively, if they exist. If we find two such paths it is easy to derive a cross in $(G, \Omega)$; and if there are not two such paths it is easy to see that $(G, \Omega)$ has no cross. (Since Shiloach's algorithm has running time $O(v\varepsilon)$, so does (8.1).) $\square$

Let $(G, \Omega)$ be a society. A subset $Z \subseteq V(G)$ is *trisecting* if $\bar{\Omega} \subseteq Z$, and for every $v \in V(G) - Z$ there are $\leq 3$ vertices $z \in Z$ such that there is a path of $G$ between $v$ and $z$ with no other vertex in $Z$. Thus, $V(G)$ is always trisecting.

A *core* $C$ in $(G, \Omega)$ is a simple graph with vertex set a trisecting subset $Z \subseteq V(G)$, in which distinct $u, v \in Z$ are adjacent if and only if there is a [p. 94] path in $G$ between $u$ and $v$ with no internal vertex in $Z$. (Note that $E(C) \not\subseteq E(G)$ in general.) If $C$ is a core, we say $v \in V(C)$ is 4-linked in $C$ if there is no separation $(A, B)$ of $C$ of order $\leq 3$ with $v \in V(A) - V(B)$ and $\tilde{\Omega} \subseteq V(B)$. A core $C$ is 4-linked if every $v \in V(C)$ is 4-linked in $C$.

(8.2) *Let $C$ be a core of $(G, \Omega)$, and let $v \in V(C)$ not be 4-linked. Let $(A, B)$ be a separation of $C$ with $v \in V(A) - V(B)$ and $\tilde{\Omega} \subseteq V(B)$, of minimum order, and let $C'$ be the simple graph obtained from $B$ by adding edges joining all pairs of vertices in $V(A \cap B)$ if they are not already adjacent. Then $C'$ is a core of $(G, \Omega)$, and each $v' \in V(C')$ which is 4-linked in $C$ is also 4-linked in $C'$.*

*Proof.* (1) *$V(B)$ is trisecting.*

*Subproof.* Let $u \in V(G) - V(B)$ and suppose that $z_1, \ldots, z_4 \in V(B)$ are distinct, and for $1 \leq i \leq 4$ $P_i$ is a path of $G$ between $u$ and $z_i$ with no vertex in $V(B)$ except $z_i$. Since $|V(A \cap B)| \leq 3$, we may assume that $z_4 \notin V(A)$. Since $V(C)$ is trisecting, some vertex of $P_1 \cup \cdots \cup P_4$ belongs to $V(A) - V(B)$. Hence there is a path of $P_1 \cup \cdots \cup P_4$ between $z_4$ and $V(A) - V(B)$ with no internal vertex in $V(C)$. But then the ends of this path are adjacent in $C$, contradicting that $(A, B)$ is a separation of $C$. This proves (1).

(2) *If $a, b \in V(A \cap B)$ are distinct, there is a path of $G$ with ends $a, b$ and with no internal vertex in $V(B)$.*

*Subproof.* By the minimality of $|V(A \cap B)|$ and Menger's theorem, there are $|V(A \cap B)|$ paths of $A$ between $v$ and $V(A \cap B)$, mutually disjoint except for $v$, and the union of two of these yields a path $P$ of $A$ with ends $a$ and $b$ and with no internal vertex in $V(B)$. For each $e \in E(P)$, let $P_e$ be a path of $G$ with the same ends as $e$ with no internal vertex in $V(C)$. Then $\bigcup(P_e : e \in E(P))$ is connected, and meets $V(B)$ in precisely $\{a, b\}$, and so it includes a path of $G$ between $a$ and $b$ with no internal vertex in $V(B)$, as required.

(3) *For distinct $a, b \in V(B)$, there is a path of $G$ with ends $a, b$ and*

*with no internal vertex in $V(B)$ if and only if $a, b$ are adjacent in $C'$.*

*Subproof.* Since every two vertices in $V(A \cap B)$ are adjacent in $C'$, we may assume by (2) that $b \notin V(A \cap B)$. If $a, b$ are adjacent in $C'$, they are adjacent in $C$ and are therefore joined by a path of $G$ as required. Conversely, suppose that $P$ is a path of $G$ with ends $a, b$ and with no internal vertex in $V(B)$. Since $b$ is not adjacent in $C$ to any vertex in $V(A) - V(B)$, it follows that $V(P) \cap V(A) \subseteq \{a\}$, and so $a, b$ are adjacent in $C$ and hence in $C'$, as required.

From (1) and (3) we see that $C'$ is a core.

[p. 95] (4) *If $v' \in V(C')$ is 4-linked in $C$ then it is 4-linked in $C'$.*

*Subproof.* Suppose that $(A', B')$ is a separation of order $\leq 3$ of $C'$ with $v' \in V(A') - V(B')$ and $\tilde{\Omega} \subseteq V(B')$. Since $V(A \cap B)$ is the vertex set of a complete subgraph of $C'$, either $V(A \cap B) \subseteq V(A')$ or $V(A \cap B) \subseteq V(B')$. In the first case $((A \cup A') \cap C, B' \cap C)$ and in the second case $(A' \cap C, (A \cup B') \cap C)$ is a separation of $C$ of order $\leq 3$ showing that $v'$ is not 4-linked in $C$, as required.

From (4), the result follows. □

From (8.2) we obtain

(8.3) Algorithm.

*Input:* A society $(G, \Omega)$.

*Output:* A 4-linked core of $(G, \Omega)$.

*Running time:* $O(\nu\varepsilon)$.

*Description:* Initially, set $C = G$; then $C$ is a core, but it need not be 4-linked. For each vertex $v$ of $G$ in turn, if $v \in V(C)$ test if $v$ is 4-linked in $C$; and if $v$ is not 4-linked in $C$, choose $(A, B)$ as in (8.2), define $C'$ as in (8.2), and replace $C$ by $C'$. When this terminates, $C$ is a 4-linked core, by the last assertion of (8.2). (For each $v \in V(G)$ this takes time $O(\varepsilon)$; and so the total running time is $O(\nu\varepsilon)$.) □

Next, we need the following.

(8.4) *If $C$ is a core of $(G, \Omega)$ then $(G, \Omega)$ has a cross if and only if $(C, \Omega)$ has a cross.*

The proof is easy and we omit it (compare [18, Theorem (11.10)]).

*Proof.* Let $Z = V(C)$. Thus $\bar{\Omega} \subseteq Z$. Suppose first that $P \cup Q$ is a cross in $(G, \Omega)$, where $P, Q$ have ends $p_1, p_2$ and $q_1, q_2$. Let the vertices of $P$ in $Z$ be $z_1, \ldots, z_n$, in order along $P$. Then $z_1 = p_1$ and $z_n = p_2$, since $p_1, p_2 \in \bar{\Omega} \subseteq Z$. For $1 \leq i < n$ the subpath of $P$ between $z_i$ and $z_{i+1}$ has no internal vertex in $Z$, and so $z_i, z_{i+1}$ are adjacent in $C$. Hence $C$ has a path $P'$ with ends $p_1, p_2$ and $V(P') \subseteq V(P)$, and similarly a path $Q'$ with ends $q_1, q_2$ and $V(Q') \subseteq V(Q)$. Then $P' \cup Q'$ is a cross in $(C, \Omega)$.

Conversely, let $P' \cup Q'$ be a cross in $(C, \Omega)$, where $P', Q'$ have ends $p_1, p_2$ and $q_1, q_2$. For each edge $e$ of $C$, with ends $x, y$ say, choose a path $J_e$ of $G$ between $x$ and $y$ with no internal vertex in $Z$, which exists since $C$ is a core. Let $U_1$ be the union of $V(P')$ and of the sets $V(J_e)$ $(e \in E(P'))$, and define $U_2$ similarly from $Q'$. We claim that $U_1 \cap U_2 = \emptyset$. First, $V(P') \cap V(Q') = \emptyset$. Next, if $e \in E(Q')$ then $V(J_e) \cap Z$ is the set of ends of $e$, which lie in $V(Q')$, and so $J_e$ is disjoint from $V(P')$. Similarly $J_e$ is disjoint from $V(Q')$ for $e \in E(P')$. Finally, let $e_1 \in E(P')$ with ends $x_1, y_1$ and $e_2 \in E(Q')$ with ends $x_2, y_2$, and suppose that $v \in V(J_{e_1}) \cap V(J_{e_2})$. Then $v \notin Z$, since $J_{e_1}$ and $J_{e_2}$ meet $Z$ only in their ends, and $\{x_1, y_1\} \cap \{x_2, y_2\} = \emptyset$. But then subpaths of $J_{e_1}$ and $J_{e_2}$ are paths of $G$ from $v$ to the four distinct vertices $x_1, y_1, x_2, y_2$ of $Z$, each with no other vertex in $Z$, contrary to the fact that $Z$ is trisecting. This proves that $U_1 \cap U_2 = \emptyset$. Now replacing each edge $e$ of $P'$ by $J_e$ gives a walk of $G$ between $p_1$ and $p_2$ with all its vertices in $U_1$, and so $G$ has a path $P$ between $p_1$ and $p_2$ with $V(P) \subseteq U_1$. Similarly $G$ has a path $Q$ between $q_1$ and $q_2$ with $V(Q) \subseteq U_2$. Then $P \cup Q$ is a cross in $(G, \Omega)$. □[F17]

We say that $(G, \Omega)$ is *rural* if it can be drawn in a disc with the vertices of $\tilde{\Omega}$ drawn in the boundary of the disc, in order according to $\Omega$. The following is related to results of [9, 18, 26, 27, 29].

(8.5) *Let $C$ be a 4-linked core of $(G, \Omega)$. Then $(G, \Omega)$ has no cross if and only if $(C, \Omega)$ is rural.*

*Proof.* By (8.4), $(G, \Omega)$ has no cross if and only if $(C, \Omega)$ has no cross. But by [18, Theorems (2.3) and (2.4)], $(C, \Omega)$ has no cross if and only if $(C, \Omega)$ is rural, since $C$ is 4-linked. □

Let $(G, \Omega)$ be a society. For a subgraph $A$ of $G$ we define $\partial A$ to be the set of all $v \in V(A)$ such that either $v \in \tilde{\Omega}$ or $v$ is incident with an edge of $G$ not in $A$. A division of $(G, \Omega)$ is a set $\mathcal{A}$ of subgraphs of $G$ such that

(i) $\bigcup(A : A \in \mathcal{A}) = G$

(ii) for all distinct $A, A' \in \mathcal{A}$, $E(A \cap A') = \emptyset$ and $V(A \cap A') = \partial A \cap \partial A'$

[p. 96] (iii) for each $A \in \mathcal{A}$, $|\partial A| \leq 3$ and there are $|\partial A|$ mutually vertex-disjoint paths of $G$ between $\partial A$ and $\tilde{\Omega}$

(iv) for each $A \in \mathcal{A}$ and all $u, v \in \partial A$, there is a path of $A$ between $u$ and $v$ with no internal vertex in $\partial A$

(v) if $A, A' \in \mathcal{A}$ are distinct then $\partial A \neq \partial A'$

(vi) for each $A \in \mathcal{A}$, some vertex or edge of $A$ belongs to no other member of $\mathcal{A}$.

[F17] **Missing proof.** The paper omits the proof of (8.4). The argument above is the one used in the formalization.

If $\mathcal{A}$ is a division, we write $\partial\mathcal{A} = \bigcup(\partial A : A \in \mathcal{A})$. Thus, $\tilde{\Omega} \subseteq \partial\mathcal{A}$. We say a division $\mathcal{A}$ of $(G, \Omega)$ is *rural* if $(G', \Omega)$ is rural, where $G'$ is the bipartite graph with vertex set $\mathcal{A} \cup \partial\mathcal{A}$ in which $A \in \mathcal{A}$ is adjacent to $v \in \partial\mathcal{A}$ if $v \in \partial A$.

(8.6) ALGORITHM.

*Input:* A society $(G, \Omega)$.

*Output:* Either a cross in $(G, \Omega)$, or a rural division of $(G, \Omega)$.

*Running time:* $O(\nu\varepsilon)$.

*Description:* First we use (8.1) to find a cross if there is one. If not, we use (8.3) to find a 4-linked core $C$. Let $\mathcal{A}'$ be the set of all subgraphs $A$ of $G$ such that either

(i) $A$ consists of an edge of $G$ with both ends in $V(C)$, and its ends, or

(ii) $A$ consists of a component of $G \setminus V(C)$, together with all edges of $G$ between this component and $V(C)$, and their ends, or

(iii) $A$ consists of a vertex $v \in \tilde{\Omega}$ incident with no edges of $G$.

We compute $\mathcal{A}'$. Finally, say $A_1, A_2 \in \mathcal{A}'$ are *equivalent* if $\partial A_1 = \partial A_2$, let $\mathcal{A}_1, \ldots, \mathcal{A}_k$ be the equivalence classes into which $\mathcal{A}'$ is partitioned by this relation, and let

$$\mathcal{A} = \left\{ \bigcup(A : A \in \mathcal{A}_i) : 1 \le i \le k \right\}.$$

Output $\mathcal{A}$, and stop.

We must prove that $\mathcal{A}$ is a rural division of $(G, \Omega)$. First, we observe:

(1) *For each $A \in \mathcal{A}'$, $|V(A) \cap V(C)| \le 3$ and there are $|V(A) \cap V(C)|$ mutually vertex-disjoint paths of $G$ between $V(A) \cap V(C)$ and $\tilde{\Omega}$.*

*Subproof.* Now $|V(A) \cap V(C)| \le 3$ since $V(C)$ is trisecting. Since $C$ is 4-linked, there are $|V(A) \cap V(C)|$ mutually vertex-disjoint paths of $C$ between $V(A) \cap V(C)$ and $\tilde{\Omega}$; and these can be converted easily to paths in $G$.

[p. 97] We deduce

(2) *For each $A \in \mathcal{A}'$, $V(A) \cap V(C) = \partial A$.*

*Subproof.* Now $V(A) \cap V(C) \subseteq \partial A$, by (1). For the converse, let $v \in \partial A$. Then $v \in V(A)$, and we claim that $v \in V(C)$. Since $\tilde{\Omega} \subseteq V(C)$ we may assume that $v \notin \tilde{\Omega}$, and so there is an edge $e$ of $G$ incident with $v$, with $e \notin E(A)$. By definition of $\mathcal{A}'$, $v \in V(C)$ as required.

From (1) and (2) it follows that $\mathcal{A}'$ satisfies conditions (i), (ii), (iii), (iv) in the definition of a division.

(3) *Let $X \subseteq V(G)$, let $\mathcal{B} = \{A \in \mathcal{A}' : \partial A = X\}$, and let $B = \bigcup(A : A \in \mathcal{B})$. Then $\partial B = X$, if $\mathcal{B} \neq \emptyset$.*

*Subproof.* Certainly $\partial B \subseteq X$. For the converse, since $\mathcal{B} \neq \emptyset$, there exists $A \in \mathcal{B}$, and by (1) there are $|X|$ mutually vertex-disjoint paths of $G$ between $X$ and $\tilde{\Omega}$. Each of these paths has no vertex in any $A' \in \mathcal{B}$ except its end in $X$, because $\partial A' = X$. It follows that each of these paths has no vertex in $B$ except its end in $X$, and so $\partial B = X$, as required.

Since $\mathcal{A}'$ satisfies (i)-(iv) in the definition of a division, it follows from (3) that $\mathcal{A}$ satisfies (i)-(v) and hence is a division. To see that $\mathcal{A}$ is rural, we observe first that $(C, \Omega)$ is rural, by (8.5). For each $A \in \mathcal{A}$, the vertices of $V(A) \cap V(C)$ form a complete subgraph of $C$, and since $C$ is 4-linked, this is either empty, a single vertex, an edge of $C$, or a triangle of $C$ bounding a region of the drawing of $C$. Since no two members of $\mathcal{A}$ correspond to the same complete subgraph of $C$ (and in particular, no two correspond to the same triangular region) it follows that $\mathcal{A}$ is rural, as required.

Thus the algorithm gives the desired output. Moreover, its running time is $O(\nu\varepsilon)$, since (8.1) and (8.3) both take time $O(\nu\varepsilon)$ and computing $\mathcal{A}'$ and $\mathcal{A}$ both take time $O(\varepsilon)$. □

## 9. Finding a Flat Section of a Wall

We begin by defining a *subwall* of a wall. First, let $J$ be an elementary wall of height $h$. Draw $J$ in the plane as in Fig. 1, and let us name the vertices of $J$ by $x, y$-coordinates, taking the origin in the bottom left of the figure. Thus, $J$ has vertex set

$$\{(x, y) : 0 \leq x \leq 2h + 1, 0 \leq y \leq h\} - \{(0, 0), (2h + 1, h)\}$$

and corners $(1, 0), (2h + 1, 0), (0, h), (2h, h)$. Let $h'$ be even with $2 \leq h' \leq h$, choose $i, j$, even, with $0 \leq i \leq 2h - 2h'$ and $0 \leq j \leq h - h'$, and let $J'$ be the subgraph of $J$ induced on the vertex set

$$\{(x, y) : i \leq x \leq i + 2h' + 1, j \leq y \leq j + h'\} - \{(i, j), (i + 2h' + 1, j + h')\}$$

[p. 98] with corners $(i+1, j)$, $(i+2h'+1, j)$, $(i, j+h')$, $(i+2h, j+h')$. Then $J'$ is an elementary wall of height $h'$. Now let $H$ be a subdivision of $J$, and let $H'$ be the corresponding subdivision of $J'$; we say that the wall $H'$ is a *subwall* of $H$. It follows that if $H'$ is a subwall of $H$, then $H'$ consists of its perimeter $C$ together with the portion of $H$ inside $C$. (Thus the wall $H'$ of (7.3) is in general not a subwall of $H$.) If $H'$ is a subwall of $H$, we shorten "inside the perimeter of $H'$" to "inside $H'$", etc. If $H$ is a wall, and $H_1, H_2$ are non-null subgraphs of $H$, we define $d(H_1, H_2)$ to be the minimum of $d(v_1, v_2)$ over all $v_1 \in V(H_1)$ and $v_2 \in V(H_2)$, where $d$ is the distance function of $H$.

Let $H$ be a wall in a graph $G$, and let $p_1, p_2, p_3, p_4$ be the corners of $H$, in cyclic order. A *cross over* $H$ in $G$ is a subgraph $P \cup Q$, where $P, Q$ are mutually vertex-disjoint paths of $G$ with ends $p_1, p_3$ and $p_2, p_4$, respectively.

(9.1) *For any integer $p \geq 0$ there exist $k, r \geq 0$ such that the following holds. Let $H$ be a wall in $G$, and let $H_1, \ldots, H_k$ be subwalls of $H$ with $d(H_i, H_j) \geq r$ $(1 \leq i < j \leq k)$, where $d$ is the distance function of $H$. For $1 \leq i \leq k$, let $P_i \cup Q_i$ be a cross over $H_i$, such that*

(i) *$(P_i \cup Q_i) \cap H \subseteq H_i$, for $1 \leq i \leq k$*

(ii) *$(P_i \cup Q_i) \cap (P_j \cup Q_j)$ is null for $1 \leq i < j \leq k$.*

*Then $G$ has a $K_p$-minor. Moreover, a model of $K_p$ in $G$ can be found in time $\leq C\varepsilon(G)$, where $C$ depends only on $p$ and the height of $H$.*

*Proof.* Let $\Sigma$ be a sphere, and for $p$ as given, let $\theta, t$ be as in [19, Theorem (4.5)]. We may assume that $t \geq 2$. Let $k = t + 1$ and let $r = 4\theta + 4$. Now let $H, G, H_1, \ldots, H_k$ and $P_i, Q_i$ $(1 \leq i \leq k)$ be as in the theorem, and let $H$ be drawn in $\Sigma$. All but one of the regions of $H$ are "bricks;" let $s$ be a point of $\Sigma$, from the region that is not a brick. For $1 \leq i \leq k$, let $d_i$ be the minimum of $d(s, x)$ taken over all $x \in V(H_i)$. We may assume that $d_1, \ldots, d_t \geq d_k$. Since for $1 \leq i \leq t$ we have $r \leq d(H_i, H_k) \leq d_i + d_k$, it follows that $d_1, \ldots, d_t \geq \frac{1}{2}r$.

Let $H$ have $n$ horizontal paths, and for each of them (say the $j$th), let $P_j$ be a subpath of length $\frac{1}{2}r - 2 = 2\theta$ with one end in the perimeter of $H$. Since $d_i \geq \frac{1}{2}r = 2\theta + 2$ for each $i$, it follows that $P_j$ is disjoint from $H_i$, for all $i$ $(1 \leq i \leq t)$ and $j$ $(1 \leq j \leq n)$. Let $\Gamma$ be obtained from $H$ by deleting all of $H_i$ except its perimeter, for $1 \leq i \leq t$. Then for $1 \leq j \leq j' \leq n$ there are $\theta$ paths of $\Gamma$ between $V(P_j)$ and $V(P_{j'})$, mutually vertex-disjoint. Let $\mathscr{T}$ be the set of all separations $(A, B)$ of $\Gamma$ of order $< \theta$ such that none of $P_1, \ldots, P_n$ is a subgraph of $A$. The result follows easily from [19, Theorem (4.5)] applied to $\Sigma, \Gamma, \mathscr{T}$. □

[p. 99] Let $H'$ be a subwall of a wall $H$ in a graph $G$. We say $H'$ is *dividing* if there is no $H$-path in $G$ with one end inside $H'$ and the other end outside, and *non-dividing* otherwise. The following result is closely related to Theorem (6.3) of [21] and its proof is along similar lines (although much easier).

(9.2) *For any $l \geq 3$, and $m, p \geq 0$, there exist $k, r \geq 0$ such that the following holds. Let $H$ be a wall with distance function $d$ in a graph $G$, and let $H_1, \ldots, H_k$ be non-dividing subwalls of $H$, with*

$$d(H_i, H_j) \geq r \quad (1 \leq i < j \leq k).$$

*Then either there is an $(l, m)$-star over $H$ or $G$ has a $K_p$-minor. Moreover,*

*there is an algorithm with running time $\leq C\varepsilon(G)$, where $C$ depends only on $l, m, p$ and the height of $H$, which given such subwalls $H_1, \ldots, H_k$, finds either an $(l, m)$-star over $H$ or a model of $K_p$ in $G$.*

*Proof.* Let $k, r$ be sufficiently large, in terms of $l, m, p$. (We shall specify later how to choose $k$ and $r$.) Then $H_1, \ldots, H_k$ are mutually disjoint.

Let $F$ be a minimal subgraph of $G$ such that for $1 \leq i \leq k$, $F$ contains an $H$-path from inside $H_i$ to outside $H_i$. Thus $E(F \cap H) = \emptyset$. We can find $F$ in time $O(\varepsilon)$, for fixed $l, m, p, k, r$. In fact the algorithm needs only a subgraph $F'$ of $G$ with the same property, of bounded size except for divalent vertices, and we find such a subgraph as follows. Let $1 \leq i \leq k$. Since $H_i$ is non-dividing, there is an $H$-path of $G$ from inside $H_i$ to outside $H_i$, and we find one, $Q_i$ say, in time $O(\varepsilon)$ by breadth-first search from the set of vertices inside $H_i$, using only edges not in $E(H)$ and continuing only from vertices not in $V(H)$, until a vertex of $H$ outside $H_i$ is reached. We build a subgraph $F'$ of $G$ in $k$ steps, starting from the null graph. At step $i$, if no internal vertex of $Q_i$ belongs to $F'$, we add $Q_i$ to $F'$. Otherwise let $z$ be the first internal vertex of $Q_i$ (from its end inside $H_i$) that belongs to $F'$, and add the subpath of $Q_i$ between its end inside $H_i$ and $z$. Then $z \notin V(H)$, since $z$ is an internal vertex of the $H$-path $Q_i$. Let $j < i$ be the first step at which a path containing $z$ was added, and let $A$ be the subpath of this path between its end inside $H_j$ and $z$. The subpath of $Q_i$ between its end inside $H_i$ and $z$ meets $A$ only in $z$, because no vertex of $Q_i$ before $z$ is in $F'$ except possibly the first, which is inside $H_i$, while the only vertex of $A$ in $V(H)$ is its end inside $H_j$. So the union of these two subpaths is an $H$-path of $F'$ from inside $H_i$ to inside $H_j$, and so to outside $H_i$, since $H_i$ and $H_j$ are disjoint. Thus at the end $E(F' \cap H) = \emptyset$, and for $1 \leq i \leq k$ the graph $F'$ contains an $H$-path from inside $H_i$ to outside $H_i$.

The algorithm uses $F'$ in place of $F$, as follows. By the choice of $z$, each of the $k$ paths added to $F'$ meets the earlier ones only in its ends. Hence the vertices of $H \cup F'$ that are neither branch vertices of $H$ (the vertices that correspond to vertices of the elementary wall) nor ends of these $k$ paths have valency 2 in $H \cup F'$, and so the size of $H \cup F'$ is bounded, except for divalent vertices, by a function of $k$ and the height of $H$. Moreover $H_1, \ldots, H_k$ are non-dividing subwalls of $H$ in the graph $H \cup F'$, and so by the first assertion of (9.2), applied to $H \cup F'$ in place of $G$, there is an $(l, m)$-star over $H$ in $H \cup F'$, or $H \cup F'$ has a $K_p$-minor. Thus at the end of the proof (page 101), where the algorithm searches $H \cup F$, it searches $H \cup F'$ instead. To do so, we call the branch vertices of $H$ and the ends of the $k$ paths *nodes*, and we replace the internal vertices of each path of $H \cup F'$ between two nodes, with no node as an internal vertex, by a single new vertex. This takes time $O(\varepsilon)$.

The new graph has bounded size, and in it the images of $H, H_1, \ldots, H_k$ are a wall and non-dividing subwalls of it, with the same distances as before. We find an $(l, m)$-star over the new wall or a model of $K_p$ in the new graph by examining all its subgraphs, in time bounded in terms of $l, m, p, k$ and the height of $H$. Replacing each new vertex by the path that it replaced then gives an $(l, m)$-star over $H$ or a model of $K_p$ in $G$, in time $O(\varepsilon)$.[F18] Since any circuit of $F$ has $\geq 2$ vertices in $V(H)$ (by the minimality of $F$) it follows that we may express $F$ as a union of trees $S_j$ $(1 \leq j \leq t)$, where $S_j \cap S_{j'} \subseteq H$ for all $j \neq j'$, and every vertex of each $S_j$ in $V(H)$ is a leaf of $S_j$, and each leaf of $S_j$ is in $V(H)$.

A vertex $v \in V(F \cap H)$ is *essential* if $v$ is inside $H_i$ for some $i$, and $v$ has valency 1 in $F$, and no other vertex of $F$ is inside $H_i$.

(1) *For $1 \leq j \leq t$, if $e \in E(S_j)$ there exists $i$ with $1 \leq i \leq k$ such that some leaf $v$ of $S_j$ inside $H_i$ is essential, and every path of $S_j$ between $v$ and another leaf uses $e$.*

*Subproof.* From the minimality of $F$, there exists $i$ $(1 \leq i \leq k)$ such that every $H$-path in $F$ from inside $H_i$ to outside $H_i$ uses $e$. In particular, every such $H$-path is a path of $S_j$, and so for $j' \neq j$, either no leaf of $S_{j'}$ is inside $H_i$ or no leaf of $S_{j'}$ is outside $H_i$. The latter contradicts the minimality of $F$, for then any edge of $S_{j'}$ could be deleted from $F$. Hence for $j' \neq j$, no leaf of $S_{j'}$ is inside $H_i$. Let $v$ be a leaf of $S_j$ inside $H_i$, and suppose that some leaf $u \neq v$ of $S_j$ is either inside $H_i$ or in the perimeter of $H_i$. Let $f$ be the edge of $S_j$ incident with $u$. There is a path of $S_j$ from $v$ to outside $H_i$, and it does not use $f$. Also, for $i' \neq i$, there is a path of $F$ from inside $H_{i'}$ to outside $H_{i'}$, and it can be chosen not to use $f$, by rerouting it to $v$ instead of to $u$ if necessary. But this contradicts the minimality of $F$. Consequently, $v$ [p. 100] is the only leaf of $S_j$ inside $H_i$ or in its perimeter. Let $u \neq v$ be a leaf of $S_j$. We have already seen that $u$ is outside $H_i$, and hence the path of $S_j$ between $u$ and $v$ uses $e$. This proves (1).

We deduce

(2) *For $1 \leq j \leq t$, at most one vertex of $S_j$ has valency $\geq 3$ in $S_j$; and if $S_j$ has $\geq 3$ leaves they are all essential, and if $S_j$ has two leaves then at least one is essential.*

*Subproof.* The first claim follows from (1) applied to an edge $e$ of the path of $S_j$ between two vertices of valency $\geq 3$. The second and third claims also follow from (1).

(3) *If $r \geq m$ then any vertex of $F$ of valency $\geq l$ (in $F$) is the center*

[F18] **Missing proof.** The paper does not say how $F$ is found, nor how $H \cup F$ is searched. The method above is the one used in the formalization.

*of an $(l,m)$-star in $G$ over $H$.*

*Subproof.* If $v$ is such a vertex and $v \notin V(H)$ then $v$ is the unique vertex of valency $\geq 3$ of some $S_j$ by (2), and we may choose the desired $(l,m)$-star from $S_j$. If $v \in V(H)$, then $v$ is a leaf of at least $l$ different $S_j$'s, each of which is an $H$-path from $v$ to some essential vertex, by (2); thus the union of $l$ of these $S_j$'s yields the desired $(l,m)$-star.

By (3) we may assume that every vertex of $F$ has valency $< l$. It follows that

(4) *For any $n \geq 0$, if $k \geq n(l-1)^2$, then there exist $n$ mutually vertex-disjoint $H$-paths $Q_1, \ldots, Q_n \subseteq F$ with ends $s_i, t_i$ $(1 \leq i \leq n)$, and we may renumber $H_1, \ldots, H_k$ so that $s_i$ is inside $H_i$, $t_i$ is outside $H_i$, and no other $s_j$ or $t_j$ is inside $H_i$ $(1 \leq i \leq n)$.*

*Subproof.* By (2), for $1 \leq j \leq k$ at most one vertex of $S_j$ belongs to $S_{j'}$ for some $j' \neq j$, and that vertex only belongs to $\leq l-2$ $S_{j'}$'s for $j' \neq j$. Thus $V(S_j \cap S_{j'}) \neq \emptyset$ for $\leq l-2$ values of $j' \neq j$. Choose $J \subseteq \{1, \ldots, t\}$ maximal such that $S_j \cap S_{j'}$ is null for all distinct $j, j' \in J$. Since each $S_{j'}$ $(j' \notin J)$ meets $S_j$ for some $j \in J$, and any $S_j$ $(j \in J)$ meets $\leq l-2$ $S_{j'}$'s $(j' \notin J)$, it follows that $t - |J| \leq (l-2)|J|$, that is, $|J| \geq t(l-1)^{-1}$. Now each $S_j$ $(1 \leq j \leq t)$ has $\leq l-1$ leaves, and each $H_i$ $(1 \leq i \leq k)$ contains a leaf of some $S_j$, and so $(l-1)t \geq k$. Thus $|J| \geq k(l-1)^{-2} \geq n$. For each $j \in J$ let $Q_j \subseteq S_j$ be an $H$-path with one end an essential vertex $v$, and the other end outside $H_i$, where $v$ is inside $H_i$; then the claim follows.

Later we shall specify five numbers $r_1, n_1, n_2, n_3, r$ depending on $l, m, p$ but not on $G$. Then $n$ will be defined in terms of $r_1, n_1, n_2, n_3, r$ and $k$ will be determined by (4).

(5) *For all sufficiently large $r_1$ and $n_1$, if there exists $I \subseteq \{1, \ldots, n\}$ with $|I| = n_1$ such that $d(s_i, t_i) \geq r_1$ for $i \in I$ and $d(t_i, t_{i'}) \geq r$, for distinct $i, i' \in I$, then $G$ has a $K_p$-minor.*

[p. 101] *Subproof.* If $i \in I$, there are at most two $i' \in I - \{i\}$ such that either $d(s_i, t_{i'}) \leq \frac{1}{2}r_1$ or $d(t_i, s_{i'}) \leq \frac{1}{2}r_1$. Thus we may choose $I' \subseteq I$ with $|I'| \geq \frac{1}{3}n_1$ such that for distinct $i, i' \in I'$,

$$d(s_i, s_{i'}), d(s_i, t_{i'}), d(t_i, s_{i'}), d(t_i, t_{i'}) > \frac{1}{2}r_1.$$

By (7.4), if $r_1$ and $n_1$ are sufficiently large, $G$ has a $K_p$-minor. This proves (5).

Choose $r_1, n_1$ to satisfy (5).

(6) *If $n_2$ and $r$ are sufficiently large, and there exists $I \subseteq \{1, \ldots, n\}$ with $|I| = n_2$ such that $d(s_i, t_i) \leq r_1$ for all $i \in I$, then $G$ has a $K_p$-minor.*

*Subproof.* Now $d(s_i, C) \geq \frac{1}{2}r - 1$ for all $i \in I$ except possibly one, where $C$ is the perimeter of $H$. Hence we may enlarge $n_2 - 1$ of the $H_i$'s to subwalls $H_i'$, say of $H$, still mutually far apart but now with $t_i$ inside $H_i'$. These $H_i'$'s have crosses as in (9.1), and the claim follows from (9.1).

(7) *If $n_3$ and $r$ are sufficiently large, and there exist $i_0$ with $1 \leq i_0 \leq n$ and $I \subseteq \{1, \ldots, n\} - \{i_0\}$ such that $|I| = n_3$ and $d(t_i, t_{i_0}) \leq r_1$ for $i \in I$, then $G$ has a $K_p$-minor.*

*Subproof.* By (7.3) we may choose a wall $H' \subseteq H$ with $t_i \notin V(H')$ $(i \in I)$ and with all the $t_i$'s $(i \in I)$ in the same component of $H \setminus V(H')$. Since $H \setminus V(H')$ has maximum valency $\leq 3$, we may pair many of the $t_i$'s by vertex-disjoint paths of $H$ disjoint from $H'$. This yields many $H'$-paths in $G$ with their ends mutually far apart, and $G$ has a $K_p$-minor by (7.4), with suitable choice of $n_3$ and $r$. This proves (7).

Choose $n_2, n_3$, and $r$ to satisfy (6) and (7). (All these choices are independent of $G$.) Finally, we observe that

(8) *If $n$ is sufficiently large, then there exists $I \subseteq \{1, \ldots, n\}$ as in (5), (6), or (7).*

We choose $n$ to satisfy (8) and $k$ as in (4). The result follows.

With $k$ and $r$ chosen as in the theorem, the algorithm to find an $(l, m)$-star over $H$ or a model of $K_p$ is trivial. First, we determine $F$. This takes time $O(\varepsilon)$. Since the size of $F$ is bounded (exept for divalent vertices) by a function of $l, m$, and $p$, we can find an $(l, m)$-star or a model of $K_p$ in $H \cup F$ in time $\leq C\varepsilon(G)$, where $C$ depends only on $l, m, p$, and the height of $H$. □

From (9.2) we obtain

(9.3) *For any $p \geq 0$ there exists $k, r \geq 0$ such that there is an algorithm as in (9.4), with running time $\leq C\varepsilon(G)$, where $C$ depends on $p, t$, and the height of $H$.*

[p. 102] (9.4) ALGORITHM. *Input:* A graph $G$, a wall $H \subseteq G$ with distance function $d$, and subwalls $H_1, \ldots, H_t$ of $H$ with $d(H_i, H_j) \geq r$ $(1 \leq i < j \leq t)$.

*Output:* Either (i) a model of $K_p$ in $G$, or (ii) a subset $X \subseteq V(G)$ with $|X| < \binom{p}{2}$ and a subset $I \subseteq \{1, \ldots, t\}$ with $|I| \geq t - k$ such that for all $i \in I$, $H_i$ is dividing in $(G \setminus X) \cup H$.

*Proof* of (9.3). Let $l \geq 4$, $m \geq 0$ satisfy (7.7), and let $k, r \geq 0$ satisfy (9.2). Now we describe the algorithm for (9.4). We set $Y_1 = \emptyset$, and perform a recursion. At the beginning of the $i$ th iteration we have a set $Y_i$ of $i - 1$ $(l, m)$-stars over $H$ in $G$, with distinct centers. We proceed as follows:

(1) If $i = \binom{p}{2}$ we apply (7.7) to obtain a model of $K_p$ in $G$, and stop.

(2) Otherwise $i < \binom{p}{2}$. Let $X$ be the set of centers of the stars in $Y_i$. Let $G' = (G \setminus X) \cup H$. Then $H$ is a wall in $G'$, and no $(l, m)$-star in $G'$ has

center in $X$ (since $l \geq 4$ and every vertex of $G'$ in $X$ has valency $\leq 3$).

(3) We test which $H_i$'s are dividing in $G'$. If at least $t-k$ of them are dividing in $G'$ we stop.

(4) Otherwise, at least $k$ of the $H_i$'s are non-dividing. We choose $k$ of them and apply (9.2). We obtain either a model of $K_p$ in $G'$ and hence in $G$, when we stop, or an $(l,m)$-star $S$ over $H$ in $G'$, when we set $Y_{i+1} = Y_i \cup \{S\}$ and return for the next iteration.

This algorithm has running time $\leq C\varepsilon(G)$, where $C$ depends only on $p$, $t$, and the height of $H$, because there are at most $\binom{p}{2}$ iterations, each involving $t$ tests of "is $H_i$ dividing?" which takes time $O(\varepsilon)$, together possibly with an application of (7.7) or (9.2). □

Let $H$ be a wall in a graph $G$, with perimeter $C$. Let $W \subseteq V(G)$ be the union of $V(C)$ and the vertex set of the unique component of $G \setminus V(C)$ that contains a vertex of $H$. We define the *compass* of $H$ in $G$ to be the restriction of $G$ to $W$. It follows that $H$ is a subgraph of its compass, and its compass is connected.

(9.5) *Let $H$ be a wall in a graph $G$. If $H_1$ is a dividing subwall of $H$ with compass $K_1$, then $K_1 \cap H \subseteq H_1$; and if $H_2$ is another dividing subwall of $H$ with compass $K_2$, and $H_1 \cap H_2$ is null, then $K_1 \cap K_2$ is null.*

*Proof.* Since $H_1$ is an induced subgraph of $H$, to prove the first claim it suffices to show that $V(K_1 \cap H) \subseteq V(H_1)$. Suppose that $t \in V(K_1 \cap H) - V(H_1)$. Since $t$ belongs to the compass of $H_1$ and not to the perimeter $C_1$ of $H_1$, there is a path of $G \setminus V(C_1)$ between $t$ and the inside of $H_1$, and hence there is an $H$-path with one end inside $H_1$ and the other end outside. But this contradicts that $H_1$ is dividing and proves the first claim.

[p. 103] For the second claim, certainly $K_1 \cap H_2$ is null, since $K_1 \cap H_2 \subseteq K_1 \cap H \subseteq H_1$, and similarly $K_2 \cap H_1$ is null. Suppose that $t \in V(K_1 \cap K_2)$. Since $t \in V(K_1)$ and $t \notin V(H)$, there is a path of $G \setminus V(C_1)$ between $t$ and the inside of $H_1$. By combining this and a similar path for $H_2$, we obtain an $H$-path from the inside of $H_1$ to the inside of $H_2$, a contradiction. Hence $K_1 \cap K_2$ is null, as required. □

A subwall $H$ of $G$ is *flat* in $G$ if there is no cross $P \cup Q$ over $H$ such that $P \cup Q$ is a subgraph of the compass of $H$ in $G$.

(9.6) *For any $p, s \geq 0$ and $h \geq 2$, even, there exists $h' \geq p^2 + 2$, even, and an algorithm as in (9.7) with running time $O(v\varepsilon)$ for fixed $p, s, h$.*

(9.7) Algorithm. *Input:* A graph $G$, and a wall $H$ in $G$ of height $h'$. *Output:* Either (i) a model of $K_p$ in $G$, or (ii) a subset $X \subseteq V(G)$ with $|X| < \binom{p}{2}$ and $s$ subwalls $H_1, \ldots, H_s$ of $H$ of height $h$, mutually vertex-

disjoint, and each flat and dividing in $(G \setminus X) \cup H$, and each disjoint from $X$.

*Proof* of (9.6). We may assume that $p \geq 2$. Let $k = k_1$, $r = r_1$ satisfy (9.3), and let $k = k_2$, $r = r_2$ satisfy (9.1) for the given $p$. Let $h'$ be some even number satisfying

$$h' \geq \left(s + k_1 + k_2 + \binom{p}{2} + 1\right) \cdot (h + \max(r_1, r_2) + 1).$$

Our algorithm is as follows. Given $G, H$ as input, we choose $s + k_1 + k_2 + \binom{p}{2}$ subwalls of $H$, each of height $h$, and mutually at distance $\geq \max(r_1, r_2)$ (this is possible from the choice of $h'$). We apply (9.4) and obtain either a model of $K_p$ in $G$ when we stop, or a subset $X \subseteq V(G)$ with $|X| < \binom{p}{2}$ and some $s + k_2 + \binom{p}{2}$ of our subwalls, each dividing in $(G \setminus X) \cup H$. Now we test which of these subwalls are flat in $(G \setminus X) \cup H$ and find crosses over them in their compasses, if possible, using (8.1). Suppose that we find crosses for at least $k_2$ of them. By (9.5), these crosses satisfy the hypotheses of (9.1). By (9.1) there is a model of $K_p$ in the union of $H$ and these crosses, and we can find it in time $O(\varepsilon)$. On the other hand, if fewer than $k_2$ of our subwalls yield crosses, then at least $s + \binom{p}{2}$ of them are flat in $(G \setminus X) \cup H$, and since $|X| < \binom{p}{2}$ we may choose $s$ of them disjoint from $X$. Again we stop. □

Let $H$ be a wall in $G$. The *society* of $H$ in $G$ is the society $(K, \Omega)$, where $K$ is the compass of $H$ in $G$ and $\Omega$ is the cyclic permutation of the corners [p. 104] of $H$ corresponding to their order in the perimeter of $H$. Let $\mathcal{A}$ be a rural division of the society of $H$ in $G$. We say that $A \in \mathcal{A}$ is *interior* if $A \cap C$ is null, where $C$ is the perimeter of $H$.

(9.8) *For all $p \geq 0$ and $h \geq 2$, even, there exists $\omega \geq 0$ such that there is an algorithm as in (9.9), with running time $O(v\varepsilon)$ for fixed $p, h$.*

(9.9) Algorithm. *Input:* A graph $G$. *Output:* Either (i) a branch-decomposition of $G$ of width $\leq \omega$, or (ii) a model of $K_p$ in $G$, or (iii) a subset $X \subseteq V$ with $|X| < \binom{p}{2}$ and a flat wall $H$ in $G \setminus X$ of height $h$, and a rural division $\mathcal{A}$ of the society of $H$ in $G \setminus X$, and a branch-decomposition of width $\leq \omega$ of each interior member of $\mathcal{A}$.

*Proof* of (9.8). We may assume that $p \geq 2$. Let $s = \binom{p}{2} + 5$. Choose $h' \geq 2$, even, as in (9.6). Choose $\omega$, by (4.2), so that every graph with branch-width $\geq \omega/3$ contains a wall of height $h'$. Now we describe the algorithm. We are given $G$ as input.

(1) We apply (7.2) to $G$. (This is possible by (7.1).) If we obtain a branch-decomposition of $G$ of width $\leq \omega$ we stop. Otherwise we obtain a wall $H$ in $G$ of height $h'$.

(2) Now we perform a recursion. We set $G_1 = G$, $Z_1 = \emptyset$, $H_1 = H$. At the beginning of the $i$ th iteration we have

(i) a subgraph $G_i$ of $G$ with $|\varepsilon(G_i)| \leq (\frac{1}{2})^{i-1}|\varepsilon(G)|$

(ii) a subset $Z_i \subseteq V(G_i)$ with $|Z_i| \leq s-3$ such that no vertex of $G_i \setminus Z_i$ is incident with an edge of $G$ not in $G_i$,

(iii) a wall $H_i$ in $G_i$ of height $h'$.

The iteration proceeds as follows. (3) We apply (9.7) to $G_i, H_i$. If we obtain a model of $K_p$ in $G_i$ and hence in $G$ we stop. Otherwise, we obtain a subset $X \subseteq V(G_i)$ with $|X| < \binom{p}{2}$ and $s$ subwalls $H^1, \ldots, H^s$ of $H_i$ of height $h$, mutually vertex-disjoint, and each flat and dividing in $(G_i \setminus X) \cup H_i$, and each disjoint from $X$. Let $K^j$ be the compass of $H^j$ in $(G_i \setminus X) \cup H_i$ $(1 \leq j \leq s)$.

(4) Since the $H^j$'s are all dividing in $(G_i \setminus X) \cup H_i$, it follows that $K^1, \ldots, K^s$ are mutually vertex-disjoint and that each contains no vertex in $X$, by (9.5). Since $|Z_i| \leq s-3$, we may choose three of the $H^j$'s, say $H^1$, $H^2$, and $H^3$, such that $V(K^j) \cap Z_i = \emptyset$ $(1 \leq j \leq 3)$.

[p. 105] Since $H^3$ has height $h' \geq p^2 + 2$ it follows that $\varepsilon(K^3) \geq \varepsilon(H^3) \geq p^2 + 2 \geq |X| + 2$. Let there be $d_j$ edges in $G_i$ with one end in $X$ and the other in $V(K^j)$, for $1 \leq j \leq 3$. Then (we recall that by definition, $\varepsilon(G) = |E(G)| + |V(G)| + 1$) we have

$$\varepsilon(K^1) + \varepsilon(K^2) + \varepsilon(K^3) - 2 + d_1 + d_2 + d_3 + |X| \leq \varepsilon(G_i).$$

But $\varepsilon(K^3) \geq |X| + 2$, and so

$$\varepsilon(K^1) + \varepsilon(K^2) + d_1 + d_2 + 2|X| \leq \varepsilon(G_i).$$

Choose one of them, $K^1$ say, so that $\varepsilon(K^1) + d_1 + |X| \leq \frac{1}{2}\varepsilon(G_i)$. (5) We apply (8.6) to obtain a rural division $\mathcal{A}$ of the society of $H^1$ in $(G_i \setminus X) \cup H_i$. (6) For each interior $A \in \mathcal{A}$, we apply (7.2) to $A$ and obtain either a branch-decomposition of $A$ of width $\leq \omega$, or a wall in $A$ of height $h'$.

(7) If for each interior $A \in \mathcal{A}$ we obtain a branch-decomposition of $A$ of width $\leq \omega$, we stop.

(8) If for some interior $A \in \mathcal{A}$ we obtain a wall $H_{i+1}$ in $A$ of height $h'$, we let $G_{i+1}$ be the subgraph of $G_i$ with vertex set $V(A) \cup X$ and edges those edges of $G_i$ with an end in $V(A)$ and the other end in $V(A) \cup X$. Since $A$ is a subgraph of $K^1$, it follows that

$$\varepsilon(G_{i+1}) \leq \varepsilon(A) + d_1 + |X| \leq \varepsilon(K^1) + d_1 + |X| \leq \frac{1}{2}\varepsilon(G_i) \leq \left(\frac{1}{2}\right)^i \varepsilon(G).$$

Let $\partial A$ be the set of all vertices $v$ of $A$ such that either $v$ is a corner of $H^1$ or $v$ is incident with an edge of $(G_i \setminus X) \cup H_i$ not in $E(A)$; then $|\partial A| \leq 3$ since $\mathcal{A}$ is a division. Set $Z_{i+1} = X \cup \partial A$; then $|Z_{i+1}| \leq \binom{p}{2} + 2 = s - 3$, since $|X| \leq \binom{p}{2} - 1$. We return for the next iteration.

We must verify that the conditions (i)-(iii) of (2) hold with $G_i, Z_i, H_i$ replaced by $G_{i+1}, Z_{i+1}, H_{i+1}$. Certainly (i) holds, as we saw above, and so does (iii). Let us verify (ii). Let $v \in V(G_{i+1} \setminus Z_{i+1})$, and let $e \in E(G)$ be incident with $v$, and suppose for a contradiction that $e \notin E(G_{i+1})$. Since $v \in V(G_{i+1} \setminus Z_{i+1}) = V(A) - \partial A$ and $V(A) \subseteq V(K^1)$ and $V(K^1) \cap Z_i = \emptyset$, it follows that $v \notin Z_i$, and so $e \in E(G_i)$. Since $e$ has one end in $V(A)$ and $e \notin E(G_{i+1})$, its other end is not in $V(A) \cup X$, and in particular, $e \in E((G_i \setminus X) \cup H_i)$. Since $A$ is internal and hence $v$ does not belong to the perimeter of $H^1$, it follows that $e \in E(K^1)$, since $v \in V(A) \subseteq V(K^1)$. But every edge of $K^1$ incident with $v$ belongs to $E(A)$, since $\mathcal{A}$ is a division of the society of $H^1$ in $(G_i \setminus X) \cup H_i$ and $v \in V(A) - \partial A$. This contradicts that $e \notin E(G_{i+1})$ as required, and completes the verification of (ii).

This completes the description of the algorithm. To estimate running time we observe that step (1) takes time $O(v^2 + \varepsilon)$. Steps (3) and (5) both [p. 106] take time $O(v\varepsilon)$. Step (6) takes time $\Sigma(Cv(A')\varepsilon(A') : A' \in \mathcal{A})$ for some constant $C$. But for $A' \in \mathcal{A}$, $\varepsilon(A') \leq 4(\varepsilon(A') - |\partial A'| - 1)$, since some vertex or edge of $A'$ belongs to no other member of $\mathcal{A}$, and hence, summing over $A' \in \mathcal{A}$,

$$\begin{aligned}\Sigma v(A')\varepsilon(A') &\leq \Sigma 4v(A')(\varepsilon(A') - |\partial A'| - 1) \\ &\leq 4v(G_i)\Sigma(\varepsilon(A') - |\partial A'| - 1) \leq 4v(G_i)\varepsilon(G_i).\end{aligned}$$

Consequently, step (6) takes time $O(v(G_i)\varepsilon(G_i))$. Thus the $i$th iteration takes time $O(v(G_i)\varepsilon(G_i))$. Since $\varepsilon(G_i) \leq (\frac{1}{2})^{i-1}\varepsilon(G)$ and $v(G_i) \leq v(G)$, the whole recursion takes time $O(v\varepsilon)$. Hence the algorithm takes time $O(v\varepsilon)$, as claimed. □

## 10. Homogeneously Labelled Walls

Our last task is to show how to make use of the third possible outcome of (9.8) when solving Folio. The following condition is a common hypothesis of several results to follow and so we give it a name.

Hypothesis C. *$G$ is a digraph, $X \subseteq V(G)$ with $|X| = q$, $H$ is a wall in $G \setminus X$, and $\mathcal{A}$ is a rural division of the society of $H$ in $G \setminus X$.*

Under Hypothesis C, let $(K, \Omega)$ be the society of $H$ in $G \setminus X$. Let $A \in \mathcal{A}$ be interior, and let $|\partial A| = k$; then $k \leq 3$. Choose a numbering $\{s_1, \ldots, s_k\}$

of $\partial A$, where if $k = 3$ we number "clockwise;" that is, we number $s_1, s_2, s_3$ so that there are three paths $P_1, P_2, P_3$ of $K$ from $\partial A$ to $\bar{\Omega}$, mutually vertex-disjoint, where $P_i$ has ends $s_i, t_i$ $(i = 1, 2, 3)$, and $t_1, t_2, t_3$ occur in this order under $\Omega$. Let $X = \{x_1, \ldots, x_r\}$; the sequence $(s_1, \ldots, s_k, x_1, \ldots, x_r)$ is called an *attachment sequence* of $A$, and we denote it by $\pi(A)$. For each $A \in \mathcal{A}$, choose a subgraph $\tilde{A}$ of $G$ with $X \subseteq V(\tilde{A})$ and with $\tilde{A} \setminus X = A$, such that every edge of $G$ with both ends in $X \cup V(K)$ is in $E(\tilde{A})$ for exactly one $A \in \mathcal{A}$. Let $T_A$ be the $\delta$-folio of $(\tilde{A}, \pi(A))$.

Now for each interior $A \in \mathcal{A}$, let $v(A) \in V(H)$ such that there is a path of $K$ from $\partial A$ to $v(A)$ with no vertex in $V(H)$ except $v(A)$. (Such a path always exists since $K$ is connected and $A$ is non-null.) This information (that is, for each interior $A \in \mathcal{A}$, a choice of $\tilde{A}$, an attachment sequence $\pi(A)$ of $A$, the $\delta$-folio of $(\tilde{A}, \pi(A))$, and a vertex $v(A)$) is called a *vision*. For fixed $\omega$ and $q$, we need the following.

(10.1) Algorithm.

*Input:* $G$, $X$, $H$, $\mathcal{A}$ satisfying Hypothesis C, and an integer $\delta \geq 0$, and for each interior $A \in \mathcal{A}$ a branch-decomposition of $A$ of width $\leq \omega$.

[p. 107] *Output:* A vision.

*Running time:* $O(\varepsilon^2)$.

*Description:* First we compute an attachment sequence for each interior $A \in \mathcal{A}$. For each $A$ this takes time $O(\varepsilon(G))$, and so we can compute all the attachment sequences in time $O(\varepsilon^2)$, since $|\mathcal{A}| \leq |E(G)|$.

For each interior $A \in \mathcal{A}$ we choose $\tilde{A}$, and find a branch-decomposition of $\tilde{A}$ of width $\leq \max(\omega, 2) + q$ (this exists,[F19] since $\tilde{A}$ is obtained from $A$ by adding $q$ extra vertices), by adapting the given branch-decomposition of $A$ in the natural way. This takes time $O(\varepsilon)$.

Third, we compute the $\delta$-folio of $(\tilde{A}, \pi(A))$ for each interior $A \in \mathcal{A}$, by (4.1). This takes time $O(v\varepsilon)$, since $\max(\omega, 2) + q$ is a constant.

Fourth, for each interior $A \in \mathcal{A}$ we compute some $v(A)$, for instance, by extending a spanning tree of $H$ to a spanning tree of its compass. This takes time $O(v\varepsilon)$ again, and completes the algorithm.

Under Hypothesis C and given some vision, a subwall $H'$ of $H$ is said to be *$h$-homogeneous* if for every internal $A \in \mathcal{A}$ with $v(A)$ inside $H'$ and for every subwall $H''$ of $H'$ of height $h$, there exists some internal $A' \in \mathcal{A}$ with $v(A')$ inside $H''$ such that $(A', \pi(A'))$ has the same $\delta$-folio as $(A, \pi(A))$. The

[F19] **Proof gap.** A graph with at most one edge has branch-width 0 by the convention of page 70, and adding vertices can raise its branch-width to 2. For example, with $\omega = 0$ and $q = 1$, adding to $A = K_2$ a vertex joined to both ends gives $K_3$, of branch-width $2 > \omega + q$. With $\max(\omega, 2)$ in place of $\omega$ the bound holds. The width is still a constant, which is all that (4.1) needs, so the algorithm and its running time are unchanged.

following lemma is proved in [24]; it is an application of the main results of [21, 22].

(10.2) *For any $q$, $\delta \geq 0$ and $h \geq 2$, even, there exists $f(h) \geq h$, even, such that under Hypothesis C, if $H' \subseteq H$ is an $h$-homogeneous subwall of height $f(h)$ then both middle vertices of $H'$ are irrelevant to the $\delta$-folio of $G$ relative to $X$.*

(The middle vertices of a wall are the two vertices corresponding to the two middle vertices of the elementary wall of which the wall is a subdivision.) We use (10.2) to show

(10.3) *For all $q$, $\delta \geq 0$ there exists $h \geq 0$, even, such that for all $\omega \geq 0$ there is an algorithm as in (10.4) with running time $O(\varepsilon^2)$.*

(10.4) Algorithm.

*Input:* $G$, $X$, $H$, $\mathcal{A}$ satisfying Hypothesis C, where $H$ has height $h$, and for each interior $A \in \mathcal{A}$ a branch-decomposition of $A$ of width $\leq \omega$.

*Output:* A vertex irrelevant to the $\delta$-folio of $G$ relative to $X$.

*Proof of* (10.3). Let $f$ be the function of (10.2) (for given $q$, $\delta$). Let $\mathcal{L}$ be the class of all rooted digraphs with $\leq q+3$ roots and with detail $\leq \delta$. Then $\mathcal{L}$ is the union of finitely many isomorphism classes $W_1, \ldots, W_l$, say.

[p. 108] For any rooted graph with $\leq q+3$ roots, its $\delta$-folio is a union of some of $W_1, \ldots, W_l$. Let

$$\mathcal{T}_1 = \left\{ \bigcup (W_i : i \in I) : I \subseteq \{1, \ldots, l\} \right\}.$$

Thus, $\mathcal{T}_1$ contains the $\delta$-folio of every rooted graph with $\leq q+3$ roots, and $\mathcal{T}_1$ is finite. Let $|\mathcal{T}_1| = n$, let $h_{n+1} = 2$, for $1 \leq i \leq n$ let $h_i = f(h_{i+1})$, and let $h = h_1$. Then $h$ satisfies (10.3), for an algorithm proceeds as follows. Given input as in (10.4), we first use (10.1) to find a vision. Second we set $H_1 = H$ and perform a recursion. At the beginning of the $i$th iteration we have $H_i$ and $\mathcal{T}_i$ such that

(i) $H_i$ is a subwall of $H$ of height $h_i$

(ii) $\mathcal{T}_i \subseteq \mathcal{T}_1$ and $|\mathcal{T}_i| = n + 1 - i$

(iii) for each internal $A \in \mathcal{A}$, if $v(A)$ is inside $H_i$, then $\mathcal{T}_i$ contains the $\delta$-folio of $(\tilde{A}, \pi(A))$.

The iteration proceeds as follows. We check if $H_i$ is $h_{i+1}$-homogeneous. If it is, we may output one of its middle vertices and stop, by (10.2). Otherwise we choose a subwall $H_{i+1}$ of $H_i$ of height $h_{i+1}$ such that some $T \in \mathcal{T}_1$ is the $\delta$-folio of $(\tilde{A}, \pi(A))$ for some internal $A \in \mathcal{A}$ with $v(A)$ inside $H_i$, and $T$ is not the $\delta$-folio of $(\tilde{A}, \pi(A))$ for any internal $A \in \mathcal{A}$ with $v(A)$ inside $H_{i+1}$. Then $T \in \mathcal{T}_i$; we set $\mathcal{T}_{i+1} = \mathcal{T}_i - \{T\}$, and return for the next iteration.

Each iteration takes time $O(v)$ (since there are only a bounded number of subwalls of $H_i$ to check) and there are at most $l+1$ iterations. Finding a $\delta$-vision initially takes time $O(\varepsilon^2)$, and so the algorithm also has running time $O(\varepsilon^2)$. □

Now we give the complete algorithm for Folio. Let $\xi$ and $\delta \geq 0$ be fixed integers.

(10.5) Algorithm.

*Input:* A digraph $G$ and a subset $Z \subseteq V(G)$ with $|Z| \leq \xi$.

*Output:* The $\delta$-folio of $G$ relative to $Z$.

*Running time:* $O(v^3)$.

*Description:* (1) Let $p$, $C$ be as in (6.5). We apply (6.5) to $G$, $Z$, and obtain a subdigraph $G' \subseteq G$ with $Z \subseteq V(G')$ such that $G$ and $G'$ have the same $\delta$-folio relative to $Z$ and such that $|E(G')| \leq C|V(G')|$.

Let $q = \binom{p}{2} + \xi - 1$; let $h$ be as in (10.3); let $\omega$ be as in (9.8). We set $G_1 = G'$ and perform a recursion. At the beginning of the $i$th iteration we have a subdigraph $G_i$ of $G'$ with $|V(G_i)| = |V(G')| - i + 1$ and with $Z \subseteq V(G_i)$, such that the $\delta$-folios of $G_i$ and $G'$ relative to $Z$ are equal. The iteration proceeds as follows. We apply (9.9) to $G_i \setminus Z$.

[p. 109] (2) If we obtain a branch-decomposition of $G_i \setminus Z$ of width $\leq \omega$, we convert it to a branch-decomposition of $G_i$ of width $\leq \max(\omega, 2) + \xi$[F20], compute the $\delta$-folio of $G_i$ relative to $Z$ by (4.1), and stop.

(3) If we obtain a model of $K_p$ in $G_i \setminus Z$ and hence in $G_i$, we apply (6.2) to find a vertex $v$ of $G_i$ irrelevant to the $\delta$-folio of $G_i$ relative to $Z$; we set $G_{i+1} = G_i \setminus v$ and return for the next iteration.

(4) Otherwise, we obtain $X$, $H$, $\mathcal{A}$ as in Hypothesis C (with $G$ replaced by $G_i \setminus Z$), where $|X| \leq \binom{p}{2} - 1$ and $H$ has height $h$. Put $X' = X \cup Z$; then $G_i$, $X'$, $H$, $\mathcal{A}$ also satisfy Hypothesis C. We apply (10.4) to $G_i$, $X'$, $H$, $\mathcal{A}$, to compute an irrelevant vertex $v$ for the $\delta$-folio of $G_i$ relative to $X'$. Since $Z \subseteq X'$, $v$ is also irrelevant for the $\delta$-folio of $G_i$ relative to $Z$, by (2.1). We set $G_{i+1} = G_i \setminus v$, and return for the next iteration.

The application of (6.5) in step (1) takes time $O(v^3)$. In the recursion, each iteration takes time $O(\varepsilon(G')^2)$. Since there are $\leq v(G)$ iterations, and $\varepsilon(G') \leq O(v(G))$, the algorithm takes time $O(v^3)$. □

We would like to say that not only does there exist an algorithm as in (10.5), but we can *construct* it, and that depends on being able to construct the various constants $p, h, \omega$ used in the course of the proof, for given nu-

[F20] **Proof gap.** The same correction as in Algorithm (10.1), for the same reason: if $G_i \setminus Z$ has at most one edge, its branch-width is 0 by convention, and adding the $\xi$ vertices of $Z$ can give branch-width 2.

merical values of $\xi$ and $\delta$. Now $p$ is easy, but $h$ and $\omega$ are defined by (10.3) and (9.8), and these are formulated as purely existential statements, containing no method for calculating $h$ and $\omega$. ("For all $q$, $\delta \geq 0$ there exists $h \geq 0 \ldots$ ") Nevertheless the proofs of (10.3) and (9.8) do yield procedures for calculating $h$ and $\omega$, except that (10.3) uses the function $f$ of (10.2), and (9.8) uses $h'$ of (9.5) and another $\omega$, from (4.2). The sequences of definitions can be traced further and further back, through other papers in this series, until they terminate; and all the proofs used to define these constants are constructive. Even, say, Theorem (3.5) of [16], which asserts that "For every matching $G$ in $\Sigma$ and every homoplasty class $\mathcal{D}$ there exists $H \in \mathcal{D}$ with $U(G) \cap U(H)$ finite" can be proved constructively, in the sense that it is easy to give an algorithm which, given explicitly some $G$ and $\mathcal{D}$, will compute an appropriate $H$. Thus, we do indeed have an algorithm which, with input numbers $\xi$ and $\delta \geq 0$, finds the algorithm of (10.5); and also the constant $C$ in the bound on its running time can be constructed.

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